Learning, Complexity, and the Paradoxes of Confirmation

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To my parents

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Contents

Abstract

Some classical problems in epistemology are essentially about learning from experience. This thesis takes two of them – Goodman's New Riddle of Induction and Hempel's Paradox of the Ravens – and suggests that insights from computational learning theory, the theory behind machine learning, can help dissolve them. I adopt two main premises: first, learning is a computational process, so insights from computational learning theory apply to how we learn. Second, for data used in a learning process to be about physical phenomena, there must be a causal connection between those phenomena and the data's being as they are.

Chapter 1 explains these two premises and uses them to argue that Goodman's New Riddle of Induction does not add a further challenge beyond Hume's old riddle. Chapters 2-4 argue for the significance of complexity for epistemology. Complexity constrains what can be learnt by learners like us. Chapters 2-3 discuss *computational complexity* and its connection to learnability and confirmation. Chapter 4 discusses *algorithmic complexity,* which is a measure of the complexity of data, and argues that some philosophical difficulties with this concept are essentially the same as those posed by Goodman's New Riddle and can be dealt with in analogous ways. Chapter 4 also develops a variant of algorithmic complexity that is specifically suited to the needs of epistemology and discusses its relation to learnability and predictability. Chapters 5-6 discuss Hempel's Paradox of the Ravens. Chapter 5 argues that the standard Bayesian approach to the paradox falls short of solving it. Chapter 6 presents my solution to the paradox, which is based on considerations of computational and algorithmic complexity explained in Chapters 2-4. Chapter 7 draws a broader methodological conclusion. Idealisation is indispensable in epistemology, but models of learning and confirmation should not idealise away the computational and representational limitations of the learner, which are essential to explaining the very phenomena being modelled.

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"The light dove, cleaving the air in her free flight, and feeling its resistance, might imagine that its flight would be still easier in empty space." Immanuel Kant, Critique of Pure Reason, A5/B8 (1929/1787, Kemp Smith translation).

Introduction

Epistemology is the study of knowledge. The process of transitioning from ignorance to knowledge is learning. What is knowable but not yet known by subjects for whom it is knowable must therefore be learnable. What is now known but was once unknown must have been learnt. A better understanding of learning and learnability may therefore be beneficial for epistemology. It is thus somewhat surprising that epistemologists have so far given relatively little attention to learning and learnability.

Not everyone will agree with the characterisation of epistemology as the study of knowledge. Some would give priority to the study of understanding, rationality, justified beliefs, accurate credences, or other kinds of *cognitive successes*, to use Steup & Neta's (2015) term from their *Stanford Encyclopaedia of Philosophy* entry on epistemology. However, it seems that whatever cognitive success or successes you have in mind as the subject matter of epistemology, achieving these cognitive successes involves what we would reasonably call learning. Cognitive success is something you gain by processing information, either new or old, in ways that lead to a better epistemic position. Not a lot of what I argue in this thesis hangs on which cognitive successes we take as more basic. I predict that one can suitably modify the arguments to apply to one's favourite cognitive successes.

Learning is usually understood as a success term, especially when we say that one *has learnt* something. That is, we say that one has learnt when one's processing of information indeed leads to cognitive success. Sometimes, however, we want to refer to *the attempt* to learn by processing information, regardless of whether it eventually leads to cognitive success or not. When I say that one *learnt that P,* I use *learnt that* as a success term. However, when I talk about *learning processes,* I usually refer to all attempts at learning, whether they involve good processes that tend to lead to cognitive success or not. Indeed, even the same process can succeed at times and fail in others, sometimes due to luck. In general, when I talk about a learning process leading to some end result, I do not wish to imply that this end result is a form of cognitive success, unless explicitly stated otherwise (e.g., when I say that a learning process ends with inferring that *P* it shouldn't imply that the inference was a good one).

Problems related to how we can *achieve* knowledge or other cognitive successes are thus essentially about learning. Famous examples of such problems (not the only ones, for sure) are Hume's Problem of Induction, Goodman's New Riddle of Induction, and Hempel's Paradox of the Ravens. These problems are essentially about why and how we are able to learn from experience. Today, we know much more about learning than we knew 80 or 280 years ago, when these problems were first introduced.[1] Analysing these problems as problems of learning can thus shed new light on them and hopefully help make progress in solving them. This thesis focuses only on Goodman's New Riddle and Hempel's paradox and indeed suggests solutions to both.

We know today more about learning not only due to empirical research on human and animal learning in psychology and biology, but also due to theoretical research in computational learning theory, the theory behind the rapidly growing field of machine learning. Computational learning theory focuses on any learning process that can be modelled by a Turing machine, and deals with more abstract questions about learning, such as which algorithms can lead to successful learning of a solution to a given problem given a certain kind of data; what the minimal computational resources required to learn a solution to a given

[1] Some would say the problem of induction was introduced already by Sextus Empiricus, around two millennia ago.

problem are; what kind and amount of data are required for learning a solution using a given algorithm. It is the abstract knowledge we have gained from computational learning theory that will play a central role in this thesis.

This thesis adopts two main premises. First, it assumes that every learning process is computational, and thus the insights of computational learning theory apply to every learning process. Every learning process can thus be modelled as including data that function as the inputs of the process and computations the learner runs on these data. The second premise is that a necessary condition for the data that are given as inputs to the learning process to encode information *about* physical phenomena, and hence for the output of the learning process to be about the physical world, is that there is a causal connection between the thing that is encoded by the data and the data being the way they are. These two premises will be explained in more detail in Chapter 1.

A common theme across all chapters is the key role our limitations play in our learning. Like Kant's dove metaphor, it is our limitations that make it possible for us to learn the way we do and be occupied with the questions we are occupied with. Creatures with zero mass can fly without air resistance, but ignoring air resistance will prevent us from understanding how massive creatures can still fly. Ignoring or neglecting our limitations might help us understand how unlimited godlike entities can learn, but it prevents us from understanding how learners like us can learn.

Limitations on what information we can encode, and how, explain why we are able to learn that emeralds are green rather than grue. Limitations on our computational resources explain why we can learn that ravens are black from seeing black ravens but cannot learn that ravens are black from seeing white shoes and green apples. Our limitations explain why learning is so complicated, and why learners like us need to make sophisticated trade-offs, test a variety of possible learning algorithms, and learn things that they didn't even consider, instead of somewhat dully updating credences of possibilities we already considered. Limitations make life harder, but in a sense, they also make life much more interesting.

Part 1: Grue

Chapter 1: Learning from Data and the New Riddle of Induction

1. Introduction: Two Riddles of Induction

All emeralds we have observed so far have been green. They have also been grue, where grue is defined as "first examined before 2050 and green or not examined before 2050 and blue". However, we tend to infer from our observations that all emeralds are green, call it *the Green Hypothesis*, and not that all emeralds are grue, call it *the Grue Hypothesis*. We also tend to think that we *should* infer the Green Hypothesis from our evidence and not the Grue Hypothesis. But why? What breaks the symmetry? This is Goodman's (1953/1983) famous *New Riddle of Induction*.

The riddle stems from a seeming symmetry between predicates like *green* and *grue* and how the Green Hypothesis and the Grue Hypothesis relate to our current evidence. Note that *grue*, of course, is only an example. There are infinitely many *gruesome* predicates that accurately describe the objects we have observed, and corresponding gruesome inferences we can make using those predicates, leading to any conclusion we want. A solution to the New Riddle should apply to all gruesome predicates and inferences, not just a specific one.

This chapter proposes a solution to the New Riddle of Induction. I argue that if (a) learning is a computational process, and (b) encoding information requires a causal connection between what is encoded and the data being the way they are, then the symmetry between the Green Hypothesis and the Grue Hypothesis breaks. The argument shows first that the learning processes of any single learner naturally discriminate between *green* and *grue* (§2). We may call it *the intra-learner challenge*. Then it deals with what we may call *the inter-learner challenge*. It shows that for each possible world *w* (or for each possible world except for some special ones whose possibility is unclear), the learning processes of all learners inhabiting *w* similarly discriminate between *green* and *grue* (§3). Assumption (a) will be the main assumption for §2. Assumption (b), which I refer to as the thesis of *data externalism*, plays at best an implicit background role in §2, but plays a crucial role in §3, so I defer discussing it to §3. Before presenting my solution, let me first make a few introductory remarks.

The name of the riddle – "The *New* Riddle of Induction" – suggests that this riddle is supposed to come in addition to, or replace, another riddle, which is, of course, Hume's Old Riddle of Induction. While the two are related, the New Riddle is presented as posing an extra challenge beyond the old one. That is, a solution to Hume's Old Riddle (which might ignore the New Riddle and work only conditionally on there being a solution to it) should not suffice for solving the New Riddle as well.

This chapter is not about Hume's Old Riddle; thus, I shall not present it in detail. For our purposes, it suffices to state the challenge posed by Hume's Old Riddle as the challenge to produce a good argument for the claim that we can learn non-deductively from previous observations about the unobserved. That is, it is the challenge of showing that, and explaining why, *some* (non-deductive)[1] learning processes, whatever they might be, are *good* learning processes, in the sense that they can be used to gain knowledge or justified beliefs about the unobserved based on previous observations.[2]

What is, then, the *distinct challenge* posed by Goodman's New Riddle? The way I suggest answering this question is as follows: The *New* Riddle seems to show that, due to the symmetry between gruesome and natural descriptions of our observations, the epistemic status of gruesome and natural inferences should be the same. Whether and to what extent it's a good idea to infer the Green Hypothesis from our observations, it should be good to the same extent

[1] This chapter focuses only on inductive learning processes, so I will not repeat this qualification every time.

[2] In principle, as Hume's riddle is stated here, a solution to it can be non-constructive, in the sense that it explains why learning inductively from observations about the unobserved is possible, without specifying which learning processes are good. Such a solution would leave the residual challenge of explaining why specific learning processes are good. Doing that will likely require relying on insights from the non-constructive solution to Hume's riddle, together with some specific characteristics of those specific learning processes that are deemed to be good. Some may argue that this residual challenge is part of Hume's riddle, so a non-constructive solution won't be complete. In any case, these two challenges are distinct from the unique challenge posed by the New Riddle.

to infer the Grue Hypothesis (and infinitely many other gruesome hypotheses). Thus, if a solution to the *Old* Riddle needs to show that some inductive learning processes are good, a solution to the *New* Riddle, to the extent it poses an extra challenge beyond the Old Riddle, needs to show how this doesn't automatically imply that infinitely many gruesome processes are equally good. That is, a solution to the New Riddle needs to show where the symmetry between natural and gruesome inferences breaks.

Of course, the problems are connected. Since *any* solution to the *Old* Riddle will tell us that some learning processes are good, and since it can't be that the Green Hypothesis and the Grue Hypothesis are both good, then if we can't break the symmetry between *green* and *grue*, the New Riddle is an obstacle we must remove to fully solve the old one. On the other hand, if there is no solution to the *Old* Riddle, and we can't show that not all inductive learning processes are equally bad, then we can't show that there can be an asymmetry in the epistemic status of the processes that led to the Green Hypothesis and the ones that led to the Grue Hypothesis.

However, the two challenges are still distinct. One can bracket the New Riddle and try to explain why some learning processes are good, with the hope that there is a way to break the symmetry and prevent the solution from applying equally to gruesome processes.[3] One can also bracket the Old Riddle and try to show that there is no relevant symmetry between gruesome and natural processes, with the hope that independent reasons will show that some processes (presumably, natural ones) are good. If the New Riddle should pose a challenge that is distinct from the challenge posed by the Old Riddle, then solving the New Riddle, to the extent that it is a distinct problem from the old one, amounts to showing that the New Riddle doesn’t add any challenge *over and above* the old one.

[3] This is a strategy explicitly adopted by Van Cleve (1984, p. 556).

This will be my attitude in this chapter. My solution to the New Riddle will show that *if* a solution to the Old Riddle *can* be found, then there is no reason to think that such a solution, whatever it might be, should be vulnerable to the extra challenge posed by the New Riddle. In the emeralds example, I will not try to show that inferring the Green Hypothesis from our observations is good, or better than inferring the Grue Hypothesis. Rather, I will try to show that whatever the epistemic status of the Green Hypothesis is (a solution to the *Old* Riddle may tell us that it is good), there is no reason to think that the Grue Hypothesis should have the same status. This is a key move.[4] My solution to the New Riddle is *not* meant to solve the old one as well.

In the next section, I explain why this is the case for a single learner, and in §3 I extend the conclusion to the case of many learners inhabiting the same world. §4 draws the conclusion from §2 and §3. In §5, I discuss how my solution relates to other solutions previously proposed to the New Riddle.

2. Grue for a Single Learner

i. Learning Is Computational

Section 2 aims to establish that it is not even clear that there *can* be symmetry in the role *green* and *grue* play in a single learner's learning processes. The main premise for §2 (and a main premise for the whole thesis) is that learning is a computational process. This assumption can be understood at various levels of strength. The required strength of the premise depends on the required strength of the conclusion that the assumption is required to support.

[4] And, as far as I can tell, a point of divergence from most of the literature about the New Riddle.

If our goal is to establish the claim that *some* learning processes can discriminate between *green* and *grue* in an epistemically relevant way, then we only need to assume that *some* good learning processes that can lead to hypotheses about *green* and *grue* are computational. If our goal is to establish the stronger claim that there is an epistemically relevant asymmetry between *green* and *grue* in the learning processes that *we use* when *we* learn about emeralds (or between natural and gruesome predicates when we learn inductively, in general), we need to assume that some of *our* relevant learning processes are computational. If our goal is to establish the even stronger claim that for *every* good learning process we can non-arbitrarily discriminate between it and its seemingly symmetrical gruesome versions, then we need the assumption to be that all learning processes whatsoever are computational.

Solving the New Riddle requires achieving the first goal, and maybe also the second. The assumption required for achieving the first goal is very weak, and the assumption required for achieving the second goal is also not especially strong. The assumption required for the third goal is indeed more contentious, but still a good working hypothesis. What it says, in effect, is that learning is done via algorithms. Coupled with the Church-Turing thesis, it implies that every learning process can, in principle, be simulated by a (learning) Turing machine.

If not every learning process can, in principle, be simulated by a Turing machine, it follows that the Church-Turing thesis is false, or that some ways to process information are not types of computation, or that some ways of learning don't involve the processing of information. None of these alternatives is plausible, so we can adopt the strongest assumption that all learning processes are computational as a working hypothesis, even though the weaker forms of this assumption should suffice for my argument to work. Adopting the strongest version simplifies the argument I'm going to make. If the assumption in its strongest form is true, we can model any learner as a Turing machine without the need for further qualifications. If the assumption in its strongest form is false, it would be interesting to see where reality deviates from it and whether these deviations are relevant to the discussion of the New Riddle in this chapter and other discussions in this thesis.

ii. A Toy Model for a Single Learner

Learning can involve getting novel information at various stages of the learning process. For this reason, in the general case, we should model the learning process as composed of stages, where in each stage, the learner can get novel information as input and run computations on this novel information as well as on information the learner received and computations the learner did prior to that stage.

However, in the settings of the New Riddle, we have a learner who already has some observational data (about emeralds) and now has the task of inferring the best hypothesis about the next unseen emerald, or all emeralds, based on these data. Thus, we don't need to complicate our model. For the New Riddle, we can model our learner's learning process as composed of a single stage, where a Turing machine gets all the observational data as inputs and, based on these data, needs to infer hypotheses. If learning is computational, then every learner who needs to infer hypotheses based on their data can be modelled this way. For simplicity, we can think of the learner's training data as composed of picture files, some of which are pictures of emeralds. Nothing crucial hangs on modelling our learner this way. To achieve the goal of §2, it suffices then to demonstrate that generally, there is no symmetry between *green* and *grue* for a single learning machine.

The starting point of the New Riddle is that for every emerald in the learner's data, the learner's data encode that the emerald is green.[5] Now, indeed, if all the emeralds observed by the learner are green, it follows, given that 2050 is in the future, that all the emeralds observed by the learner are also grue. However, and here is a key point, we should be careful and *not*

[5] For simplicity, I assume that emeralds that were green when observed are still green.

infer too quickly from this that if the learner's data *encode* for each emerald in the data that it is green, it follows that the learner's data also *encode* for each of the emeralds in the learner's data that it is grue, or that an emerald being green and an emerald being grue are encoded in the learner's data symmetrically. In fact, as we shall see, the latter doesn't follow from the former.

To see that, let us have a closer look at our learner's data. I defer discussing what it means for data to encode properties of emeralds to §3. For simplicity, let's assume that the data come in the form of binary strings, i.e., sequences of bits; each bit can have either 1 or 0 as its value. If each emerald in the learner's data is encoded as being green, it means that some bits in the learner's data encode that. If the learner's data is composed of picture files, we can say that in each picture of an emerald, some bits encode the fact that the emerald is green. For simplicity, let us suppose for now that whether an emerald in a picture is green is encoded by a single bit[6] in each picture file, call it *the Green Bit*, which is 1 if the emerald in the picture is green and 0 otherwise.

If an emerald being green is encoded by the Green Bit being 1, what does it mean for the learner's data to *encode* that all the emeralds in the learner's data are grue? For simplicity, let us slightly change the definition of *grue* and, as is common in the literature, replace the word 'blue' with the words 'not green'. The dramatic role *blue* plays in Goodman's riddle is that we assume things cannot be green and blue at the same time, and *not green* gives us that as well.[7]

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[6] Thinking of *green* as encoded by a single bit only makes the argument less cumbersome. Nothing crucial changes in the argument if *green* is encoded by many bits.

[7] Some philosophers take the fact that objects cannot be both green and grue to be a crucial feature of the New Riddle (see Freitag, 2015, 2016 and especially Zinke, 2018; 2024). For these philosophers it is crucial that we know about the next emerald that it is not yet observed and that we know that it is not both green and grue. I think this is mistaken. Think of the predicate 'guare', where 'square' takes the role of 'blue'. Obviously, things *can* be both green and square, including the next emerald we see (or the first one we observe after 2050). However, presumably, it would still be a bad idea to infer that the next emerald we see will be both green and square, given that all emeralds we saw before a given time were both green and guare. Moreover, knowledge that some emeralds are unobserved is not crucial for the New Riddle. Ignorance about this fact suffices. Even if we happen to have observed all emeralds, and in such a case both the Green Hypothesis and the Grue Hypothesis are true, if we don't *know* that we have observed all emeralds, it would still be weird to infer that all emeralds are grue just from the fact that the emeralds we observed so far were grue. Supposedly, even if by inferring that, in this case, we come

Thus, in this new definition, x is grue if and only if x has been examined before 2050 and is green or has not been examined before 2050 and is not green.

What, then, does it mean for an emerald in the learner's data to be encoded as grue? On the face of it (I revisit this claim in the next section), given that being green is encoded in each picture by the Green Bit, some other bits in each picture need to encode the time the pictured emerald was first examined. Let us refer to the bits that encode the time of first examination as the *timestamp*. If the Green Bit is 1 and the timestamp encodes a time before 2050, or the Green Bit is 0, and the timestamp encodes a later time, then the emerald is *encoded* as grue. To simplify further, we can think of the timestamp as a single bit,[8] which is 1 if the emerald was first examined before 2050 and 0 otherwise. Then, being grue is encoded in our model as:

x is encoded as grue if and only if the timestamp in x's picture is 1 and the Green Bit is 1, or the timestamp of x's picture is 0 and the Green Bit is 0.

Here is the first important observation. Not all data that encode objects being green (assuming *green* is encoded by the Green Bit) necessarily come with timestamps. Thus, while as a matter of logic if all emeralds observed so far, before 2050, were green, then all emeralds observed so far were grue, it can still be the case that the learner's data encode each emerald in the learner's data as green, but do not encode it as grue.[9] However, suppose the learner's data do come with accurate timestamps for each emerald. Then all emeralds encoded in the learner's data were not only green and grue but also *encoded* in the learner's data as both green and grue.

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to believe truly that all emeralds are grue, we can't *know* that all emeralds are grue without the further knowledge that all emeralds were already observed (and therefore were observed before 2050). If we *do* know that all emeralds were already observed and they are all green, then we *can* know that they are all grue, but we learn this deductively, not inductively, so it's a whole different matter. There is still the riddle of why we can learn that all emeralds are green *inductively* but cannot learn that all emeralds are grue *inductively*. Furthermore, animals don't do mineralogy, but they do seem to learn that grass and other things are green, not grue, even if they don't necessarily have the concept of a thing being observed or the knowledge that a thing cannot be both green and blue. Supposedly, the New Riddle applies to inductive learning in general, human, other animals, and non-living learners alike.

[8] Again, thinking of the timestamp as a single bit only simplifies the argument and doesn't change anything crucial.

[9] Moreover, if we allow the learner's data to include false timestamps, then even now, in 2026, the learner's data can encode some emeralds as green but not as grue. Whether we allow timestamps to be false depends on considerations discussed in §3.

However, while both are encoded in the data, *green* and *grue* are not encoded symmetrically. Being green is encoded by a single bit (the Green Bit), and being grue is encoded by an interdependence between two bits – the Green Bit and the timestamp.[10]

Thus, under the assumptions of this model, there is no reason to think that any learning process treats being green and being grue symmetrically. In datasets with no timestamps, *grue* is not even encoded, so nothing can be learnt about which objects are grue from the data. In data with timestamps, inferring from the learner's data that all emeralds are green requires a different process from the one required to infer from the same data the seemingly symmetrical hypothesis that all emeralds are grue. At a minimum, in the first case the head of the Turing machine needs to read one bit from a picture file for the emerald being green to take part in the learning process, whereas in the latter case the head of the Turing machine needs to read two bits from a picture file and compare them for the emerald being grue to take part in the learning process. Thus, the states of the Turing machine, and therefore its learning algorithm, must be different in the two cases.

The hypothesis that all emeralds are grue might appear symmetrical to the hypothesis that all emeralds are green, perhaps due to our natural language describing our data only in a coarse-grained manner. However, at the level of the bits and bytes, and the specific stages of the learner during the learning process, there is nothing symmetrical in how they are inferred. Therefore, if *some* processes are good for learning about the unobserved, there is no reason to think that *these* processes will have trouble discriminating between green and grue, or that if a process leading to the inference that all emeralds are green is good, then another process that leads to the inference that all emeralds are grue must be equally good.

[10] Note that the architecture of this toy model does make it the case that *green* and *being observed before 2050* are encoded symmetrically. However, this is just an artefact of the specific way we simplified our model. The New Riddle doesn't take *green* and *being observed before 2050* as symmetrical, so if this emergent symmetry bothers you, there is nothing ad hoc in encoding the timestamp using two bits instead of one, say. I thank Jeremy Goodman for this point.

iii. Symmetrical Scenarios and Scenarios of Symmetry

A natural worry might arise at this point. What entitles me to say that *green* is encoded by the Green Bit and *grue* is encoded by an interdependence between the Green Bit and the timestamp rather than the other way around? Can't it be that *grue* is encoded by a single bit ("The Grue Bit") and green is encoded by an interdependence between the Grue Bit and the timestamp? It is still correct in this scenario that the learner's data encode that all observed emeralds are green, as the New Riddle requires, but in this symmetrical scenario, the same process that earlier led to inferring the Green Hypothesis would now lead to inferring the Grue Hypothesis and vice versa.

However, note that this symmetric scenario, like the scenario above, is not a scenario of symmetry. As in the above scenario, *green* and *grue* are not encoded symmetrically. They might have switched roles, but still, if there are good learning processes, they can easily discriminate between *green* and *grue*, and if one of those good learning processes leads to inferring the Green Hypothesis, there is no reason to think that there must be another good learning process that leads to inferring the Grue Hypothesis from the same data (and vice versa).

One might worry that if, in the above scenario, the Green Hypothesis is justified and the Grue Hypothesis is not, then in this symmetrical scenario, the Grue Hypothesis would be justified and the Green Hypothesis would not. I shall discuss this worry later (in §3.2), after I discuss the case of mirror learners and the thesis of data externalism. For now, however, let me just state that my goal is to show that the Green Hypothesis and the Grue Hypothesis need not have the same epistemic status, rather than saying which of them has a better standing.[11] We

[11] However, it should be clear how my solution can be used to block an argument that concludes that the Grue Hypothesis is in good standing from the premise that the Green Hypothesis is in good standing. In this regard, see fn. 19.

can see that in the symmetrical scenario, where *grue* is encoded by the Grue Bit, and *green* is encoded by an interdependence between the Grue Bit and the timestamp, the symmetry breaks as well. That is, if there is a solution to the Old Riddle, there is no reason to expect that this solution would have any trouble discriminating between the Green Hypothesis and the Grue Hypothesis, in any of the scenarios discussed so far.

Is there a way to devise a *scenario of symmetry* where *green* and *grue* are encoded symmetrically, such that it's not clear how a good learning process can discriminate between *green* and *grue*, or how a solution to the Old Riddle can discriminate between two processes, one leading to the Green Hypothesis and the other to the Grue Hypothesis? I believe the answer is probably negative, but even if there is a way, it wouldn't be too relevant for the specific challenge posed by the New Riddle. Supposedly, the New Riddle should arise in every possible world, and it arises because logically, our observations can be described in many gruesome but accurate ways. We needn't entertain specific sceptical scenarios for the New Riddle to arise, and in the case of the New Riddle, it's not clear that such scenarios of symmetry are indistinguishable from what we take to be the actual case. I discuss the issue in more detail in §3.2.

3. Grue for Many Learners

As Goodman already noted, the greater challenge of the New Riddle is not how a *single* learner can discriminate between *green* and *grue* (the intra-learner challenge), but how we can discriminate between *two learners*, for one of whom *green* is basic and *grue* is derivative, and for the other, *grue* is basic and *green* is derivative (the inter-learner challenge). This is exemplified in Goodman's famous remark on a language that uses *grue* and *bleen* as primitive concepts (Goodman, 1953/1983, pp. 79-80). We define *bleen* as follows:

x is bleen if x has been examined before 2050 and is blue,[12] *or has not been examined before 2050 and is green.*

Someone who speaks a language in which *grue* and *bleen* are primitive would define *blue* and *green* in terms of *grue* and *bleen* and a moment in time, in an exactly analogous manner, and seemingly any reason we have to prefer *green* over *grue* is a reason for the *grue-bleen English* (Griblish?) speaker to prefer *grue* over *green*. I shall deal with this worry in this section of the chapter.

i. Mirror Learners, Content Externalism, and Data Externalism

The worry about a grue/bleen language can be redescribed as a worry about the existence or the possibility of *mirror learners*. By 'mirror learners' I mean learners who are just like the original learner in terms of how their learning algorithm makes computations on their data, as they are encoded for them, but they mirror the way *green* and *grue* are encoded for the original learner. If for the original learner, *green* is encoded by the Green Bit and *grue* is encoded by the Green Bit and a timestamp, for the mirror learner, *grue* is encoded by the Grue Bit, and *green* is encoded by the Grue Bit and a timestamp. The original learner infers that all emeralds are green. The mirror learner can run the same learning processes on *their* data and infer that all emeralds are grue. Why should we prefer the hypothesis of the first learner over that of the mirror learner?

Here is the place to introduce the thesis of data externalism, which is a natural extension of content externalism, and the second major component of my argument. Content externalism is the view that the content of one's mental state can constitutively depend on external factors from one's environment that are not necessarily accessible to one solely by introspection (commonly captured by the slogan that "meanings just ain't in the head"). This idea was

[12] Or 'not-green' if we use the corresponding variant of 'grue'.

influentially defended by Putnam (1973; 1975) and Burge (1979), and though still controversial, is now widely endorsed. Putnam famously illustrates his point via the story of Twin Earth. On Twin Earth, the chemical formula of the liquid that fills the lakes and the oceans that the locals *call* 'water', and that at normal temperatures and pressures is indistinguishable from water, is not H_2O but rather a liquid with a complicated formula that we can abbreviate as 'XYZ'. But other than that, Twin Earth is a planet very similar to Earth. We may even assume that each of us has a doppelgänger with the same internal constitution.[13]

However, although we and our doppelgängers can be internal duplicates of each other, our word 'water' (it seems) refers to H_2O, and their word 'water' refers to XYZ. The difference, it seems, is that there is a causal chain between liquid H_2O and our use of the word 'water' here on Earth, but no such causal chain exists on Twin Earth. On Twin Earth, there is a causal chain between our doppelgängers' use of the word 'water' and liquid XYZ.[14]

By *Data Externalism,* I refer to a similar view (some might say it's ultimately the same view) that data fed into machines (or data that are given as inputs to learners in general), especially data concerning physical phenomena, can themselves be *about* something, but what they are about can constitutively depend on factors external to the data. In a nutshell, what the bits encode cannot be read solely from the bits. To borrow another famous example from Putnam (1981), a binary string can encode a picture of Winston Churchill, but the fact that it is a picture *of* Churchill is determined by factors that are external to the mere combination of 1's and 0's in the string. It depends on a causal process that led to the creation of the string, which includes, among other things, some causal connection to Churchill and some facts about how pictures are encoded to binary strings in the specific file format. This very string can, in principle (by a coincidence of a cosmic scale, as appropriate), also be read as (say) the first

[13] In such a scenario, the cells in our doppelgängers' brains would be full of XYZ instead of H_2O, but I ignore that.

[14] The mere existence of some causal chain might not suffice. We might need to specify some additional conditions that make it a causal chain "of the right kind". In what follows, I only assume that some causal chain is necessary.

chapter of the Bible in another readily available text file format, but in such a case, we wouldn't say that the string encoded information about the creation of the world.

If data externalism is correct, what data can encode, and how, depends on the physics of the universe. It is the physics of the universe that determines what can cause data to be one way or another, e.g., what can cause a specific bit to be 1 or 0. For the Green Bit to encode that an emerald is green, there needs to be a causal chain from a physical property of the emerald to the Green Bit being 1 instead of 0. For the Grue Bit to encode that an emerald is grue, there needs to be a causal chain from a physical property of the emerald to the Grue Bit being 1 instead of 0.

However, the physics of each possible world in which encoding is possible restricts which causal chains are in accord with the physics of that world. For each such world, the physics of that world determines the fundamental causal interactions in that world, and thereby the permissible causal chains and their structure. Thus, the physics of each world restricts what can be encoded and how, and two properties can be distinct in whether and how the physics of a world allows their encoding. Some properties might not be encodable in some worlds, others might be encodable but only derivatively via a complex causal chain that involves the encoding of other, more fundamental properties. Thus, we cannot just stipulate that a property is encoded in one way or another. Some ways of encoding are physically permissible, and others are not.

We talked about the Green Bit encoding *green* and the Grue Bit encoding *grue*, but we need to ask *in virtue of what* the Green Bit and the Grue Bit encode what they encode. We should not be misled by speaking about them as single bits. In fact, encoding *green* and *grue* by single bits, the Green Bit and the Grue Bit, can require whole mechanisms full of encodings of more basic properties, and the Green Bit and the Grue Bit encode an emerald being green or grue only in virtue of encoding something about these whole mechanisms. The encoding of an emerald being green with a single bit or an emerald being grue with a single bit is itself part of a learning process that includes these whole mechanisms in virtue of which the Green Bit and the Grue Bit can encode. Crucially, though, there is no reason to think that the process in virtue of which the Green Bit encodes *green* is symmetrical to the process by which the Grue Bit encodes *grue*. The physics in each world can enforce an asymmetric process in encoding *green* and *grue*.

To see that, suppose, for the sake of the example, that the actual world is roughly how current physics suggests it is. According to our current understanding of physics, fundamentally, there are four types of causal interactions: electromagnetic, gravitational, weak nuclear, and strong nuclear. Every data-encoding mechanism can encode only in virtue of a causal chain that is composed of interactions of these four types. To encode the properties of an emerald, there must be a causal chain from the emerald to our data encoding mechanisms. Suppose (as we believe to be the case) that the interaction is mainly electromagnetic. Electromagnetic interactions are mediated by photons, and there are only a finite number of quantum properties (e.g., wavelength, polarisation) that determine the effect of a photon on what it interacts with.

The property of emitting photons mainly with wavelengths in a range of roughly 500-570 nm in normal lighting conditions can be directly encoded in normal lighting conditions by mechanisms that are sensitive to photons of these wavelengths (e.g., because they are built from matter that changes its electric potential by absorbing such photons). Suppose that whether people use the *word* ‘green’ is determined by whether an object has this property. We can say that the word ‘green’ refers to this property because there is a suitable causal chain between objects having this property and people’s use of the word ‘green’ to describe the object (as in Putnam’s H_2O example).

However, if being grue is either having this property that people refer to using the word ‘green’ and being first examined before 2050 or not having this property and not being examined before 2050, then the physics of the actual world does not allow for a symmetrical encoding of *grue*. To encode *grue,* a data-encoding mechanism needs to encode an interdependence between the wavelengths of photons and the time of first examination, which is another physical property that is encoded in another way (e.g., by photons emitted by a clock simultaneously with photons emitted by an emerald). There is no basic causal interaction that records both the wavelength and the time of first examination. That is, while there might be a causal chain that lets a learner encode *grue*, encoding *grue* must be done in a non-symmetrical way to encoding *green*.

Thus, if our actual world is like current physics suggests it is, then mirror learners are not physically possible. There might be a mirror learner in another possible world with a different physics, but a learner and a mirror learner usually don't inhabit the same possible world. It might be that in one world *green* is basic and *grue* is derivative, and in another *grue* is basic and *green* is derivative, but for learners inhabiting the same possible world, only one is more basic than the other. Thus, in each possible world, the physics of that world can break the symmetry between *green* and *grue* because the processes required for encoding *green* and *grue,* which are part of the processes required for learning about greenness and grueness, are different.[15]

[15] Can't we say that even though the causal process in virtue of which a learner encodes *green* and *grue* is not symmetrical, their encoding can still be symmetrical, as just two labels 'green' and 'grue'? Supposedly, when we refer to two people, Moses and Einstein, the *names* 'Moses' and 'Einstein' can be treated symmetrically as just labels, regardless of the distinct causal chains that let us refer to Moses and Einstein. My point, however, is not about the *words* 'green' and 'grue', which as mere labels can indeed be treated symmetrically. There is nothing of particular interest that the word 'green' teaches us and the word 'grue' does not. Rather, my point is that the process by which we encode a property (or a person) is part of the learning process and is relevant to what we can *learn* about the property (or the person). We can decide to treat only the word, or the final bit, as what does the work of encoding. Then we would need a different name for the process by which the word or the bit refers, which is part of the learning process and relevant to what can be learnt about what the word or bit refers to. I don't think, however, that it changes anything important in the argument.

ii. Scenarios of Symmetry

It is now time to discuss the scenarios of symmetry mentioned in section 2. iii. We can get a scenario of symmetry in two ways. First, there can be two distinct but entirely symmetrical data encoding mechanisms, one for encoding *green* and the other for encoding *grue*. For simplicity, we can think of them as two bits, the Green Bit, which encodes *green*, and the Grue Bit, which encodes *grue*, with no timestamps involved. Second, there can be one data encoding mechanism that somehow encodes both *green* and *grue* in a symmetrical way. We can think of it as a single bit, the Green-Grue Bit, which encodes some symmetrical combination of *green* and *grue*, or that encodes both *green* and *grue* for objects first examined before 2050 and something else for objects not examined before 2050.

If data externalism is correct, it is not entirely clear that such scenarios of symmetry are even *coherent*. If the word 'green' refers to a specific physical property in our actual world (e.g., the property of emitting mainly photons with wavelengths of 500-570 nm in normal lighting conditions), what would it mean for a possible world to have physics that allows encoding *green* and *grue* symmetrically either in separate bits or in a single bit?

In the separate bits case, it would mean that there are two separate properties, *green* and *grue*; one can cause our data to have the Green Bit being 1, the other can symmetrically cause our data to have the Grue Bit being 1, and these properties necessarily coapply to everything first examined before 2050 and do not coapply to everything not examined before 2050. On the face of it, this requires something like physics that includes two causal forces, we can call them "the electromagnetic force" and the "gruelectromagnetic force", such that these forces allow for interactions that can be recorded, and at least one of these forces is sensitive to whether an object was first examined before 2050, such that its interactions are recorded in the opposite way for objects examined or not examined before 2050. Furthermore, we need to assume that there is no marker on an object that shows whether it was examined before 2050 or not. Otherwise, this marker functions like a timestamp and the symmetry breaks. So, one of the forces needs to change the way it works on our encoding scheme for objects not examined before 2050, without any change or marker in the object. The coherence of such a scenario is far from clear.

However, even if it is coherent, in such a scenario the learner need not have any clue that green and grue (as they are encoded by the learner) must (not) coapply for objects (not) examined before 2050. It's not clear why, in such a scenario, the learner would not infer simply that all emeralds are both green and grue, and after 2050 discover to their surprise that the predicates *green* and *grue* don't coapply for newly examined objects. This is not usually the kind of scenario we have in mind when we discuss the New Riddle. In the New Riddle, we need to explain why, now, before 2050, it is so clear that the Green Hypothesis and the Grue Hypothesis don't have the same epistemic status (assuming it's not known that all emeralds will be observed before 2050). A sudden, inexplicable change in how the world appears or behaves is the business of the Old Riddle, not the new one.

In the single-bit case, we discussed two options: either the Green-Grue Bit encodes a property that is a combination of *green* and *grue*, or the Green-Grue Bit encodes both *green* and *grue* for objects first examined before 2050 and something else for objects not examined before 2050. It is not entirely clear what both options would mean, given data externalism. The first option would imply that some property, which is a combination of *green* and *grue,* causes the Green-Grue Bit to be 1. It's not entirely clear what such a property could be, but if that's the case, then the learner's data don't encode that all emeralds are green but that all emeralds have this property, which is a combination of *green* and *grue*, unlike what is required for the initial settings of the New Riddle.

The second option would imply that the Green-Grue Bit functions like a kind of AND gate for objects first examined before 2050. For such objects, it is caused to be 1 only by objects having both the property *green* and the property *grue*. For objects not examined before 2050, it is caused to be 1 in another way (e.g., by objects being only green). However, it is unclear whether it's coherent for the physics of a world to allow this while keeping the symmetry between *green* and *grue* (i.e., without the time of first examination being causally relevant to the underlying architecture of the AND gate). Furthermore, in such a scenario, we again expect the learner to infer that all emeralds are both green and grue, rather than treating the Green Hypothesis and the Grue Hypothesis differently.

There is some physical difference in the causal powers of objects not examined before 2050, which changes the reference of the Green-Grue Bit when it takes part in encoding *their* properties. The learner might not notice this change of reference, but it is there, nonetheless. If the word 'green' was used by the learner to refer to the conjunction of *green* and *grue* encoded by the Green-Grue Bit, then the statement "all emeralds are green" uttered by the learner is incorrect in such a scenario. As above, cases of unexpected changes in how the world behaves are the business of the Old Riddle, not the new one.

In section 2. iii, I have mentioned the worry that for learners for whom *grue* is encoded by the Grue Bit and *green* is encoded by an interdependence between the Grue Bit and the timestamp (what I called "a symmetric scenario" rather than a "scenario of symmetry"), the Grue Hypothesis would be justified, and the Green Hypothesis would not. This seems to me more of a feature rather than a bug. It is not clear that in such a world it is wrong to infer that all emeralds are grue rather than that all emeralds are green.

As in Putnam's Twin Earth example, think of learners inhabiting Grue World. Grue World is exactly like we believe our world to be, except that the property that causes us the same experience as when we see green in the actual world is grueness rather than greenness, whatever that means (we can also add to the story that *blue* and *bleen* switch roles if needed). It might be because in Grue World, there is no electromagnetic force but rather a "gruelectromagnetic" force or something like that. There is no reason why people in Grue World should not speak the same language as we do, except that their word "green" refers to the property of being grue instead of the property of being green.

Grue World people infer the hypothesis that is expressed in their language as "all emeralds are green", which means that all emeralds are grue.[16] Is this a problem? It seems to me that in a Grue World, where the physics is such that the property of being grue can causally interact directly with how the learner's data is recorded and the property of being green cannot, it can be justified to infer that objects have the causally efficacious property rather than the causally inefficacious one. Compare: in Twin Earth, people infer the hypothesis that is expressed in their language as "all oceans are filled with water", which means that all oceans are filled with XYZ, and they are right to do so. A different world might call for different inferences. The way that data is encoded is part of what allows computations on the data to lead to learning.

iii. Does Adopting Data Externalism Beg the Question?

One might worry that appealing to data externalism begs the question, at least partially, because I must assume that the world is in fact one way rather than another for my solution to work. This worry is doubly unwarranted. First, even if it were the case that I need to assume that the world is in fact some way rather than another for my solution to work, it wouldn't beg the question. Second, and more importantly, I don't need to assume any such thing.[17]

To see the first point, note that the New Riddle is fundamentally an epistemic problem. It is not concerned with how the world is, but with how, regardless of how the world is, we can learn anything about the unobserved, given that there are infinitely many competing

[16] In Grue World, the atoms are also different, because they are held stable by the gruelectromagnetic force rather than the electromagnetic force, and what determines the chemical elements is the number of gruelectrons rather than the number of electrons. Thus, the word "emeralds" in Grue World refers to gruemeralds, so to be more accurate "all emeralds are green" in Grue World should be translated as "all gruemeralds are grue".

[17] There are some parallels with points that were made regarding whether Putnam (1981) offers a good reply to the sceptic. See particularly Müller (2001).

descriptions of what we have observed, that are equally accurate but can lead to any arbitrary prediction. That is, even if emeralds are in fact green, it is *still* a riddle how we learn that emeralds are green instead of inferring that they are grue, and why we are justified in doing so.

Therefore, assuming that emeralds are in fact green, or that the world is in fact in one way rather than another, does not beg the question. It might restrict the scope of the solution only to worlds that are one way rather than the other, but if our world is in fact like that, it's not necessarily a problem. Of course, assuming that we *know in advance* that emeralds are green or that the world is in one way or another *does* beg the question, as this kind of knowledge is cast into doubt by the New Riddle. However, this assumption is distinct from the assumption that emeralds are in fact green, or that the world is in fact in this or that way (even if we might not know that).

Second, I don't need to assume anything about how the world in fact is. In common terminology, borrowed from Williamson (2000), my argument is not intended to work only in the good case. According to data externalism, how the world is partially determines what is encoded by a given binary string (or any other encoding scheme). However, it doesn't imply that the truth of the thesis of data externalism depends on how the world is. It might be analytic that part of what it means for data to encode is for what is encoded to causally contribute to how the data encode it.

It follows from this thesis that worlds need to be a certain way for encoding to be possible in them, and that different physics induce different privileged encoding schemes and different hierarchies regarding which properties can be encoded directly and which must be encoded in terms of other, more basic properties.[18] However, *I* don't need to assume that our world enables encoding. The settings of the New Riddle do (by assuming that, according to our data, all

[18] We can devise artificial properties, which can be called *hyper-gruesome properties*, that cannot be encoded directly regardless of the physics of a world. For instance, we can have a property that, for each world w where encoding is possible, selects those objects that have the property that is gruesome in w. Such a property is gruesome in every possible world. I thank Jack Spencer for this observation.

observed emeralds were green). Moreover, I don't need to assume that the physics of our world makes *green* more basic than *grue*. For the New Riddle not to pose an extra challenge beyond the Old Riddle, all I need from the physics of our world is not to allow encoding *green* and *grue* symmetrically (and thus not allow learning about *green* and *grue* symmetrically), which is the general case. Entertaining (doubtfully coherent) scenarios where that's not the case is not part of the extra challenge posed by the New Riddle.

4. Upshot: Solving the New Riddle

Let us take stock. We have established that in most, if not all, possible worlds, inferring the Grue Hypothesis at those worlds requires different learning processes from inferring the Green Hypothesis at those worlds, since *green* and *grue*, at least in most worlds, cannot be encoded symmetrically. Thus, the greenness of objects is likely not encoded in our data in the same way that the grueness of objects is, and different processes are needed to infer the Green Hypothesis and the Grue Hypothesis from our data. If different processes are used for inferring two hypotheses, which are about properties that are encoded in our data in different ways, there is no reason to expect that the epistemic status of the two hypotheses will be the same, as the data and processes we use to infer hypotheses are usually relevant for the epistemic status of the hypotheses.

Therefore, *if* there is a solution to the *Old Riddle*, that is, if there is a way to justify that *some* learning processes let you learn from observational data about the unobserved, then there is no worry that *these* processes (whatever they might be) would fail to discriminate between the Green Hypothesis and the Grue Hypothesis. Thus, it is not hard to explain why we infer the Green Hypothesis and not the Grue Hypothesis from our data. Inferring each hypothesis requires different processes, and doing the processes required for inferring the former is not specifically likely to lead to inferring the latter. Furthermore, if there is a solution to the Old Riddle, then it is clear that if the processes required to infer the Green Hypothesis from our observational data happen to be among the processes that *can* lead to knowledge or justified beliefs, then there is no reason that the different processes required to infer the Grue Hypothesis must enjoy the same status.[19]

The predicate *grue* was only an example of a gruesome predicate. However, if a predicate is gruesome, that is, a gerrymandered combination of other predicates, there is no reason to think that the gruesome predicate is encodable in a symmetrical way to the non-gruesome ones it was gerrymandered from. In fact, this asymmetry of encoding seems to be built into what makes a predicate gruesome. If so, the conclusion generalises. There is no reason to think that there should be an epistemic symmetry between inferences about gruesome predicates and natural ones. This amounts to the main conclusion I argue for in this chapter: The New Riddle doesn't pose an extra challenge over and above the Old Riddle, or, in other words, the New Riddle, insofar as it should pose a distinct challenge beyond the old one, is solved.

[19] My aim in this chapter is to show that there is no symmetry between *green* and *grue*, and not to show that the Green Hypothesis is epistemically *better* than the Grue Hypothesis (this requires saying something substantive about the Old Riddle). However, one can pick one's favourite theory of knowledge/justification and use my solution to explain how it is that the Green Hypothesis is justified/known and the Grue Hypothesis is not (or at least, one can do it if one feels one's favourite theory needs to say something about the New Riddle). For instance, process reliabilists can say that the process that led to the Green Hypothesis is reliable and the process that led to the Grue Hypothesis is not. Since *green* and *grue* are not encoded symmetrically, the two processes are distinct and should not be treated symmetrically. If one thinks, for instance, that a hypothesis is justified for a learner if it is likely given the learner's data, then the fact that the learner's data contain (say) the Green Bit rather than the Grue Bit might be relevant to the likelihood of the Green Hypothesis and the Grue Hypothesis. Since there is no symmetry in how we learn about *green* and *grue*, there is no reason to think that any theory of knowledge/justification that judges the Green Hypothesis positively would need to treat the Grue Hypothesis the same.

5. Other Solutions to the New Riddle

In the introduction to his (1994) comprehensive anthology on Goodman's New Riddle, Douglas Stalker remarks: "[t]here are now something like twenty different approaches to the problem, or kinds of solutions, in the literature… None of them has become the majority opinion, received answer, or textbook solution to the problem." (ibid. p. 2). I hope my solution will enjoy a better fate, though inductive reasoning suggests modesty. Stalker has also done a huge service and provided an extensive 180-page annotated bibliography with practically everything that was written about the New Riddle until 1994 (see also Elgin, 1997). The literature today is, of course, even vaster than it was in 1994.

I cannot discuss the relations between my solution and all other solutions within the scope of this chapter, although it's an interesting discussion.[20] However, judging from experience, it is worth discussing the relations of my solution to two families of solutions: solutions involving the notion of natural kinds and the so-called *independence solutions*.

i. Natural Kinds Solutions

According to natural kinds solutions (see Quine, 1969; Lewis, 1983), some predicates, like *green*, but not others, like *grue*, denote natural kinds, or natural properties. Natural kinds, to use Quine's (and Plato's) words, "carve nature at the joints". Only natural kind predicates are legitimate for inductive learning, so only the greenness of emeralds and not their grueness is projectible from our current data.

My main problem with this line of solutions is that they seem to me more like renaming the problem than offering a genuine solution to it. If nature is carved into kinds, it just raises

[20] Some notable omissions that have interesting connections to my solution are Hesse (1969), and Salmon (1963). Gärdenfors (1990) and Suppes (1994) are examples of early attempts to analyse the New Riddle in terms of learning.

the question of why, according to our data, we can learn that *green* rather than *grue* denotes the natural kind. On the other hand, if by "natural kinds" we refer to the kind of classifications we tend to make in our learning processes, it raises the question of why we should think these classifications tend to track something in nature.[21] In some respects, the New Riddle can roughly be *restated* in terms of natural kinds, so many solutions to the New Riddle can be restated in terms of natural kinds, too.

For this reason, it's hard for me to say whether my solution belongs to the natural kind family or not. On the one hand, my solution doesn't assume or imply that some predicates are not legitimate for inductive learning. To explain how our learning should be restricted, we need to focus on how learning is done, not on declaring that this predicate is forbidden and that one is allowed.

On the other hand, we can cast my solution in terms of natural kinds if we equate the notion of a natural kind with direct encodability of a property, or if we equate the comparative notion of degrees of naturalness with some ordering that starts from directly encodable properties and continues with properties that are encodable only based on the encoding of other properties that come earlier in the ordering.

Note, however, three important points. First, and most importantly, it is clear by this account *why* a property being more or less natural is relevant to *epistemology*. The degree of naturalness corresponds to how a property can be encoded, and how a property is encoded affects what can be learnt about it and by which processes. It also explains how natural kinds, understood this way, not only play a role in our learning processes but can also provide knowledge about the world. The physics of a world, by determining what can be encoded and how, is responsible for why more natural properties are encodable in different ways from less

[21] For related points, see Beebee (2011); Weatherson (2013); Slater (2015); Martinez (2017); Kendig & Grey (2019).

natural ones, and thus why our learning processes treat them differently. By treating them differently, our learning processes capture something about the physics of the world, so the discrimination of properties based on how they are encoded is not arbitrary. Thus, my solution can be seen as a form of *explanation* or as a basis of natural kind solutions.

Second, as it should be, this kind of naturalness is world-dependent, or, more accurately, physics-dependent. Worlds with different physics invoke different hierarchies of encodability, and thus different measures of naturalness. Naturalness is a property of predicates relative to worlds, or relative to classes of worlds sharing the same physics.

Third, if naturalness is understood in terms of encodability, then nothing in the solution I presented shows that more natural properties are *better* for inductive learning than less natural ones, or that there is something *problematic* in less natural properties. We don't get that for free merely by defining naturalness in terms of encodability. We only get that learning about more natural properties requires different processes from learning about less natural ones, which is what is required to show that the New Riddle doesn't add a distinct challenge beyond the old one. Which processes are good for learning, and why, is more of the business of the Old Riddle, and at some level, maybe, of science. However, thinking about these questions in terms of encodability and learning might lead to progress in answering them.

ii. Independence Solutions

A family of solutions that has gained increasing popularity in recent years is what Warren (2023) has grouped under the label of *Independence Solutions*. He includes in this group early versions by Jackson (1975), Jackson & Pargetter (1980), Wilkerson (1973), and Moreland (1976), and more recent developments by Okasha (2007), Godfrey-Smith (2003; 2011), Schramm (2014), Freitag (2015; 2016)[22], and Warren's own (2023) solution. To this, we should

[22] See also the response by Dorst (2018) and replies by Freitag (2019) and Warren (2021).

add Zinke's (2020; 2024) development of Freitag's solution. Warren has done a great job in presenting these various solutions and their problems. It's useless to repeat Warren's work here, so I shall just recommend Warren's paper, which also includes some very good points about the New Riddle in general and the desiderata for any satisfying solution to it.

What the independence solutions have in common is that they highlight some independence requirement for the legitimacy of a conclusion of an inductive inference and show that the inference to the Green Hypothesis is independent, while the inference to the Grue Hypothesis is not. The independence requirement can be about the world, e.g., some counterfactual or causal independence of the property projected in the inference (Godfrey-Smith) or about the learner's way of forming their belief in a hypothesis (Okasha, Schramm, Freitag, Zinke) or both (Jackson, who requires *knowledge* of a counterfactual by the learner, and Warren himself, as I understand him, who emphasises independence of sampling and observation methods).

I shall not get into the details of the various independence solutions here. Instead, I shall discuss how my solution can resemble the independence solutions and the limits of this resemblance. My solution can be thought of as a kind of independence solution, where the relevant dependence relation is dependence in encoding or dependence in learning.[23] The predicate *grue* can be seen as dependent on *green* (if the world is roughly how we think it is) because to encode *grue*, a learner first needs to encode *green*. Arguably, encoding that an object is green is thereby learning that it is green, so to learn anything about the grueness of objects, a learner first needs to learn about their greenness.

However, note that my solution is not committed to *grue* being dependent on *green*, or to the thought that independence enjoys some privileged status. The main issue is breaking the symmetry between *green* and *grue*. The symmetry breaks in Grue World as well, where

[23] This kind of dependence might partly *explain* other kinds of dependence that were noted by the various independence solutions.

supposedly *green* is dependent on *grue*. To show that the New Riddle doesn't add a further challenge beyond the old one, I argue, we need only show that there is no reason to think that the two hypotheses need to have the same epistemic status.[24]

[24] It's also worth noting, first, that unlike some independence solutions, my solution should apply to any physical learner, rather than only sophisticated ones who can reason counterfactually, or have some advanced concepts like *derivative defeat* or *biased sampling methods*. Second, unlike some solutions (e.g., Freitag's and Zinke's), my solution addresses not only what Warren calls *the internal challenge* but also what he calls *the external* challenge. That is, given my solution, it should be clear how the Green Hypothesis and the Grue Hypothesis can have distinct epistemic status not only in terms of how subjectively rational it is to adopt them, but also in terms of how likely they are to be true.

Part 2: Complexity

Introduction: Know Your Enemy

A country called Grueland is constantly threatening its neighbouring country, Greenland. One night, one of Greenland's intelligence agencies intercepts an internal Grueland communication. The first half of the message appears in Figure 1 below.

Lkwu5G/mcd7/y82PM4+qReNIM25YaEV+dFRDjl7GU0lfhngDIYmiwzaYk+Qjynhx0+Z
VPDZbXnYtSmJEL7Moe/mPnPWoP3hZo+7uUzolL37RlLfzF3R+wikwBzxKcggmxJZIig
uMndsyKtDQoNbmmoGVF8pnEEWUHJ0/X0gUQg8RF8IHnYDzNQaIXwM3JisAYIBL
1WCnnUhNTeZCmX7Rre/KFylAfqe8Z9FrIoErb5abzrDcweqa2Ydvv9GTE7CdQokjGGu
DAPJbCKHSYMjcRpS3dE21ZlHolGvZNKJRWZNz9tIU12I3UTf0TOuWqJbzdbZxqIAa
0oIfOo3MUxevgVqVmQYf+FQ4M0aozaZ8Kugq9+EY+qALqVgGWv9Gb3QFG3ATvu6
DGJDt6BaVFD3K93jD+QFz1vBaFKZbuIvWK415THnH63A8BAnoIpSTPsC5V/BM3Ug
aISX4FjfvfjwYzrm7j0tt5YxSA/NSlG24Rmay5xkFirZWMQYd74vacP9ZMe7W7vqMxf/i/
kValxM11Y3eq1Ef14EQJHNJeZhxsEaMwj5Q2utSSfzj73zRulOO4GaE4gsdb74XvH0KM
Aex8WHNCr+MTFWQrvETsql9wTNP+5HI2+a31RQVFMewgWtPtGZ9LN0gJkag1gBD
hkpYYhjleb8K3x+aMOvocpwyQqc=

Figure 1: First half of the intercepted message

Imagine you are a Greenland intelligence analyst who first sees this weird-looking message. What would you do? How would you update your beliefs about Grueland's intentions based on it? Which course of action would you recommend? These questions sound a bit off. The message above seems no more than complete gibberish, just random characters one after the other. No information can be gained from it about Grueland's intentions. If this is what you think, you are not entirely wrong. In a sense, as I explain in the next chapter, you are most likely right. You might as well throw it in the trash (or the digital recycle bin).

Now imagine an alternative scenario. You are the same intelligence analyst, but instead of the message in Figure 1, you get the message in Figure 2 below:

To my absolutely tremendous generals (the best in history, everybody says so, nobody has ever seen anything like it):

At dawn tomorrow, we are sending a beautiful armada toward so-called “Greenland”. People tell me it’s very cold, very icy, not very green – a total disaster – but it’s got fantastic strategic value, really incredible value, and we are going to make Greenland green again.

Signed,
Bleenald Grump
President of the Gloriously Great Republic of Grueland

Figure 2: Message by the president of grueland. The message was generated with the help of AI (Perplexity Pro), after convincing it that I’m not really going to invade Greenland.

What would you do? This question is as easy as the previous one. You should act now! Grueland is going to attack at dawn, and we must prepare. All soldiers to their positions immediately. From a more distant philosophical point of view, we can state the obvious truth that this message provides strong confirmation for the hypothesis that Grueland is planning to attack at dawn. This, in turn, justifies updating the likelihood you ascribe to other propositions, e.g., the proposition that Greenlandian soldiers will die or get injured the next day.

So far, what I said should strike one as at least apparently trivial. The twist comes when we look at the second half of the message from Figure 1, which is presented in Figure 3 below:


```
-----BEGIN PUBLIC KEY-----
MIICIjANBgkqhkiG9w0BAQEFAAOCAg8AMIICCgKCAgEAgmOqMN/gJPC69T02HD
laRHEkKHFqv3OHqg3bmV+i3HBWyiqwRTzw/mcGkRwNpbNYybjLJoFh1ytCl27ilkzlb
LM7sf3vwezxf9cYt4pJkdUWihAJ4SBklB+BMZdzTH7kVSCywJmUKLPxbEXgCh8p1K
XTIJ0QcgGL8QLOqafhLok5I8nhGmAfqZj7yKrvBs0UeT+GVnN1r2M4Le5MFo0myVV
MiwNreQDjy+GK9kVL1qYbrKXde1jOktB9cAGA5N7PY5HnaEniZXZ5H61+rER027eQ
WUeYt+RxpZMoedtChHaDR+3FUAo69e+r+6RyKKO+L2dfkKno40op/JabYiPCnifsXm1
EN4DMDBKS7Q7PBLgRO2/T5B9ty9gQkAgA29f0MRsFyhjibaOs7NIj5x7BJLI8clpk6C
QJZd05hFyRPgIzTGl8MOz+6sdyjnmlLErkkCWTY1rUl5FaZwdEeFHqtY/KE1ISdyMjA0
ImegMMC1W+6voEbkMQIRmPCm7b/gL6oVm+iGWuIg1pJa0Jbii9BkP+qr2pp6Jc/bmlV
dEfK7Dz8JbjIxYuYuDWP4n2SAx8dzPu4h8/W8UmkwYKu+WJcnhF1weDt+eW6V5526
mTWLjkFqzQ3lOI7k38Id54DadQHSyj+KR1j0aq/Vm63gmJkzqdo8G3W0hD8P0XeWNr
1rUCAwEAAQ==
-----END PUBLIC KEY-----
```


Figure 3: The second half of the message from Figure 1

Figure 3 indicates that the weird message in Figure 1 is not exactly random gibberish. It is encrypted, using the RSA cryptosystem, with the public key presented in Figure 3. In the RSA cryptosystem, we use two keys (often called a 'key pair') instead of one. One of the keys, known as the public key, is used to encrypt messages and is not kept secret. The other key, known as the private key, is used to decrypt messages encrypted by the public key, and it should be kept secret from everyone except for those who are meant to read the decrypted messages. For every public key, there is a unique private key that corresponds to it and vice versa, since they are derivable from one another mathematically. Consequently, for every encrypted message (ciphertext), there is a unique decrypted message (plaintext) that corresponds to it.

What you didn't know (though you may have guessed by now) is that the message in Figure 1 is an encrypted version of the message in Figure 2. I know it because I encrypted it.[1] Thus,

---

[1] More accurately, I asked Online RSA Encryption / Decryption Tool | AnyCript to encrypt it for me. As a proof, here is the private key:
-----BEGIN RSA PRIVATE KEY-----
MIIJQQIBADANBgkqhkiG9w0BAQEFAASCCSswggknAgEAAoICAQCCY6ow3+Ak8Lr1PTYcOVpEcSQo
cWq/c4eqDduZX6LccFbKKrBFPPD+ZwaRHA2ls1jJuMsmgWHXK0KXbuKWTOVsszux/e/B7PF/1xi3ikmR1
RaKEAnhIGSUH4Exl3NMfuRVILLAmZQos/FsReAKHynUpdMgnRByAYvxAs6pp+EuiTkjyeEaYB+pmPvIq
u8GzRR5P4ZWc3WvYzgt7kwWjSbJVUyLA2t5AOPL4Yr2RUvWphuspd17WM6S0H1wAYDk3s9jkedoSeJld
nkfrX6sRHTbt5BZR5i35HGlkyh520KEdoNH7cVQCjr176v7pHIoo74vZ1+QqejjSin8lptiI8KeJ+xebUQ3gMw
MEpLtDs8EuBE7b9PkH23L2BCQCADb1/QxGwXKGOJto6zs0iPnHsEksjxyWmToJAll3TmEXJE+AjNMaX
ww7P7qx3KOeaUsSuSQJZNjWtSXkVpnB0R4Ueq1j8oTUhJ3IyMDQiZ6AwwLVb7q+gRuQxAhGY8Kbtv+A
vqhWb6IZa4iDWklrQluKL0GQ/6qvamnolz9uaVV0R8rsPPwluMjFi5i4NY/ifZIDHx3M+7iHz9bxSaTBgq75Yl
yeEXXB4O355bpXnnbqZNYuOQWrNDeU4juTfwh3ngNp1AdLKP4pHWPRqr9WbreCYmTOp2jwbdbSEPw/
Rd5Y2vWtQIDAQABAoICABeuHo1YR1amIqohxwc6skZeTu0KZhLlmjSb5i81Y3hJ7bUzKga1EeZ+Ev79G4
xlpUEjbr+sPTbs7B/g+0VFK2CBr5x8jBPJ4rUvPRWuR3y0+K4zyfuJHMMa8qVtRjsVC4ZbVSNt8O6weW7K
xOzJqpjsUcrpQ4CU/9OSSTKeIuMG+Gnjwt49qVpthoKz6ouvaKG5k/3046Fp8VzEG6z3zxy/AVL42Malzvkq2
cgunx6ZVcrV58K8Ru36b+GM4FoQjhUc/SwfSP7Y1JgYc/o4x4lUzeItAtMY6bvinTX+SFcSx/Zi/BP0e69JKxC
dVb9EN6jXyw8UkI9g2328Y3VgHbrFlCNkVggf6EzNdo1vTqENbAiVBwC2BVzDY7PQY0J8HsItrSu9TGA0
5VlDbf55Rj6cl6nuevYYS6pV2vsuDaJLIbnY0uLN9MkkYwYph+GvqV4n72WTJjlrdj3Eal0necfgm3Va2OQ/vj
7quL4wz0FWNJtXMp/4swb6NDWIx6QG0IqvwGoA8so3ZxuJiiQwNxz36D+BCfwyqMldFlJXXIa5t5wFlUn6
e1m/y/Yzs2wGDETMOsJoHh04nBxx5UtYiKWaStsKHc8R1Ft/2JP2weOXtCssftyNnNv7swxoQOY/6XNxCeP
lfawfstdda9xVCOTl3UcX8GMpuvnXRsLwf4YpAoIBAQC3KTZ4m8Kx/ewYZBjeFTTiPYdb4Q1owKIPppBPt
kG9EZFl1CLpPASiYOs57PNscWtfQB7cWm66gCsmpdnHiY22I5boo5nPMdlRwGxOasLld9co9dobxVTwkJX
nL/Mth/KUCSGMmYCwfOqm3l47bHsulQI04OIGqbx9cb6eKS27t9MdcKgjQKUE6hwbIvacu1aRtIzcnrRsd/8
Qc/0amANsPLRvYcJV/2r5N3o7gzU9cjbtoSHqCoowgWKo0RpvwNmWexo8+y56MJcH9o99wqgaoOKP6/Bs
EvdiSDvskgg4QWdc3CQUT6KQocROv0Uvm9aGiuaWnwleoAwc0XHCVoeZAoIBAQC2PgITB5xj5gsTeYZb
/ScZhJLMKtpDEQKviOhddDJkuAUQE36ZpaLBSUrwJvOgUJjzsmf5466Xiud5hf3UiFnEluP/kcEp4ww9fu5sa
2tsJRvE29Xb1vRuBlTk3US+X0fKUIEEXRHFhBURsYjVEpqO/1UFeL1OWrBFmmPbTGZDhFpUm0/gKNyj
zsw2seELQsBGEe11CbKMIMF4yggkuXQBaIkUseI8rxb7P2FhMQg0p/4dSSPB40nlA/nS4IbBbIQmKcFK20az
EqK2AlYgtY4DrM7IWUt6Q4qEM/UwHxrVX+h4SDdX9gMM66cQbixIp7ZPUAtoB3Mf612J9oTTKUl9AoI
BADxesB6GsGB8YjIT5AJnGPws7Y2CGypYu9HOw6gSWY6Y2MSobC1V4+0lZsK77+vhYlONzd4xTGfaU
N63MZXMJ9JN3rKlIdAiDRAeDVS3aa+6lUdh6VbX7swvy8jcjBmIwpK6g71U5w7EzY7colCRy/QplwWYzcc
GcSMjt4Fn7zP9Pe9pL3IXH40EIOls174HHVx/FyKsiKlrz707jcQTMKWfSJWhsi0oN7kW6QnukEDFgYdHYe
dDPBMKwlsEOON2vAEWaCqLW+FYXJ96TAdWfpPwTVzEloT4p2NUjbPG2kC4hfoccSPkr59eQlb2QjaT9d

as a matter of mathematical necessity, if a ciphertext is like the one in Figure 1 and the public key is like the one in Figure 3, then the plaintext must be the one in Figure 2. Here is a question: without access to any information about the private key, is the ciphertext in Figure 1, together with the public key in Figure 3, evidence for the hypothesis that Grueland will attack at dawn? Is the hypothesis confirmed by the fact that the message that was intercepted has exactly the content presented in Figures 1 & 3?

Some philosophers might be tempted to say yes. After all, the statement that Grueland sent an encrypted message with the content of Figure 1 that was encrypted using the key in Figure 3 is mathematically equivalent to the statement that Grueland sent an encrypted message, the plaintext of which is the one in Figure 2. The statements are true in the same possible worlds. In a standard Bayesian framework, the credence of every proposition given the former should be the same as the credence given the latter. Some would say that if one knows the former, then one knows the latter, albeit one might not recognise that the latter is one of the guises of the proposition referred to by the former.

There is, however, a clear sense according to which having the plaintext is somehow more epistemically valuable than having only the ciphertext. Otherwise, intelligence agencies around the world wouldn't have spent billions of dollars on breaking encryption. There is a sense in which plaintext messages can play a role in confirming hypotheses in a threat assessment report that ciphertexts cannot. Most people, I believe, will simply say that the plaintext provides knowledge and confirms hypotheses that the ciphertext does not.

x4sdIZbzwqInPpMMauQJkCggEALSMekGgKLBQGGkS3KoGA9akIUtkqRgIokCFwq/Tjpl6NonWxnYyQyx7
EGMcNAF/2AiXl6tahn5cfzdGKFOHQmghlYB9RoHZDydduZhwU9ywZvKBWT6Tl1Xpr/GNxa1VeCpX13ob
OqAUW0tMN+Pgqn9BN1pUZDhcYIaUjpmU9zusxFFODbRmxzECkA8f63TcjODeogJxuRJa4VJp1eeMVAG
RLz3xfdHhJUOFZ1SC3z5LsNYSv/o2096wMbFaACdg+wtUEsu2sjR3IfxwFCQTur6pVbPnw389qt2PHDqwlk
UO36FK6omakSsbzEfa3sScY2HwS1ylBMl6xREJyk3peYQKCAQAZjjBhfSdsN22SRF13DZCRT8AF2sRHzv
kJKQr406apiTcK4jQyzGl1vCWTbPui2gQ/KiwHcYp6poo0k1Y+dem2bhA3HjPwIb3vOGf+Bz5HMgBRGl0vh
T6+8tTi3SB0Wh8f9ydh5jxLyDNeVf1qpGXl5P3UvmSGmXrym9NCDDbUv2vfi8czMO9enQHFA5amHqfd5Q
ZA8O0KKVRT8X/CIZpGFuo87uZN0J2xIf2ZS4Dvo9pcGc/KSw6l3DSDAqpstu2PuO+ARB4b0g0iLClL2y8a9
8w0Jon8xsGyhwyo5+aQYk+hlYXIb41OZOLlcFzTwVzTWUPPw95aVdDb0WU92Bx9
-----END RSA PRIVATE KEY-----

This concept of confirmation is the one I am trying to analyse in this thesis and is the one, I believe, that is normally employed in science, as well as in everyday life. One might choose to reserve the word "confirmation" to describe what a godlike entity that can break encryption instantly would consider as more likely based on the evidence. I'm not going to argue about terminology. At this point, however, let me say that this godlike version of confirmation can be shown to be a special case of a more down-to-earth concept that applies to both gods and lesser creatures.

All this, of course, relates to discussions of Frege's puzzle and the problem of logical omniscience. Some thinkers have suggested some ways of dealing with agents that are not logically omniscient within the Bayesian framework.[2] However, the current discussion of logical omniscience seems to miss something central. The discussion usually focuses on complex logical truths and mathematical statements, as if the explanandum is the fact that the learners we are trying to model don't know all of logic and mathematics. One might get the impression that the problem of logical omniscience is somewhat peripheral. We have a great framework to model how we should reason about most matters, inductive matters for sure, that may encounter some problems with logic and mathematics, or their harder parts. These are rare and niche cases, and we can add some patches to our framework to deal with them.

I suggest that this is a misleading picture. Unlike the case of remote logical truths that are usually of little importance to anyone, in our example, the results of ignorance have a huge effect on the real world. It's literally a matter of life and death, the kind of matter it's crucial to learn about. Furthermore, part of what Greenland intelligence doesn't know is not a logical truth. The statement that Grueland sent an encrypted message, the plaintext of which is the one in Figure 2, is an empirical truth, not a logical one. It doesn't even follow logically merely from

[2] As a non-exhaustive list, see Hacking (1967); Garber (1983); Hintikka (1989); Bjerring & Skipper (2019); Skipper & Bjerring (2022); Pettigrew (2021); Elga & Rayo (2022). I don't discuss the details of these suggestions here.

the statement that Grueland sent the message in Figures 1 & 3. Grueland might have written the exact same text, not as a result of an encryption algorithm. The knowledge that Figures 1 & 3 are an encrypted version of another message is inductive, not deductive. So, we see that the problem is not only about remote abstract truths in logic and mathematics.

At this point, some might contend that the case of encrypted data is also a niche case, relevant only to intelligence agencies. However, even if we ignore the fact that intelligence agencies employ millions of people, spend many billions of dollars annually and affect the lives and deaths of many people, this kind of response is no longer available. Practically all the data flowing on the internet today is encrypted. What guards your money in your bank account is encryption, and every time you make a non-cash payment, you trust encryption to prevent others from stealing your credit card details.

Unlike hypothetical betting scenarios, in this case, we are literally willing to risk *all our money every day* on the truth of the statement that what can be learnt from encrypted data is fundamentally different from what can be learnt from these data when they are not encrypted. There aren't many statements we are willing to risk so much money on their being true. You might say that most people are not aware of this role of encryption, but you are now aware, so unless you radically change your behaviour, you'll be willing to risk all your money every single day on the truth of that statement. You will not be willing to do that merely because we think people sometimes don't reason rationally. You should be very, very sure that they *can't* reason in ways that can lead them to break encryption, no matter which norms guide their reasoning.

Thus, the encryption case is not niche, and it reveals something that we are very sure about. Note that the whole point of modern cryptography is to conceal information *by presenting it in a mathematically equivalent way*. If modern cryptography is correct, then a successful theory of confirmation should make it very clear how this is possible. The current suggestions of how to amend the Bayesian framework so that it can discriminate between plaintexts and ciphertexts seem more like patchwork on a framework that was not built to handle such cases than true explanations. While I happily bet all my money every day on the truth of the statement that even very rational people can't break encryption, I would hardly bet any of my money on any of those patches to the Bayesian framework being what truly explains why the message in Figure 2 confirms the hypothesis that Grueland is going to attack at dawn, and the message in Figures 1 & 3 doesn't. As we shall see in later chapters, there are other cases besides cryptography that the Bayesian framework is not built to handle naturally.

Instead of amending a framework that was not built to handle these cases, I suggest we should adopt a better framework that handles these cases naturally. There is a readily available, straightforward answer for why the learners we are interested in cannot learn from encrypted data, which is at the centre of modern cryptography: breaking encryption is too computationally complex. It requires more computational resources than the universe can supply.[3] If learning is computational, then physical learners cannot learn how to decrypt an encrypted message only from the ciphertext and the public key.[4] If confirmation is essentially about *what can be learnt from evidence*, then nothing about the content of the plaintext is confirmed for physical learners by the ciphertext.

Viewed this way, it becomes clear that there is nothing special specifically in complex deductive learning. Like deductive learning processes, inductive learning processes also require computational resources. Some inductive processes, like some deductive processes, are too complex to be executed by physical learners. If learning *inductively* that a hypothesis H is more likely based on evidence e is too computationally demanding for physical learners, then

[3] Today's encryption schemes are designed to be too complex for current computers to break, but with relatively little investment it can be made too complex for any physical computer, as will be explained in the next chapter.
[4] Yes, RSA is vulnerable to quantum computers. If this bothers you, you can replace it with lattice-based cryptography, which is believed to be quantum resistant.

e does not confirm *H* for physical learners, even if *e* might have confirmed *H* for learners with more computational resources.

Explaining why we don't learn, can't learn, and shouldn't be expected to learn, *either deductively or inductively*, things that are too complex should be at the heart of any theory of learning and confirmation. This, we shall see in Chapter 6, is the key to solving Hempel's Paradox of the Ravens. However, before we can discuss the Paradox of the Ravens, I should explain what I mean by *computational complexity, strong encryption, learning processes, computational resources* and *physical learners*. I do that in the next two chapters. Besides the computational complexity of learning processes, there is a sense in which the data the learner learns from can themselves be simple or complex, and this also affects learning and learnability. Chapter 4 discusses an important measure of the complexity of data called *algorithmic complexity*.

Chapter 2: Computational Complexity – The Basics

Computational Complexity Theory is a highly important and constantly evolving field in computer science and mathematics. It evolves quite naturally from Gödel's and Turing's work on provability and computability. However, unlike Gödel's and Turing's work, it has not received a lot of philosophical attention. This needs to be amended, I believe, as issues related to computational complexity are highly relevant to epistemology, or at least so I hope to convince you in this part of the thesis. The reason computational complexity is highly relevant to epistemology is that it is highly relevant to learning and learnability, which in turn are relevant to what is known and what is knowable.

Some of the topics in this chapter and the next are textbook material. My presentation of these topics is largely adapted from Valiant (2013, especially Chs. 3 & 5) and Wigderson (2019), but it is more philosophically minded and expands more on the philosophical significance of the matters presented. Other useful sources were Harman & Kulkarni (2007), Arora & Barak (2009), and Aaronson (2017). I will not cite these sources each time I present common knowledge that is part of every introductory class on computational complexity or computational learning, but some more advanced claims will be backed by specific references.

1. What Is Computational Complexity

To get an idea of what computational complexity means, consider the problem of finding the sum of the following two pairs of numbers:

(a) $5 + 3$

(b) $314{,}159{,}265{,}358{,}979 + 271{,}828{,}182{,}845{,}904$

Finding the sum of the pair of numbers in (a) is easy. I'm sure you know it at first glance. Finding the sum of the pair in (b) is harder. Not much harder, though. You might find it challenging to do it in your head, but with some memory-extending devices like pen and paper, you have probably been able to solve it in about a minute since at least when you were in third grade, using the long addition algorithm as in Figure 4. You write one number above the other, add every digit on the bottom separately to the digit on the top, sometimes carry one, write the sum of each pair of digits (plus 1 if needed) in the right place, and you get the solution. It is basically like solving the problem in (a) but doing it multiple times. For our purposes, we can treat problems like (a), which can be solved immediately, as requiring one computational step. The problem in (b) requires many computational steps like the one needed in (a). The number of computational steps you need to perform to find the sum of two numbers using the long addition algorithm depends on the length of the numbers you add. The number of computational steps needed for a computational process is called the *computation time* of the process. We can restate what we said above as the claim that the computation time is correlated with the length of the input.

$$\begin{array}{r} \scriptstyle 1\quad 1\ \ 1\ \ 111\quad 1\qquad \\ 314{,}159{,}265{,}358{,}979 \\ \underline{271{,}828{,}182{,}845{,}904} \\ 585{,}987{,}448{,}204{,}883 \end{array}$$

Figure 4: Long addition

Now, suppose that instead of finding the sum of the pair of numbers in (b), you want to find their product. This seems significantly harder. You probably did not know how to do it in third grade! However, you probably learnt how to do it in fourth grade, when you learnt the long multiplication algorithm. You just write one number on top of the other, multiply each bottom digit by every top digit separately, write the result in the right decimal place, and sometimes carry a digit to the left, and you get a staircase-like structure of numbers as in Figure 5. You add all the resulting numbers together, and you have the solution.

```
                314159265358979
               ×271828182845904
               1256637061435916
              0000000000000000
             2827433388230811
            1570796326794895
           1256637061435916
          2513274122871832
          628318530717958
        2513274122871832
        314159265358979
      2513274122871832
      628318530717958
    2513274122871832
    314159265358979
  2199114857512853
  628318530717958
8539734222673541815039977 2016
```

Figure 5: Long multiplication (source: Valiant, 2013, p. 32)

This process of long multiplication, while easy enough for children to learn and implement successfully, is still harder than long addition. We used the same numbers (i.e., the same input) as in the long addition case, and we needed to make significantly more computational steps (i.e., it took a longer computation time). This illustrates that the computation time needed to solve a problem is not only a function of the length of the input, or even the specific input, but also a function of the kind of problem we want to solve.

Like addition and multiplication, some problems are recurring problems that we want to solve many times. Thus, it is helpful to find *algorithms* that solve every case of the problem, no matter the input, as we did with the long addition algorithm for addition and the long multiplication algorithm for multiplication. Now, not all algorithms are created equal. Some are more efficient, requiring fewer steps than others. Some are more well-suited for one type of input than for another. It is a good idea to have a way to compare them. The *computational* *complexity* of an algorithm measures the computation time (the number of computational steps) it requires to solve a problem. The computation time depends on the length of the input, as we saw in the addition and multiplication cases, so the computational complexity of an algorithm is defined as a function of the length of the input.

The algorithm is judged as a tool to solve a general problem (like addition) rather than a specific special case of the problem, but it might take a different amount of computation time for each special case. We should thus be clear about what we count when we say that an algorithm requires this or that computation time. In most cases, when people talk about 'computational complexity' without any further qualifications, they refer to *Worst-Case Complexity*. The worst-case complexity of an algorithm is the computation time the algorithm requires in the special case where it does the worst. That is, it is the computation time the algorithm requires to *guarantee* that, in the end, we have a solution to the problem. Following the common practice, I also refer to worst-case complexity when I talk about 'computational complexity' simpliciter.

For some purposes, we might want to talk about the complexity of an algorithm in a *typical* case, i.e., about its *Average-Case Complexity*. Defining what we mean by that is not a trivial task (see Wigderson, 2019 §4.4, Arora & Barak, 2009 Chs. 10, 15), but for some problems, the typical case might require significantly less computation time than the worst-case. Average-case complexity will also be important later, but for now, let's focus on worst-case complexity.

Proving that a given algorithm is guaranteed to solve a general problem usually comes with proving that for an input of length n, the algorithm solves the problem in at most $f(n)$ steps, where $f(n)$ is some function of n. The larger the value of $f(n)$, the more computation time the algorithm demands for the worst-case inputs of length n. The input length n is usually measured in bits, as modern-day computers usually compute on binary strings composed of sequences of 0's and 1's (the term 'bit' is a portmanteau of the words 'binary digit'). Nothing crucial hangs, though, on whether we measure the input length in binary digits or decimal digits, so for convenience, when I discuss the complexity of addition and multiplication below, I will measure the input length in decimal digits, although in most other cases I will measure it in terms of bits.

As it turns out, the complexity of the long addition algorithm is *linear* in *n* (Valiant, 2013, p. 37). That is, the number of steps it requires for an input of length *n* is at most $an+b$, where *a* and *b* are constants (for large enough values of *n*, the contribution of *b* becomes negligible). You can get an idea of why this is the case when you look at Figure 4. We added a 15-digit number to a 15-digit number (that is, a total input length of 30),[1] and we needed to add a bottom digit to a top digit 15 times (which is half of 30) and also carry one sometimes (but no matter what the numbers are, we never need to carry one more than 15 times).

The complexity of the long multiplication algorithm is *quadratic* in *n* (Valiant, 2013, p. 31). That is, the maximal computation time we need is approximately proportional to n^2, at least for large values of *n.* A look at Figure 5 can illustrate why. We started again from an input of length 30 and got 15 stairs, each composed of 15 digits. Thus, we needed to multiply one digit by one digit 15^2 times (which is $\frac{1}{4} \cdot 30^2$). We then need to add these 15^2 digits to one another in the right way. Sometimes we need to carry digits and add them as well, but we never need to do that more than 15^2 times. This provides a formal basis for our intuitive feeling that addition problems are easier than multiplication problems, at least in the ways we are used to solving them.

It turns out, however, that it is still an open question whether multiplication is inherently harder than addition (Valiant, 2013 §3.5, Wigderson, 2019, p. 263). With advances in computer

[1] In real life it will be more than 30 because we need at least a separation sign between the two numbers to know when the first number ends and the second starts. Of course, if the numbers are represented in binary the input will be longer and representing numbers in decimal form does not shorten the input for free. I omit such technicalities.

science, it turned out that what people took for granted, probably for millennia, was incorrect, and when it comes to multiplication problems, you can do much better than the long multiplication algorithm. The best algorithm known today requires very close to, but still more than, linear time (see Valiant, 2013, p. 37, Wigderson, 2019, p. 51).[2]

This highlights an important fact: algorithms can be surprising. The fact that we have not yet *found* a more efficient algorithm for a problem than the one we have does not mean there isn't any. Let's define the complexity of a *problem* as the complexity of the algorithm that solves it most efficiently. That is, the complexity of a problem is the complexity of the least complex algorithm that solves it. Recall that 'least complex' here refers to worst-case complexity. That is, for every algorithm that solves the problem, take the worst case, and pick the algorithm for which the worst case requires the least amount of computation time among all the worst cases of all the algorithms that solve the problem. The computation time required in the best among all worst cases is the (worst-case) complexity of the problem.

We can see that there is an asymmetry between finding an *upper bound* for the complexity of the problem, i.e., showing that a problem can be solved with *at most* a certain number of steps, and finding a *lower bound* for the complexity of a problem, i.e., showing that solving a problem requires *at least* a certain number of steps (in the worst case). Finding an upper bound requires only finding an algorithm that solves the problem within that bound. You don't need more steps than the algorithm you found requires. Finding a lower bound requires showing that among *all possible algorithms,* none is guaranteed to solve the problem in fewer than a certain number of steps. That is, finding lower bounds is much harder than finding upper bounds. In other words, finding an algorithm that solves a problem (and thus finding an upper bound for

[2] The current best algorithm we are familiar with, discovered in 2019, requires computational time proportional to $n \log n$ (Harvey, D., & Van Der Hoeven, J., 2021).

the complexity of a problem) is much easier than showing that the algorithm we have found is the most efficient (i.e., that the upper bound we have found is the *least upper bound*).

In the case of multiplication, the question of the least upper bound might be less urgent. Even if we use only the long multiplication algorithm, solving multiplication problems is feasible. To get an idea, suppose that for some bizarre reason, we want to add or multiply two numbers with a combined length of *n=1,000,000* digits. The *computation speed* of processors in current laptops, i.e., the number of steps they can perform in a second, is in the order of billions. *Adding* the million-digit numbers requires computational time in the order of *1,000,000* steps. It will take my laptop about a millisecond to do that. *Multiplying* these numbers using long multiplication requires about $1{,}000{,}000^2$ steps, that is, an order of magnitude of one trillion steps. It will take my laptop about 1,000 seconds, or less than 17 minutes, to solve it. I can have coffee and wait for that long if it is so important to me. If I cannot, I can always pay some money to a cloud service provider and have them solve it for me with their computing resources in less than a second.

Things are entirely different when it comes to other problems for which we do not have such efficient algorithms. For some problems, for instance, the most efficient algorithms we know of have complexity in the order of magnitude of 2^n rather than n^2, that is, they require computation time *exponential* in *n*.[3] Suppose we want to solve one of these problems with such algorithms, but now, let's not be greedy and instead of an input of length *n=1 million,* we will start with a very modest input of length *n=100*. In the worst case, the computation time required will be proportional to $2^{100} = 1{,}267{,}650{,}600{,}228{,}229{,}401{,}496{,}703{,}205{,}376$ steps, or about 10^{30} steps. Even if my laptop were 1,000 times faster, it would still need more than the current age of the universe to compute the solution.

[3] Such is the case when to solve a problem we need to go through all the possibilities one-by-one. The number of possible binary strings of length *n* is 2^n.

You might think that this is a problem for engineers. We just need faster computers! In any case, it is only a quantitative difference, and philosophers need not bother with such earthly matters. A laptop a trillion trillion times faster than mine would solve the problem in a millisecond. But there is an inherent limit to such reasoning (cf. Valiant, 2013, §3.4). Ask yourself, what happens if we give this super laptop an input of length (say) $n=200$ instead of $n=100$? An input of length $n=200$ is still very short. The answer is that it will take the super laptop much, much more than the current age of the universe to compute the solution. The reason is that trillion trillion, or 10^{24}, is roughly 2^{80}, so a modest addition of just 80 bits suffices to cancel all the improvement we gained with the super laptop. That is, an enormous improvement in computation speed compensates only for a tiny increment in input length.

In addition, the resources in a physical universe such as ours are bounded. Suppose that we get so technologically advanced that we can make a tiny transistor out of every elementary particle in the universe and run them in parallel (let's also completely ignore this whole annoying issue of the speed of light). The estimated number of elementary particles in the universe is about 10^{90}. Suppose that we can make the particles compute at enormous speed. Many physicists today believe that no physical action can take less than a Planck time unit, which is about 10^{-43} a second, a tiny fraction of a second indeed! Let's suppose then that each elementary particle runs at an unbelievable computation speed of one step per Planck time unit, that is, around 10^{43} steps a second. The universe, it seems, has a built-in theoretical bound on computation speed of about 10^{133}, or 2^{442}, steps a second. The age of the universe is about 13.7 billion years, or 2^{55} seconds. Thus, even if the universe as a whole were busy from its inception until today just solving a problem with exponential complexity, it is not guaranteed to have solved an exponentially complex problem with input length longer than 497 bits, a very modest input length still.

Thus, what might seem to be only a quantitative distinction turns out to be also a *qualitative* distinction. Some computations are physically possible. Others are not.[4] This is a matter of philosophical interest rather than a technical issue for engineers. Of course, in reality, the bound on what is physically computable is much lower. It is believed that no signal can travel faster than light, so only a tiny minority of the particles that are not too far away can be used in parallel for computation. Moreover, of course, it is neither clear that elementary particles can function as stable and fast transistors, nor that we can use a significant number of them for this purpose. This emphasises even more how limited the physically computable is when it comes to problems with exponential complexity.

Of course, learners of the kind that interests us were not born in the Big Bang and are not going to live forever. In most cases, they are interested in answers they can get or estimate in a limited amount of time. This narrows the kind of problems for which there is a feasible way to come up with a solution even more. Indeed, shortly after people built the first physical computers, they discovered that for every practical purpose, the question of whether a solution to a problem is Turing-computable should be replaced with the much harder question of whether a solution to a problem can be found using *feasible* time and computational resources.

We are interested, then, in a way to distinguish between problems that are feasible to solve, that is, problems for which there is a realistic way to find a solution in a relatively short time, and *infeasible* problems, that is, problems for which any realistic way of finding a solution will take too much time. This distinction itself is, of course, vague, since notions like "too much time" are vague. However, it does not imply that no precise distinction can capture approximately the vague boundary between feasible and infeasible problems. A successful,

[4] Andréka et al. (2018) have suggested the concept of a *relativistic computer*, that can perform infinitely many steps in a finite amount of time. The idea is for the programmer to program the computer and then go live happily ever after in the area between the outer horizon and the inner horizon of a rotating black hole (or travelling from this area to another universe, in some versions). This way, supposedly, the journey can be arranged such that in a short amount of programmer time the computer runs for an infinite time and sends the results to the programmer during the process. It is currently too far into the realm of science fiction so I ignore this suggestion here.

precise distinction can function as a negotiable border. We mark the border precisely and clearly on the map, but we allow for minor border corrections to come later on when we look closely at specific points in the complexity terrain. The distinction between *polynomial* and *superpolynomial* complexity, presented in the next section, is intended to serve as such a border.

2. Polynomial vs. Superpolynomial Complexity

A polynomial function *P(n)* is a sum of powers of *n*. For instance, $n, 2n, n^2, 7n^3 - 3n^2 + n - 8$, $n^8 - n^5$, are all examples of polynomials. A problem is said to be *solvable in polynomial time* or to have *polynomial complexity* if its complexity is bounded by a polynomial function of the input length. That is, if there is an algorithm that is guaranteed to solve the problem in at most *P(n)* steps, where *P(n)* is a polynomial in *n*. Algorithms also have their associated *space-complexity*, which is the maximal number of memory cells (as opposed to computational steps) they require for the computation. Sometimes people emphasise that they are talking about polynomial *time-complexity* or *polytime complexity*, although when people talk about 'computational complexity' without further qualifications they almost always mean time complexity. The *complexity class* of all problems with polynomial time-complexity is called $\mathcal{P}$. As we saw, addition and multiplication are members of $\mathcal{P}$.

In contrast, a problem of complexity 2^n is not bounded by any polynomial. Indeed, for every polynomial *P(n),* no matter its degree (i.e., no matter what the highest power of *n* in *P(n)* is), there is a natural number *k* such that for every $n > k$, $2^n > P(n)$. A problem of complexity that is not bounded by any polynomial is said to have a *superpolynomial complexity*.

The distinction between polynomial-time algorithms and superpolynomial ones, that is, between problems in $\mathcal{P}$ and problems not in $\mathcal{P}$, has proved to be extremely fruitful in computer scientists' attempts to demarcate between feasible and infeasible problems. At least at the extensional level, there is a striking correlation between problems that we take to be feasible to solve and problems for which we have found a polytime algorithm that solves them.[5]

The reasons to think that the polynomial vs. superpolynomial distinction captures something very close to feasibility are not only empirical. There are some theoretical reasons to think that the correlation between feasible problems and the complexity class $\mathcal{P}$ is not coincidental. First, we saw that addition and multiplication, which are members of $\mathcal{P}$, are feasible. We tend to think that, usually, running two feasible algorithms in a sequence, or using one as a subroutine of another, is also feasible.

There is a limit, of course, to how many iterations of sequential computation or how many calls for sub-routines, keep the computation feasible, but let's defer discussing that for a few paragraphs. The complexity of doing two computations in a sequence, one with complexity $f(n)$ and the other with complexity $g(n)$, is the sum $f(n)+g(n)$. The complexity of calling a subroutine of complexity $f(n)$ inside a loop that runs $g(n)$ times is $f(n)*g(n)$. Since the sum or product of every two polynomials is itself a polynomial, the complexity of running two polytime algorithms in a sequence or one as a subroutine of the other is also polynomial. Thus, we expect that, generally, polytime algorithms should be feasible.

On the other hand, since feasible algorithms should generally maintain a kind of closure under running in sequence or calling other feasible algorithms as subroutines, we expect the class of feasible computations to correspond to a class of functions that maintain something

[5] There is some subtlety here. A problem taken to be feasible even in the worst case is usually one for which we have found an algorithm with worst-case polytime complexity that solves it. If finding a solution with high probability is taken to suffice for feasibility, the relevant correlate is usually a probabilistic polytime algorithm. If only a subclass of the instances of a problem is taken as feasible, the relevant correlate is usually a polytime algorithm for solving instances in that subclass, and so on.

akin to a kind of closure. We saw that problems with exponential complexity are infeasible. However, there aren't many classes of functions that have this kind of closure and grow slower than exponentials, but still faster than polynomials. Thus, if any approximately adequate boundary between feasible and infeasible problems can be drawn, we don't expect it to lie *very* far from the polynomial vs. superpolynomial distinction. To motivate expanding the boundary beyond $\mathcal{P}$, it seems we need to come up with an example of a clearly feasible problem with superpolynomial complexity *f(n)*, and then describe a class of functions that contains *f(n)* that includes only functions that describe feasible computation times, and that will maintain the kind of closure needed under sequential computation and calling sub-routines. There aren't currently many candidates for the job.[6]

Another thing to note is the robustness of the polynomial vs. superpolynomial distinction across different computational models. A major feature in favour of the model of a Turing machine is that it is robust under changes in the computational model. Turing-computable (incomputable) problems can be shown to be computable (incomputable) in other computation models that were offered, like multi-tape Turing machines, lambda calculus, Boolean circuits, and cellular automata (see Wigderson, 2019, p. 120). This gave rise to the famous Church-Turing thesis, which states that every computable problem is computable by a Turing machine.

The complexity class $\mathcal{P}$ has a similar kind of robustness. The membership of a problem in $\mathcal{P}$ is independent of the specific model for computation we choose (see Valiant, 2013, p. 36, Wigderson, 2019, p. 43). Thus, membership in $\mathcal{P}$ seems to capture a fundamental property of a problem, rather than a contingent feature that depends on the specifics of human modelling. Figuratively, we can say that the partition of the set of problems into problems in $\mathcal{P}$ and

[6] One complexity class that attracts some attention is the class of so-called *quasi-polynomial* complexity (not to be conflated with *quasi-polynomial functions* from combinatorics, which are another thing). Problems with quasi-polynomial complexity require computation time of less than $kn^{\log n}$ or in general $kn^{\log^c n}$ where k and c are constants.

problems not in $\mathcal{P}$ seems to carve the complexity landscape at its joints. Whether feasibility should be characterised only by the polynomial vs. superpolynomial distinction or not, it does seem clear that membership in $\mathcal{P}$ is at least not coincidentally correlated with feasibility.

3. One-Way Functions and Strong Encryption

Some functions are easy to compute, some are harder. Suppose an injective function f is easy to compute. That is, for every input x, computing $f(x)$ is easy. Here is a question: what about the inverse function f^{-1}? If for every input x, it is easy to compute $f(x)$, is it also the case that for every output y of f, it is easy to compute $f^{-1}(y)$? In a moment, we will make the question more precise, but before that, let's start with an example that gives an intuition of why the answer is most likely negative.

Consider the case of multiplication. As we saw, computing the product of two numbers is relatively easy. As we know, every number is the product of its prime factors. For convenience, let's focus only on numbers that are products of two primes. Given two prime numbers, it is relatively easy to find their product. What about the inverse problem of *prime factorisation?* Suppose you are given a number n. How hard is it to find the prime factors of n? For very small numbers, it's easy. We immediately see that if n=15, the solution is 3 and 5, and if n=77, the solution is 7 and 11. However, for even slightly bigger numbers, the task becomes very hard. Try it for yourself. Let n be 277,351. What are the prime factors? Here is a safe bet: without a calculator, it will probably take you more time to find them than to read the rest of this chapter and the next one. Even if you use a calculator, it will probably take you quite some time.

Compare this to the inverse problem of multiplying the two primes in the footnote[7] and verifying that they are the correct ones, which will probably take you less than 30 seconds.

Why is it so hard to find the prime factorisation of a big number? Essentially, the reason is that there are too many options we can't rule out in advance. Naïve solvers need to examine all the options until they find the correct one, and examining all the options requires exponential time.[8] Professional mathematicians can do a bit better, but not much better. They still need exponential time, though with a somewhat better exponent.[9] There seems to be no path we can take that can guarantee a solution that doesn't include examining a considerable portion of the options.

We can describe the situation as follows: computing the function that gets two prime numbers as inputs and outputs their product requires only polynomial time, but computing its inverse that gives the prime factorisation of a given number seems to require superpolynomial time. In other words, multiplying primes is in $\mathcal{P}$, but prime factorisation does not seem to be in $\mathcal{P}$. If we identify $\mathcal{P}$ as the class of functions that are easily computable, we get that multiplying primes is easy, but its inverse, so it seems, is hard.

As we saw, finding lower bounds to the complexity of a problem is very difficult. Thus, it shouldn't come as a surprise that up to now, no one has been able to *prove* that prime factorisation is not in $\mathcal{P}$. The reason we think that the conjecture that prime factorisation is not in $\mathcal{P}$ is most likely true is that so many very smart people spent years trying to refute it by finding a polytime algorithm that solves it. After all, finding such an algorithm amounts to breaking the RSA cryptosystem. In the RSA cryptosystem, the encryption algorithm involves

[7] 337 and 823.

[8] Well, not *all* the options, because at least one of the prime factors is not larger than $\sqrt{n}$, so a naïve algorithm is of complexity exponential in $\sqrt{n}$ rather than n. Exponentials with smaller exponents than n are sometimes called *sub-exponentials*.

[9] The currently best-known algorithm, GNFS, is believed to require computational time exponential in $\sqrt[3]{n}$ (Valiant, 2013, p. 38).

the product of two big prime numbers, and the decryption algorithm requires factoring the resulting product back to the primes. Naturally, many people are motivated to find a quick way to do that. Currently (as far as we know),[10] no one has.

The discussion leads to defining the concept of *one-way functions*. For this, we first need to define the concepts of *negligible functions* and *negligible probability* (see Goldreich, 2001, §2.2.1). A function $\mu(n)$: $\mathbb{N} \rightarrow \mathbb{R}$ is called *negligible* if for every positive polynomial[11] $P(n)$ and for every sufficiently large number, that is, for every number $n > n_P$ (where n_P depends on $P(n)$), $\mu(n) < \frac{1}{P(n)}$. That is, at the limit, $\mu(n)$, decreases faster than the reciprocal of every positive polynomial.

An algorithm *A* has a *negligible success probability of computing f* if the probability that *A* succeeds in computing *f* is bounded by a negligible function of the length of the input. The idea behind the definition is the following. Suppose *A* is an algorithm with a negligible success probability to compute a function *f*. Then the success probability of computing *f* by running *A* $P(n)$ times, where $P(n)$ is any positive polynomial of *n*, is also negligible.[12] One cannot breach the negligibility threshold if one has only polynomial resources.

A bijective[13] function $f: X \rightarrow Y$ is called a *one-way function* if computing *f* is easy and computing f^{-1} is hard, not only in the worst case but *on average*, in the following sense: Given $x \epsilon X$, and not given y, computing that $f(x) = y$ requires at most polynomial time (i.e., $f \epsilon \mathcal{P}$). However, given y, and not given x, computing that $f^{-1}(y) = x$ not only requires

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[10] Since this problem is of great interest to intelligence agencies, it's possible that some agencies have found a solution but didn't share it with the philosophical community.

[11] $P(n)$ is positive if $P(n) > 0$ for every *n*.

[12] Let the success probability of *A* be bounded by the negligible function $\mu(n)$ and let $P(n)$ be a positive polynomial. According to Boole's Inequality (also known as the Union Bound), the success probability of running *A* $P(n)$ times is bounded by $P(n) \cdot \mu(n)$. Now let $Q(n)$ be any positive polynomial. Define: $R(n) = P(n) \cdot Q(n)$. $R(n)$ itself is of course a positive polynomial. Thus, if $\mu(n)$ is negligible, then there is an n_R such that $\mu(n) < \frac{1}{R(n)} = \frac{1}{P(n) \cdot Q(n)}$ for every $n > n_R$. Therefore, $P(n) \cdot \mu(n) < \frac{1}{Q(n)}$ for every $n > n_R$. Therefore, $P(n) \cdot \mu(n)$ is negligible, so repeating *A* $P(n)$ times has negligible success probability.

[13] I simplify here. One-way functions need not be bijective, but some very useful ones are. Bijective one-way functions are called *one-way permutations*. This is not important for what follows.

superpolynomial time in the worst case, but also for every (possibly probabilistic) polytime algorithm A, the probability that A computes f^{-1} for a randomly chosen input y is negligible (ibid., §2.2).

It is straightforward why one-way functions can be useful for cryptography. Suppose we want to encrypt a message *m*. For cryptographic purposes, *m* can be treated as a number, e.g., the number associated with the binary string that encodes the content of *m*. If *f* is a one-way function, computing *f(m)* is easy, as it requires only polynomial time. The resulting output, *f(m)*, can function as the ciphertext (when we take *f(m)* to represent letters, we get a seemingly random sequence of letters, as in the ciphertext in Figure 1). Decrypting the ciphertext requires computing $f^{-1}(f(m))$, which requires superpolynomial time. It is here that the demand for *average-case* hardness becomes clear. If in many cases the ciphertext can be broken easily, it is of little consolation if there exist some cases in which it is hard to break. We need the success probability of any feasible algorithm to crack the encryption to be very small.

For some cryptographic purposes, we need functions that are not only one-way but also possess a *trapdoor*. These are functions that are easy to compute and hard to invert, but for which there exists a piece of privileged information *K*, such that, *equipped with K*, the function is easy to invert. For instance, in the RSA cryptosystem, if one knows the correct prime factorisation *in advance*, then one can easily decrypt the resulting ciphertexts (i.e., invert the encryption function). The correct prime factorisation functions as the privileged information *K*, usually referred to as the *private key* or the *decryption key.*[14]

As one may expect from the case of prime factorisation, it is not yet proven that one-way functions exist. However, like the case of multiplication and prime factorisation, other functions used in cryptography are conjectured to be one-way, and many very smart people

[14] In reality, the private key is not exactly the correct prime factorisation, but it can be easily computed from the prime factorisation (and vice versa).

have spent thousands of hours attempting, in effect, to refute the hypothesis that they are. From a philosophy of science standpoint, the conjecture that one-way functions exist has withstood severe testing and refutation attempts that don't fall short of our most strongly supported scientific discoveries. It is very reasonable, then, to adopt the assumption that one-way functions exist, and as I said above, we literally risk all our money every day on the truth of this assumption.

In any case, the assumption that one-way functions do exist is one of the central assumptions of modern cryptography. The other central assumption is that the participants in the communication can "generate unlimited number of independent, unbiased coin tosses" (Wigderson, 2019, p. 204), that is, that one can generate random numbers. Together, they imply that *strong encryption* is possible. A "strong" encryption scheme is an encryption scheme that has some desirable properties. Not all of these desired properties are important for the purposes of this thesis, but it's worth listing those that are.

One such property that we have already mentioned is the inability to guess the key from the ciphertext with a non-negligible probability. Note that we can always just make a random guess, so the probability of making a successful guess at random must itself be negligible, and no deterministic or probabilistic polytime algorithm can do non-negligibly better than random guessing. Let me emphasise (as this will be important later), that it does *not* follow that we *do* have a systematic way to guess with probability even *negligibly* better than random upon seeing a ciphertext. It is just that it is *proven* that if one-way functions exist, then the best we can hope for when we use a polytime algorithm for guessing the key based on the ciphertext is to do at most negligibly better than random guessing.

I shall mention two more properties that will be relevant later: immunity to *distinguishing attacks* and immunity to *chosen plaintext attacks*. An encryption scheme is vulnerable to distinguishing attacks if ciphertexts can be feasibly distinguished from random data with non-negligible success probability. Thus, in an encryption scheme that is immune to distinguishing attacks, no feasible process can distinguish between encrypted data and random data. For an agent with polynomially bounded computational resources, for all the agent can tell, encrypted data appears to be random (see Wigderson, 2019, ch. 7, 18 and especially §7.3.1).

Chosen plaintext attacks are cases in which a party that wants to break the encryption has access to the encryption algorithm (either by knowing the algorithm or by physical access to the encryption device). Then the attacker creates plaintexts and encrypts them, resulting in pairs of plaintexts and their corresponding ciphertexts. If the attacker can use these pairs to (feasibly) get information about how to *decrypt* ciphertexts, we say that the encryption scheme is vulnerable to chosen plaintext attacks. Immunity to chosen plaintext attacks implies immunity to *known plaintext attacks*, which are the same as chosen plaintext attacks, except that the attacker doesn't choose which plaintexts to encrypt. If one-way functions exist, then we can build encryption schemes that are immune to chosen plaintext attacks. This last result, as we shall see in the next chapter, poses limits on what is learnable.

4. Objections to Identifying Polytime Complexity with Feasibility

Some natural objections, or at least calls for clarification, immediately come to mind. We can call them *the High Degree Polynomial/Low Exponent Objection, the Asymptotic Objection,* and *the Approximation Objection*. Let's consider each objection in turn. The first two are related, so part of the reply to the first objection comes in the reply to the second.

The High Degree Polynomial/Low Exponent Objection: Polynomials can come in any degree. A problem with complexity of n^{100} (which is a polynomial of degree 100) will take

100^{100} steps for an input of length 100, much more than the 2^{100} steps required for the same input for a problem of complexity 2^n. Clearly, problems of complexity n^{100} are infeasible, even though they have polynomial complexity. Similarly, exponents can be of any size. Clearly, a problem with complexity $2^{0.000001 \cdot n}$ is feasible for all practical purposes, as it remains very close to linear for long inputs.[15] Thus, the polynomial vs. superpolynomial distinction is both too permissive and too restrictive when it comes to feasibility.

Reply: Mathematically, this is of course correct, but the conclusion is too quick. While it is true that exponents can be of any size and polynomials can come in any degree, it doesn't follow that problems, and especially problems we are interested in solving, are evenly distributed with complexities of any polynomial degree or any exponent size. As a matter of fact, we rarely encounter problems with complexity of a high-degree polynomial or a very low exponent. As Valiant (2013) puts it:

> "…the majority of algorithms that are known for important problems conveniently dichotomize between feasibly low degree polynomials, such as quadratic, $O(n^2)$, and proper exponentials, such as 2^n. Hence, for reasons that are not understood, this polynomial versus exponential criterion is more useful in practice than its bare definition justifies" (ibid., p. 34).

Wigderson (2019, p. 18) states in this regard that "there seems to be a law of small numbers at work". That is, while the definition of class $\mathcal{P}$ by itself does not suffice to demarcate between feasible and infeasible problems, the terrain of the problems we are interested in solving is built (for reasons we don't yet understand) such that the set of feasible problems fits very nicely with the low-degree part of class $\mathcal{P}$.

[15] $2^{0.000001n}$ reaches *2n* only at around $n = 10^7$. If you find 10^7 not very long, change the example to $2^{0.000000001n}$.

The Asymptotic Objection: All this talk about complexity in general and polynomial vs. superpolynomial complexity in particular only applies at the limit, when *n* approaches infinity. Therefore, it doesn't say much about the feasibility of real problems with finite input lengths that people are actually trying to solve.

Reply: This objection is highly connected to the first and can be seen as a generalised version of it. Again, the premise here is correct, but the conclusion is too quick. As a matter of fact, we don't even need to think of high-degree polynomials to see that this is the case. Even for algorithms with linear complexity, that is, algorithms with complexity $an+b$ where a and b are constants, the constants a and b of those algorithms can be so big (e.g., 10^{100}), that they are completely infeasible to execute. Indeed, even tasks with complexity that is independent of the input length, like "count to 10^{100}", can be completely infeasible (the complexity of counting to 10^{100} is independent of the length of the description of the task). So granted, the behaviour of an algorithm at the limit does not give full information about the behaviour of the algorithm for reasonable input lengths.

However, the force of this objection is limited. First, as in the previous objection, while theoretically, it's true that the asymptotically most efficient algorithm for a problem of interest can contain enormous constants, in practice, this is hardly ever the case. Such algorithms are called *galactic algorithms*. Their better efficiency compared to other algorithms is only manifest for input lengths that are too big to be of practical interest. A notable example is the best-known algorithm for multiplication, mentioned in fn. 2[16]. In practice, we currently use other algorithms for multiplication. However, such examples are rare, and it is not unlikely that algorithms with better constants will be found in the future, as is usually the case.

More importantly, this objection will not be relevant for my solution to the Paradox of the Ravens. My argument in Chapter 6 is based on some tasks being infeasible, not on tasks being

[16] References to footnotes are always to footnotes in the same chapter.

feasible. Thus, I shouldn't worry whether I'm too *permissive* in calling a task feasible when in practice it isn't. Rather, I should worry whether I'm too *restrictive*, calling a task infeasible when it is feasible in practice, and seems infeasible only due to its behaviour at the limit. Note, however, that in practice, algorithms with high complexity at the limit usually show their infeasibility *very early on*, not only in the faraway limit. We saw, for instance, that an algorithm with complexity 2^n becomes infeasible for very modest input sizes. Thus, if a task is of exponential complexity, and the exponent is not *very low*, it is safe to treat it as infeasible for all practical purposes.

A useful and more systematic way to analyse the situation is by introducing the term *feasibility range*. For each problem Q, let's define the feasibility range of Q as the maximal input length for which Q is feasibly solvable. What counts as feasibly solvable depends on what we count as the relevant computational resources. Let's call an input length *practical* for a problem Q if there is any situation in practice in which we are interested in solving Q for an input of at least that length. We can call a problem *feasible* if its feasibility range exceeds any practical input length, *partially feasible* if its feasibility range contains a significant portion, though still not all of the practical input lengths, and *practically infeasible* if its feasibility range contains only a tiny portion of the practical input lengths. We can also devise a quantitative measure of practical feasibility if needed.

The points above can be summarised with the claim that, except for rare cases, the feasibility range of problems of interest that are members of $\mathcal{P}$ is much greater than the feasibility range of problems of interest that are not members of $\mathcal{P}$ and that, in general, problems in $\mathcal{P}$ that are of interest are in most cases feasible, while problems not in $\mathcal{P}$ are at most partially feasible.

Here is another way to think about the issue. Consider all problems of interest coupled with all suitable inputs of practical length. Among all these instances, some are feasible to solve, and others are not. Only a tiny minority among the feasible cases are instances of problems not in $\mathcal{P}$. Thus, this objection might call for what I called 'minor border corrections', but it doesn't change the big picture. Class $\mathcal{P}$ seems to capture a lot of what we mean by feasible problems.

The Approximation Objection: The discussion so far was about the time required to find an exact solution to a problem. But surely, in most cases, finding an approximate solution suffices, so the discussion should not be about the complexity of solving a problem but about the complexity of finding an approximate solution to a problem.

Reply: It's true that in some cases an approximate solution can satisfy our needs (although there are cases in which tiny initial errors can evolve into very large ones). In such cases, we might say that the problem of interest is finding an approximate solution rather than an exact solution, and the problem of finding an approximate solution can be feasible while the problem of finding an exact solution is not. The main moral to draw from that is that the fact we weren't able to find an exact solution is not necessarily an indication of the hardness of finding an approximate solution, and in general, it is always good to be very clear about the problem we are trying to solve.

However, one should not be tempted to think that approximation can be a kind of panacea for every problem for which finding an exact solution is infeasible. Not only is there no reason to think that an approximate solution can be feasibly computed for most problems, but there is a *theorem,* called the *PCP theorem*, which states that under the not yet proved but extremely likely conjecture that $\mathcal{P}\neq\mathcal{NP}$,[17] there are many problems for which not even an approximation can be found in polynomial time (see Aaronson, 2017, §3, Wigderson, 2019, §4.3 and §10.3).

[17] $\mathcal{NP}$ is the class of problems for which it is possible to verify in polynomial time whether a candidate solution is correct. Obviously, $\mathcal{P}\subseteq\mathcal{NP}$ since computing a solution is a way to verify it, but there are strong reasons to think that the converse doesn't hold. Some problems require superpolynomial time to find a solution, even though

Thus, I grant that the hardness of finding an exact solution does not by itself suffice in any case to declare that all hope is lost. Sometimes an approximate solution is easy to find and can satisfy our needs as well. However, the importance of the distinction between polynomial and superpolynomial complexity remains intact. It is the possibility of computing an approximation in polynomial time that makes finding an approximation feasible. While it might be that finding an approximation is in $\mathcal{P}$, but finding an exact solution is not, this doesn't take away from the importance of $\mathcal{P}$ as a mathematical object that characterises which tasks are feasible, and which are not. The objection serves as a warning to define our problems accurately and say explicitly whether we look for an exact solution or only an approximate one. Showing that the first is infeasible doesn't show that the latter is, too. But if we know which task we are interested in, the fact that this task requires superpolynomial time is usually enough to make it infeasible.

5. Conclusion

This chapter explained the basics of computational complexity theory. The last section dealt with some common reservations people tend to have about the polynomial/superpolynomial distinction. We can treat this chapter and the next as two halves of a larger chapter, and we have now reached the half-time break. The next chapter discusses the connection of computational complexity theory to epistemology. Applying the theory to specific learners who are usually engaged not with a general problem but rather with concrete instances of it may raise further worries and objections. The next chapter begins by dealing with these further possible worries.

once we have a candidate solution at hand we can check whether it is a correct solution in polynomial time. Prime factorisation is an example of a problem believed to be in $\mathcal{NP}$ but not in $\mathcal{P}$.

Chapter 3: Complexity and Learning

1. Preliminary Remarks

How is computational complexity related to learning? In short, if learning is computational, then learning something amounts to performing a computational task. If the task to be performed requires more computational resources than the learner has, then the learner cannot perform the task, and thus cannot learn the thing to be learnt. This might immediately raise some objections for some readers, especially those who are already familiar with computational complexity theory. We can call these objections *the Specific Case Objection*, *the Classification Objection*, and *the Ad-Hoc Algorithm Objection*. It's a good idea to start by clearing the table of these initial objections before we say anything more systematic.

The Specific Case Objection: The computational complexity of a problem is defined as the complexity of the worst case within a class of cases rather than of specific cases. A problem can be very complex, but some instances of it can be very simple. E.g., when you multiply by 0, you can immediately know that the solution is 0 regardless of the length of the input. The fact that a learner needs to solve an instance of a problem with high complexity doesn't mean that the instance the learner is trying to solve is too demanding. In a sense, the complexity of solving each specific instance using a specific algorithm is a constant, rather than a function of the length of the input, and we cannot know which constant applies to which instance in advance.

Reply: Some instances of complex problems can indeed be much simpler than the worst case. It's hard to know how far each instance is from the worst case. It is here that the example of cryptographic problems becomes particularly relevant. If one-way functions exist, then some problems in cryptography are *hard on average*. That is, recall, for every polytime algorithm A, the probability of A solving a randomly chosen instance of the cryptographic problem is negligible for long enough inputs. Thus, if a learner with bounded resources needs to solve a randomly selected instance of a cryptographic problem with a long enough input, we can predict with high certainty that the learner will not be able to do that.

The Classification Objection: Classifying an instance as falling within a specific problem rather than within another is arbitrary. We might gather instances of what we usually take to be different problems, group them in a gerrymandered (or gruesome) way, and treat them as instances of the same problem. Thus, saying that an instance falls within a class that has some computational complexity is arbitrary.

Reply: Again, focusing on cryptographic problems, which are hard on average, can be useful to see why this objection is ineffective. Let's take a specific instance of the prime factorisation problem: find the prime factors of 52,322,940,983,667,496,651. This is actually a pretty small number for cryptographic purposes, and online calculators can find its prime factors. The point, however, is that unless today is your lucky day and you spend it factoring primes instead of buying a lottery ticket, or unless some prior knowledge of the answer found its way to your algorithm, you will need to perform a very long calculation (conditional, of course, on the assumption that multiplication of primes is a one-way function). This remains the case regardless of how we decide to group specific computational tasks into general problems.

The Ad-Hoc Algorithm Objection: OK, but as you acknowledged in your last reply, for every specific instance, some algorithms solve it instantly, namely, algorithms that contain the specific instruction to output the specific solution of the instance when given the specific input of the instance. For example, you can have an algorithm with whatever instructions, which also contains the instruction that, for the specific input 52,322,940,983,667,496,651, output the pair of numbers 5,463,458,053 and 9,576,890,767. This algorithm solves this specific instance of prime factorisation in one step, or a very small number of steps if we count reading and writing each digit as one step. This shows that there is no meaning in ascribing computational complexity to an instance.

Reply: We might call these algorithms *ad-hoc algorithms* or algorithms that contain *cheats.* Before replying to the objection, let me warn you against a possible mistake one can make. One might be tempted to think that no specific task is too computationally demanding for a learner who has a suitable cheat, and thus, nothing is too complex to learn. This is too strong, because the input or the required output can be so long that it takes the learner too much time to read the input or write the output. It's important not to overlook this fact because the answers to some questions we find interesting, for instance, what is the optimal strategy for chess, may be too long for us to read or store in any physical memory device. Thus, it's not clear that we can store a cheat for solving chess in our memory.

However, the important point is that the theory of computational complexity doesn't seem to give us tools to speak meaningfully about the complexity of specific instances and thus the possibility of finding solutions to them by bounded learners. Relatedly, one might worry that since computational complexity theory discusses only the asymptotic behaviour of problems and algorithms at the limit where the length of the input approaches infinity, there is not much that follows for finite learners encountering specific instances with finite input lengths.

Regarding the last point, we have started to deal with this worry in the last chapter when we discussed the asymptotic objection. The rest of this chapter deals with applying the insights of computational complexity theory to finite learners trying to solve specific instances, so let's now focus only on the previous point. Granted, at least for many instances of interest that seem very complex, there is a cheat that allows a bounded learner to solve the task very quickly.

This, however, still doesn't change the fact that for an extremely large majority of the instances of (say) prime factorisation, a bounded learner will not have a cheat and will thus need a very long calculation to solve them. If we use 4096-bit RSA, which is the mode of encryption I used for the message in Figure 1, there are roughly 2^{4096} possible keys,[1] or possible different instances of prime factorisation, that can be associated with decrypting this message. A learner might have a cheat for some of those possible instances, but to have a cheat, the learner needs to store the cheat in their memory.

Even if we assume that for any instance of prime factorisation, the learner uses only one bit of memory to store each cheat,[2] we still get that a physical learner, and even all physical learners combined, can at best store cheats for an extremely tiny fraction of instances of prime factorisation. It's estimated that there are about $10^{90} (\approx 2^{300})$ elementary particles in the universe, so even if every elementary particle is used to store a different cheat, physical learners will still need to run very long calculations to solve many orders of magnitude more than 99.999999999999999999999999999999999% of the instances.

Cryptography is only an extreme example that illustrates a pattern. Physical learners cannot cheat their way into survival in the world. They encounter too many instances of problems, many of which could have easily been different instances. Pre-memorising all possible solutions is physically impossible. If there is a way to group some instances and find one feasible algorithm, both in terms of computation time and the memory required to store it, that

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[1] According to the prime number theorem, there are about $\frac{N}{\log(N)}$ prime numbers that are not greater than *N*, for large *N*'s. Not all keys are made of primes with exactly equal lengths, so there are actually considerably more options, but if we focus on primes with equal length, we need a pair of primes of length 2048. There are roughly $\frac{2^{2048}}{2048 \log 2}$ of those, so $\left(\frac{2^{2048}}{2048 \log 2}\right)^2$ ways to pair them, which is roughly 2^{4096}.

[2] In fact, considerations of algorithmic complexity that will be discussed in the next chapter show that the vast majority of cheats will require roughly 4096 bits instead of one, but even without it the bottom line remains the same.

solves them all, it might be very beneficial, but a necessary condition for finding such an algorithm is that such an algorithm exists.[3]

I conclude that although computational complexity theory deals with asymptotic behaviour of general problems that have infinitely many instances, it is too quick to infer that the theory is not relevant for the analysis of what finite learners can and mainly *cannot* learn in specific cases. If learning is computational, computational complexity theory may indeed tell us quite a lot about the limits of learnability by physical learners. We shall now examine more systematically the lessons we can draw.

2. Realistic Learners

This section aims to provide a model for realistic learners. By 'realistic', I don't necessarily mean that the modelled learners actually exist. It's an idealised model of learners who are computationally bounded. If we put high enough bounds, the capacities of the modelled learner would exceed those of any physically possible learner. However, even such an idealised realistic learner is still much more limited than other idealised learners, e.g., Bayesian ones.

The emphasis would not be on what realistic learners can learn, but rather on what they cannot. Idealised realistic learners can function as upper bounds on actual learners. If something is not learnable even for highly idealised realistic learners, then it is not learnable by actual ones. The lower we set the bounds in our idealisation, the more that will be deemed unlearnable. However, the lower we set the bounds, the less we can be sure that no actual learner can break them. In any case, the arguments in this thesis should work even for models

[3] And that there is a feasible learning process that leads the learner to find it. More on that below.

of realistic learners that are so idealised that their abilities clearly exceed those of any learner in our physical universe.

A learner doesn't just solve problems or perform specific tasks according to pre-programmed algorithms. A calculator might be used for learning, but we don't think of the calculator itself as a learner. Learners regularly get novel information and process this information to discover novel ways to solve a problem or perform a task. We wouldn't call it learning if the learner were pre-programmed to start solving the problem in a new way, regardless of the information the learner was given. Rather, to count as learning, these novel ways of solving the problem should result from the learner processing the specific information that was fed into the learning process.[4]

Thus, even assuming that all learning processes can be simulated by a Turing machine, not every Turing machine can be classified as a learner. The pre-programmed algorithms of the learner must include at least some algorithms that are *learning algorithms*, that is, algorithms that can process novel information the learner gets and output novel or updated ways to solve problems or perform tasks based on that information. A model of a learning process needs to include the information the learner gets throughout the process.

Since the learner often needs to select which algorithm to execute among the ones at the learner's disposal, it is useful to think of the learner's learning algorithms as one big "über-algorithm" that also includes instructions on which "smaller" algorithms to call and when. The learning process will be modelled as divided into stages or moments, and at each moment the learner can (but doesn't have to) get novel information and run some computations.

[4] Many times, learning involves teachers. Teachers can enhance the learning process in many ways, by offering the right advice at the right moment or labelling the information we process (e.g. pointing out that the object to the left of us is a cat), and providing us with knowledge by testimony, thus saving us from learning something from scratch. In what follows, though, I shall treat the input of teachers just as another part of the input processed by the learner. Regarding testimony, a lot of what we learn is learnt via testimony, but for testimony to transmit knowledge, at least the first link in the testimonial chain needed to have learnt what is to be learnt in a non-testimonial way, and the discussion in this chapter is intended to apply to this non-testimonial learning process.

Another thing to note is that learning should be different from mere guessing. An algorithm might randomly but successfully guess a method to solve a problem. For instance, a learner can run an algorithm that, for an input of a 20-digit number, selects two 10-digit prime numbers randomly, and outputs them. If this algorithm happened to select 5,463,458,053 and 9,576,890,767, then this algorithm would yield knowledge of how to factor 52,322,940,983,667,496,651. By using multiplication, the learner may *learn* that the two prime numbers give the correct factorisation. However, we wouldn't want to say in this case that the learner learnt the factorisation *from* the input. The input didn't play any role in making this successful wild guess.

Furthermore, the correct prime factorisation was learnt in this case by a fluke rather than by processing the learner's information methodically. Epistemology focuses more on what can be learnt methodically than on what can be learnt by flukes. At the very least, there is an important epistemic difference between the two categories. If one learning algorithm can yield successful results methodically or reliably, then it is not epistemically on a par with an algorithm that can yield successful results but only by a fluke. When I talk about what is learnable, I usually mean what is learnable methodically from various kinds of inputs, rather than what can be learnt only conditionally on making an extremely lucky guess.

Here is a general model: Let l be a learner and T a task to be performed by the learner.[5] A learner l can be modelled as an ordered quintuple $l=\langle A,m,t,n,T_0\rangle$, where A is l's learning algorithm, m is l's memory; (little) t is l's time to live (measured in number of computational steps);[6] n is the maximal input length l gets at each stage of the process; and (capital) T_0 is l's initial algorithm for performing T. If our learner has relevant background knowledge, we can model our learner as an ordered sextuple that includes the background knowledge as well. At

[5] The task can be of various levels of generality.

[6] We can measure it in physical time instead of computational time, and add another variable s for computation *speed*, and deduce the number of computational steps available for the learner from them.

each stage of the process, *l* gets inputs for *l*'s learning process. Let *I(k)* be a *k*-tuple denoting the inputs *l* got at each stage up to the *k*-th one. *l*'s state after *k* stages, getting the information *I(k)*, is modelled as the ordered quintuple $l(I(k))=<A,m,t,n,T(I(k))>$, where *T(I(k))* is *l*'s algorithm for performing *T* after running *l*'s learning algorithm *A* on *I(k)*, using *l*'s computational resources.[7]

l's learning process as a whole can be modelled as the succession of *l*'s states from stage 0 to the final stage of the learning process. Let's call *l*'s algorithm for performing *T* at the final stage of the learning process *T(I(end))*. For the learning process to be *incrementally successful*, the output of the process, *T(I(end))*, should be an algorithm that is better at performing *T* than T_0. For the learning process to count as *successful* (period) or as resulting in learning, *T(I(end))* must be good enough for the learner to be said to know how to perform *T*, or for the learner to be able to gain knowledge by applying *T(I(end))* to *T*.[8] For *A* to be a successful learning algorithm, it should reliably yield successful learning processes, at least for the variety of inputs the learner usually gets.

So far, the model doesn't say anything about limitations on *l*'s memory, the amount of information *l* can process, or about *l*'s time to live, and hence time limits to *l*'s learning process or the running time of the algorithm *T(I(k))*. The model, as it is, allows for the values of the variables *m,t,* and *n* to be infinite, so in its current form, the model is still too general to model realistic learners specifically. Realistic learners are limited in their resources. They have limited memory, limited access to information, and limited time to live. They need to be able to use the products of their learning during that limited time.[9]

[7] If we want, the variables *A, m, t,* and *n* can be functions of which stage of the process *l* is in, but it adds nothing to the arguments in this thesis as far as I can tell.
[8] Or, if we take the cognitive success that is the aim of the learning process not to be knowledge, then we can say that a learning process is successful if *T(I(end))* is, or can be used to achieve, the required cognitive success.
[9] We don't have to think of learners as individual organisms. We can model the learning process of a community of learners where the products of the process are intended to serve the later generations of the community. Still, we assume the community doesn't live forever and that the products need to be useful for the later generations, so the running time of *T(I(end))* must not be too long.

This requires both that the learning process provide results quickly and that those results can be used quickly. Realistic learners also need their learning processes and the products of those processes to be general enough to be useful for a variety of situations. This includes both that the learning process will be useful for a variety of inputs and that the product of the learning process is useful for a variety of cases, including cases that are different from the ones the learner has seen so far.

To model a realistic learner, we need to impose suitable finite bounds on *l*'s memory, *l*'s time to live, and the maximal input length *l* can get at each stage. We can think of these bounds as specific numbers: m_0, t_0, n_0, which are upper bounds on the number of memory cells *l* can use, the computational steps *l* can perform, and the maximal input length *l* can get per turn. The number t_0 is related both to the number of steps in the learning process *and* the runtime of the algorithm *T(I(end))*. For a learning process for performing *T* to be successful for a realistic learner, the number of steps of the learning process *plus* the runtime of *T(I(end))* must be less than t_0. Otherwise, the learner cannot perform the task *T* successfully.

We need to note, further, that for a realistic learner *l* to have a learning algorithm *A* or an algorithm for performing the task *T(I(k))* at their disposal, *A* and *T(I(k))* need somehow to be encoded in *l*'s memory. I don't distinguish here between the part of *l*'s memory that is dedicated to computation and the part of *l*'s memory that stores learning algorithms and task-performing algorithms. They will both be taken as part of *l*'s memory regardless of whether they use distinct physical configurations. Thus, encoding *A* and *T(I(k))* must use less than m_0 memory cells and only what's left of *l*'s memory can be used for further computations.

For instance, if $m_0 < 2^{4096}$, *l* cannot have a learning algorithm that already includes a giant table with 2^{4096} solutions to 2^{4096} prime factorisation problems and *l*'s final task-performing algorithm *T(I(k))* cannot include such a table either. Such considerations put a limit on the amount of cheating a realistic learner can do by having the answers in advance. We can call this *cheating by memorising the solution manual*. Some solution manuals are just too big to be memorised by a realistic learner.

Another kind of cheating occurs when a learner's learning algorithm A either completely ignores the inputs of the learning process and simply outputs an algorithm $T(I(end))$ that happens to solve T specifically. Similarly, A can encode specific possible inputs which coincidentally happen to be l's actual inputs, together with specific instructions about what to output for those inputs, such that l happens to output a good $T(I(end))$ for the specific inputs l got. For instance, A can tell the learner something like: whatever the input is, after k steps, output that the prime factorisation of 52,322,940,983,667,496,651 is 5,463,458,053 and 9,576,890,767. Or A can say something like: if you happen to get inputs $I_1, I_2, \ldots, I_k$ (which coincidentally happen to be l's actual inputs), output that the prime factorisation of 52,322,940,983,667,496,651 is 5,463,458,053 and 9,576,890,767.

We can call this *cheating by inexplicably successful wild guesses*. Realistic learners can be lucky enough to make a number of good wild guesses, but as I said above, not even all realistic learners combined can have algorithms that successfully guess more than a negligible fraction of the cases. Encoding each wild guess requires some memory, so l cannot encode too many wild guesses. In what follows, I assume that realistic learners don't have any cheats at all. With a very high probability, this is correct in at least most of the cases, so we can make this assumption to analyse the general case.

What should the bounds m_0, t_0, and n_0 be? A simple answer can be that they should correspond to the maximal resources available for a physical learner in our universe. This way, we get to model an ideal physical learner. Such a learner will be more talented than any learner that can ever exist in our universe, so everything that lies beyond the capabilities of this idealised learner is also beyond the capabilities of any physical learner.

It is useful at this point to define the notion of a *class of learners*.[10] For many purposes, we might not be interested in a specific learner with their specific learning algorithm A and initial task performing algorithm T_0. We may be interested in focusing on the limits of learnability posed by the bounds m_0, t_0, and n_0. We can say that these bounds define a class L_{m_0,t_0,n_0} that includes all learners that are bounded by m_0, t_0, and n_0. For the concept of a class of learners to be useful for discussing the limits of learnability, we need to either focus on tasks T that are of a suitable level of generality (e.g., solve any randomly selected prime factorisation problem, as opposed to factor the specific number 52,322,940,983,667,496,651) or to exclude cheaters from the class. Otherwise, for each specific instance there can many times be a member of the class who solves the problem by cheating. In what follows, when I speak about a class of learners, I assume that we exclude cheaters or that the task in question is suitably general.

If m_0, t_0, n_0 denote the maximal bounds on any physical learner in our universe, the class L_{m_0,t_0,n_0} denotes the class of physically possible learners. If we want to model more modest physical learners, like human learners, or humans with access to 2026 technology, then we can use suitably lower bounds. If we still put the bounds safely above the current capacity of humans with 2026 technology, we would still get an upper bound to what is currently learnable by humans. We can even, for some purposes, choose bounds that aim to model the average or median human learner.

Of course, the closer you want the bound to be to the *least upper bound*, or the closer you want your model to be to specific cases of actual learners, the messier and more empirically informed your model needs to be, and the likelier it becomes to make empirical errors that prevent your model from being an upper bound on the learning abilities of the actual learners you're trying to model. However, the arguments in this thesis should work even if we focus on any physical learner whatsoever, so we can let the physical universe set our bounds. When I

[10] This notion will play a major role in Chapter 7, specifically in §7.7.

talk about *realistic learners,* I will refer to learners who are *at most* as talented as an ideal physical learner, even though an ideal physical learner is a very high bound.

For mathematical proofs, though, it can sometimes be more convenient to think of the bounds not as specific numbers, but rather as functions of the length of the input. We can upgrade our ideal physical learners to what we can call an ideal polynomial learner (cf. Valiant, 2013). We can think of our learner as having a maximal input capacity *n,* which is not smaller than the input capacity of any physical learner. The number of possible inputs the learner can get is 2^n, which is gigantic. Let's suppose that the learner's memory and time to live *m(n), t(n)* must be polynomial functions of *n*, and all the algorithms this learner runs, at least above some threshold of input length, are polynomial in their time complexity, space complexity, and the amount of memory required to store them.

The learning abilities of polynomial learners exceed those of ideal physical learners. For physical learners, the 'learnable from' relation is not transitive. It might be that A is learnable from B, because the complexity of learning A from B is within the bounds; B is learnable from C for the same reason, but A is not learnable from C because learning A from C (e.g., by first learning B from C and then learning A from B) requires a number of steps that exceeds the learner's time to live. For a polynomial learner, learnability maintains a kind of closure, because if the complexity of learning A from B is polynomial in *n* and the complexity of learning B from C is polynomial in *n,* then the complexity of learning A from C is also polynomial in n.

This shows that even the model of a polynomial learner is too idealised to capture some important features of physical learners. However, polynomial learners are still far from being logically omniscient, because the closure of what is learnable by them does not include everything that logically follows from their knowledge. We should be careful not to conclude that a physical learner can do something because a polynomial learner is shown to be able to do it. However, ignoring the possibility of cheats in a negligible fraction of the cases, if something is not learnable by a polynomial learner, it is also not learnable by a physical learner. The idea of a polynomial learner brings us close to the concept of *PAC-Learnability*. This important concept in computational learning theory will be explained in the next section.

3. PAC-Learnability

So far, the desired cognitive success of the learning process has not been defined formally. There may be some advantages to that. For instance, we might take the desired cognitive success, e.g., knowledge, to be primitive, not requiring any further analysis, and interpret what we said so far based on the way we understand knowledge. The model is compatible with such an attitude. However, for some purposes, we might want to have a mathematically workable definition of the cognitive success sought.

The concept of *Probably Approximately Correct Learnability,* or in short, *PAC-Learnability* (as it's usually called), can be used for this purpose, at least for a specific type of learning called *supervised learning*. In supervised learning, the information the learner gets is in the form of *labelled examples*, that is, examples of the problem the learner needs to solve, together with the solution for this specific example. For instance, if the learner needs to classify whether pictures depict cats, a labelled example can be a picture of a cat with the label 'yes' or a picture that does not depict a cat with the label 'no'.[11]

The model of a polynomial learner presented above, while somewhat more general and loosely defined, is heavily influenced by the definition of PAC-Learnability. A problem is said

[11] Don't the labels make the process rely at least partially on testimony (namely, the supervisor's testimony that a given picture depicts a cat) that I said I'm going to bracket? In general, yes. But if there are ways for a realistic learner to detect that there is something common to cats or other types of examples, then the supervisor is not necessary to the process. When we discuss the Paradox of the Ravens, we shall see that what plays the crucial role is not the labels but the content of the examples.

to be PAC-learnable, loosely, if there is a feasible learning algorithm that can, with arbitrarily high probability, approximate a solution to the problem to an arbitrary accuracy given a relatively small number of labelled examples.

More formally: let ε and δ be positive numbers smaller than 1, which can be as small as we like. The δ will be associated in the definition with the 'probably' and the ε with the 'approximately'. Let *n* be the size of each example in terms of input length. A problem is called *PAC-Learnable* if there is a learning algorithm *A* such that for every probability distribution on the set of possible examples, it produces, with probability greater than $1-\delta$, a solution to the problem with an error smaller than ε, where the running time of *A* and the number of examples are at most polynomial in $\frac{1}{\delta}$, $\frac{1}{\varepsilon}$, and *n* (see Wigderson, 2019, p. 193; Valiant, 2013, Ch. 5).[12] That is, the more confidence and accuracy we aim for, the larger the number of examples the learner is allowed to examine and the time the learning algorithm *A* is allowed to run, but the number of examples and the needed computation time don't grow uncontrollably.

In the case of classifying cats, we can say that there is a function to be learnt, *f*, namely, a function that outputs 'yes' for all and only inputs that depict cats. The labelled pictures of cats and non-cats can be seen as *examples* of *f* (that is, examples of inputs with their corresponding outputs under *f*). The function *f* is PAC-learnable if a learner with polynomial resources (in $\frac{1}{\delta}$, $\frac{1}{\varepsilon}$ and the length of the input) can, with probability $1-\delta$, learn a function *f** that agrees with the classification of *f*, except perhaps for at most a measure ε of the cases.

[12] Sometimes PAC-learnability is defined as a weaker notion and the notion defined here is referred to as *efficient PAC-learnability* (see Kearns & Vazirani, 1994). Note that for every δ, ε, $n=a\delta=b\varepsilon$ where *a* and *b* are constants, so any polynomial in $\frac{1}{\varepsilon}, \frac{1}{\delta}$ is also a polynomial in *n* (with different constants). Thus, strictly speaking, the requirement for the runtime to be polynomial in *n* is redundant, because it follows from its being polynomial in $\frac{1}{\delta}, \frac{1}{\varepsilon}$. Additionally, processing examples takes time, so the requirement that the number of examples is polynomial in $\frac{1}{\delta}, \frac{1}{\varepsilon}, n$ is also redundant.

For realistic learners, we can, on the one hand, strengthen the requirements and demand that the required resources don't exceed the specific bounds (i.e., specific numbers) set by the physics of the learner. On the other hand, we need not require the definition to hold for any $\delta, \varepsilon > 0$, but rather for any $\delta > \delta_0, \varepsilon > \varepsilon_0$, where δ_0, ε_0 are the certainty and accuracy standards that are needed in the case at issue for the learner to achieve the cognitive success sought.

How PAC-learnability relates to various kinds of epistemic successes, e.g., knowledge, is an interesting philosophical question. I shall not discuss this question in depth here. However, to prevent a possible misunderstanding, I shall only note that successfully PAC-learning something is not equivalent to successfully guessing a lottery case. In a lottery case with one winning ticket out of a million, we can guess whether a ticket is a losing ticket with an error rate of 1/million, but we wouldn't say we know that a ticket is a losing ticket.

However, suppose that classifying whether an object is a cat is PAC-learnable from labelled pictures of cats and non-cats. That is, a polynomial learner that runs a suitable learning algorithm would, with high probability, be able to classify cats with a small error rate after seeing a reasonable number of examples. The reason the learner could classify cats is that the learner interacted in the right way with pictures of cats (or just with cats, given that the pictures are causally connected to actual cats). The "probably" doesn't mean that what the learner has *learnt* acts like a lottery, but rather that with a small chance the learner could have been unlucky during the learning process, and seen such a biased sample that the learning process wouldn't have been successful. The "approximately" doesn't mean that the classification works like a lottery, but rather that the learner's classification ability is not infallible.

We usually think that humans can know how to classify cats even if they're fallible in their classification. We also usually count humans as having knowledge even if it was improbable but possible for them to get bad evidence that would not lead to knowledge, instead of getting good evidence (compare: you can know that an actual fair coin is fair after seeing enough tosses, even if it's improbable but possible to get a sequence of heads that would not yield this knowledge). Thus, the fact that PAC-learnability only guarantees that a suitable algorithm would yield results that probably would be approximately correct does not by itself show that PAC-learnability falls short of knowability.

Even if PAC-learning doesn't suffice for achieving knowledge, and thus doesn't *suffice* for learnability simpliciter, when 'learning' is understood as coming to know, it still seems to be necessary, at least when we focus on learnability by realistic learners. To come to know something methodically, or reliably, rather than by luck, it seems we need to use a method that tends to give us at least an approximation of the truth. If this is correct, then showing that something is *not* PAC-learnable shows that it is also not learnable methodically by realistic learners. Readers who are reluctant to interpret learning as coming to know or as PAC-learning may treat my uses of "learnability" as placeholders for their favourite concept. It would be interesting to see whether the conclusions I draw about the limits of learnability would still apply to these other concepts, and if not, where the points of departure are.

4. The Limits of Learnability

If strong encryption is possible (that is, if some functions useful to cryptography are one-way), and if learnability is bounded by what is feasibly computable, then not all of $\mathcal{P}$ is PAC-learnable. Here is a proof (cf. Valiant, 2013 §5.8): suppose E is a strong encryption scheme and therefore immune to known plaintext attacks. In known plaintext attacks, recall, the attacker sees pairs of plaintexts and their corresponding ciphertexts, and is able to use that to crack the decryption algorithm or extract some information from ciphertexts about their corresponding plaintexts. Let D be the decryption algorithm. We include the private key k as part of D.

Decrypting messages using D (i.e., decrypting while possessing the private key) is a polytime process. Thus, D is in $\mathcal{P}$. If all of $\mathcal{P}$ is PAC-learnable, then so is D. That is, there is a polytime process that, by seeing a reasonable number of labelled examples of inputs and outputs of D, that is, pairs of ciphertexts and their corresponding plaintexts, is probably able to approximate D. That is, from seeing enough examples of pairs of plaintexts and their corresponding ciphertexts, one would, with high probability, be able to guess the plaintext from the ciphertext correctly most of the time. Therefore, E is vulnerable to known plaintext attacks, contrary to our initial supposition. Thus, if strong encryption is possible, then not all of $\mathcal{P}$ is PAC-learnable.

Now, of course, even if D is not learnable *from examples*, D is learnable in other ways, namely by getting to know the private key in other ways. For instance, we can be the ones who computed the private key by picking two big prime numbers, or we can learn by other means that the person who computed the key is keeping it in a drawer under their desk, and steal it, or the person who created the key can share it with us voluntarily. Thus, infeasibility of learnability in one way need not imply infeasibility of learnability in other ways. This highlights a general point: what is learnable, whether inductively or deductively, from one kind of data, may not be learnable from another kind of data.

Two comments are in order, however. First, learning from examples is an important mode of both day-to-day and scientific learning. Arguably, the data we learn scientific theories from are composed of examples of the laws of nature in action. This mode of learning is also the one discussed in the Paradox of the Ravens. Thus, it is important to note the inherent limitations of learning from examples. Second, even if sometimes access to privileged information can compensate for the hardness of learning from examples, we cannot assume without an argument that the required privileged information for every problem in $\mathcal{P}$, or even every problem in $\mathcal{P}$ we see examples of, can be found somewhere.[13]

5. Learning and Confirmation

Confirmation is essentially about what can be learnt from evidence (and background knowledge). I take it as a starting point. I shall examine here and in later chapters what follows from this starting point if we further assume both that learning is computational and that data externalism is true. Given that I take it as a starting point, I can't really offer an argument for it. If one offers an account of confirmation where a piece of evidence *e* confirms a hypothesis *h* even if nothing can be learnt from *e* about *h,* then we are most likely talking about different concepts. I offer some motivating remarks below, though.

I estimate that most readers will consider this starting point quite trivial in other contexts. Some readers might take the claim that confirmation is essentially about what can be learnt from evidence to be much more substantive than it actually is, given that the assumptions that data externalism is true and that learning is computational are substantive. However, it is data externalism and the claim that learning is computational that do the heavy lifting. One can adopt different assumptions about learning and data encoding, but still acknowledge that confirmation relates to what can be learnt from evidence.

As a minimal adequacy condition, we can say that a piece of evidence *e* provides novel confirmation for a hypothesis *h* for a learner *l* with background knowledge *K,* only if *l* should

[13] Even if we think, as some do (see Wigderson, 2019, §3.10), that nature is incapable of performing superpolynomial computations, that is, that the fundamental equations that govern dynamical processes must be easy to compute, it doesn't follow that we can methodically learn all of nature's secrets. Some processes might be easily executed but infeasible to learn from just observing them in action. We have no guarantee that our world is not like that. It might be that the world is simple, but learning that it is simple is not.

assign h a better epistemic status given background knowledge K and evidence e than given K alone. "Better epistemic status" usually stands for h being taken as more likely given $K\&e$ than given K alone. For reasons explained below, let's keep talking about "epistemic status" for now. Note that there is a sense in which e can confirm h even if l shouldn't take h to be more likely given e. For example, l may already be certain of h given l's background knowledge K. Hence my use of the word 'novel'. I shall not discuss this non-novel sense of confirmation, but I hint at how we can do that in the footnote.[14] I will not repeat this qualification every time.

The *should* is important. *Confirmation* is a normative concept. It's about the epistemic state one *ought* to be in, given one's evidence, rather than about one's actual, sometimes bad, learning processes. Equally important, and in some tension with the former claim, confirmation is not about which hypotheses are true *regardless* of background knowledge and further evidence.[15] One is not expected to simply know all truths. Rather, one is expected to respond to one's evidence and background knowledge. Learners whose total evidence is different can be rationally required to have different judgments regarding the likelihood of some hypotheses.

Thus, the concept of confirmation involves some idealisation, but it still takes into account the *limitations* of the learners, like their access to information. If one cannot know all truths, one is not required to be in the same epistemic state as someone who *does* know all truths, or even as someone who *can* know all truths. This is a form of an ought-implies-can principle,[16]

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[14] We want e to confirm h if, given l's knowledge K, e should raise the likelihood of h, unless l already has background knowledge that makes e redundant. Thus, we need to say that e confirms h if e provides novel confirmation for l relative to a part of K, K^*. It cannot be just any proper part. For instance, if l already knows that h, then l can learn the material conditional $e \to h$ from K. Then for any part K^* that includes $e \to h$, e would confirm h, and this applies to any possible evidence e. The way to deal with this problem, I think, is by modelling the learner's background knowledge as a directed graph, where inputs fed to the process are sources, computations performed on them are descendants, and computations on descendants are also descendants. An appropriate K^* would be such that if a node is taken out of K then so are all its descendants. Then, e can be said to confirm h if e provides novel confirmation for h relative to an appropriate K^*. A further complication is that the parts that were removed from K must not include defeaters for taking e as confirming h. Formalising this exactly is harder and I don't have a fully worked out account yet.

[15] I don't take a position on the E=K debate, i.e., on whether evidence just is knowledge. By separating between evidence and background knowledge I don't intend to suggest that the evidence is not part of the learner's knowledge, but rather that we focus on the difference a specific piece of evidence makes and take the rest of the learner's knowledge as part of the background conditions.

[16] I discuss whether 'ought' implies 'can' in epistemology in Chapter 7.

and it strikes me as almost trivial in the context of the analysis of *confirmation*. *Of course* we should take one's inherent limitations regarding access to information into account when we talk about confirmation. This is what confirmation is all about, isn't it?

Consider again the minimal condition above: a piece of evidence *e* provides (novel) confirmation for a hypothesis *h* for a learner *l* with background knowledge *K*, only if *l* should assign *h* a better epistemic status given background knowledge *K* and evidence *e* than given *K* alone. If 'ought' implies 'can', here is a natural way to rephrase this adequacy condition:

> *Adequacy Condition for Incremental Confirmation:* a piece of evidence *e* provides novel confirmation for a hypothesis *h* for a learner *l* with background information *K,* only if *l can learn* from *K&e* that *h* should be assigned an epistemic status that is better than the epistemic status that *l* can learn to assign to *h* from *K* alone.

Denying the latter requires *l* to make judgments about the epistemic status of *h* that cannot be successfully learnt by *l* from *l*'s evidence.

It strikes us as obvious to take *l*'s inherent limitations regarding access to information into account. Here is a suggestion: it's obvious because it's so clear to us that different things can be *learnt* from different data, so of course we should relativise confirmation to *l*'s evidence and background knowledge. *l*'s having different evidence and background knowledge (naturally) affects what *l* can learn from *l'*s evidence and background knowledge, so *l*'s evidence and background knowledge are essential to what is confirmed for *l*.

But if learning is computational, then the computational resources of *l* also limit what *l* can learn from what. Given the previous discussion, it should strike the reader as equally natural to take into account *l*'s *other* learning limitations besides *l*'s lack of access to all the information. At least, it should come naturally if the learner *l* in question is (or can be modelled as) a *realistic* learner. If 'ought' implies 'can', then a realistic learner cannot be rationally required to learn from the evidence something that is not realistically learnable. Thus, in the same way that confirmation is defined relative to a learner's access to information, it should be defined relative to the learner's computational resources. Since we are interested here in realistic learners, the kind of confirmation we're interested in is about what can be realistically learnt from the evidence.

Note that, taking computational limitations into account, it cannot be assumed that a realistic learner has a verdict regarding the likelihood of every hypothesis *h* given their background knowledge *K,* or given *K* and an additional piece of evidence *e*. For the learner to have a verdict, there must be a feasible learning process that leads from *K* or *K&e* to a verdict about the likelihood of *h*. As we shall see in the following chapters, this is not always the case (and in any case, there are too many possible hypotheses for a realistic learner to have a verdict about them all).[17]

Thus, for *e* to realistically confirm *h* for a (realistic) learner *l* with background knowledge *K*, *e* need not necessarily increase the likelihood *l* already ascribes to *h* given *K. e* can also make it feasible for *l* to form a novel verdict about a hypothesis *h* not considered by *l* before, such that now, after receiving *e*, *l* takes *h* to be likely. In such a case, I would still say that *l* should take *h* to have a better epistemic status given *K&e* than given *K* alone, because having a good epistemic status like being likely is epistemically better than not having any epistemic status at all. This is the main reason I used "better epistemic status" above instead of "more likely".

If we agree that having a likelihood above some threshold is considered to be more likely than not having any likelihood assigned at all, then we can change our condition simply to: a piece of evidence *e* provides confirmation for a hypothesis *h* for a learner *l*, only if *l* can learn

[17] This might suggest a distinction that may be important to epistemology and is usually overlooked. A piece of evidence *e* can be said to be *neutral* regarding hypothesis *h* if the learner can compute the implications that *e* bears on the likelihood of *h* and see that *e* keeps the likelihood of *h* intact. *e* can be said to be *irrelevant* to *h* if no feasible learning process can lead the learner from *e* to any verdict regarding the likelihood of *h*.

from *K&e* that h should be taken as more likely compared to what l can learn about the likelihood of h given K alone.

By requiring that h be taken as "more likely", we characterise a very minimal form of confirmation, which we have called *incremental confirmation*. According to this condition, for every $\varepsilon > 0$, no matter how small, if h's likelihood should increase by ε upon receiving *e*, then *e* confirms h. This is a very weak condition. It is compatible with one living one's life while constantly getting only evidence of the same kind of *e* and still ending one's life with as tiny an increment as we like in the likelihood of h. Usually, when we think evidence *e* confirms hypothesis h, we mean that there is a realistic way in which collecting evidence of the same type as *e* can convey non-negligible confirmation for h.

The cryptographic notion of negligible functions can be of use here. A function is negligible, recall, if at the limit it decays faster than the inverse of any polynomial. Note that the definition here applies only to the limit. If $\mu(n)$ is negligible, then for every polynomial $P(n)$, there *exists* an *n* after which $\mu(n) < \frac{1}{P(n)}$. However, this *n* can be too big to be of any practical significance. Furthermore, for every n_0, there *exists* some polynomial $P_0(n)$ for which $\mu(n_0) > \frac{1}{P_0(n)}$, so there is no *n* for which $\mu(n) < \frac{1}{P(n)}$ holds for *every* polynomial $P(n)$.[18] One might wonder, then, why the concept of negligible functions is significant in practice.

However, the definitions of negligible functions and negligible probabilities can be adjusted for realistic learners. Following the rationale that led to the definition of negligible functions, we can say that if a learner has a time to live of t_0, then events with a probability of less than $\frac{1}{t_0}$ are not likely to happen in the learner's life. If we take t_0 to be pretty big (though not unrealistically big), we get that events with probability $\left(\frac{1}{t_0}\right)^2$ are unlikely to happen, and a

[18] This is a pattern that we shall see throughout this thesis. We can say that while expressions of the form $\mu(n) \cdot P(n)$ converge to zero for every polynomial $P(n)$, they don't converge uniformly.

difference of $\left(\frac{1}{t_0}\right)^2$ in probability is unlikely to change the number of occurrences of an event throughout the learner's life.

We can say that if getting evidence of type e t_o times provides an increment in the likelihood of *h* of less than $\left(\frac{1}{t_0}\right)^2$, then *e* provides negligible confirmation for *h*. We can think of each step in the learner's learning process as an attempt to learn or at least to find a true or approximately true hypothesis. If each attempt adds only a negligible success probability, then the learning process has a negligible probability of significantly improving the epistemic state of the learner, which defeats the purpose of the process. This seems to motivate an adequacy condition for this stronger notion of confirmation, which we can call *non-negligible confirmation*:

> *Adequacy Condition for Non-negligible Confirmation*: a piece of evidence *e* provides non-negligible confirmation for a hypothesis *h,* relative to a learner *l* with background knowledge *K*, if *l* can learn from *K&e* that *h* should be taken as non-negligibly more likely compared to what *l* can learn about the likelihood of *h* given *K* alone.

Non-negligible confirmation is stronger than merely incremental confirmation, but still quite weak. In ordinary cases, it seems that we have something stronger in mind. We may want to say that *e* confirms *h* only if *e* can play a significant role in learning that *h* (in case *h* is true), or if *e* gives a *significant* increment to the likelihood of *h*, such that *h* is now a salient hypothesis to be considered. The 'significant' increment is understood here in absolute rather than relative terms. If the learner with background knowledge *K* takes *h* to have a likelihood of $\frac{1}{2^{1000}}$, and upon receiving *e* the learner should update the likelihood of *h* to $\frac{1}{2^{500}}$, the likelihood of *h* has increased by a factor of 2^{500}, which is gigantic. However, *h* would still be so improbable that it would not be a salient hypothesis that should be taken seriously. We wouldn't say that *e* *significantly* confirms *h* in the sense I have in mind. Let's call this stronger sense of confirmation *robust confirmation*. If we are interested only in cases of robust confirmation, we might want to restrict our attention to these.

The kind of confirmation we are looking for in science is usually of this latter, robust kind. Science usually aims at providing knowledge or high-confidence predictions, rather than at showing that some hypothesis is a bit more likely (in the absolute sense) than we initially thought. Science might reach this aim by aggregating many pieces of evidence that together provide this robust kind of confirmation, but then it is usually this large body of evidence that is said to confirm a hypothesis. Focusing on robust confirmation motivates the following adequacy condition:

> *Adequacy Condition for Robust Confirmation*: a piece of evidence *e* provides robust confirmation for a hypothesis *h,* relative to a learner *l* with background knowledge *K,* if *l* can learn from *K&e* that *h* should be taken as significantly more likely (in the absolute sense) compared to what *l* can learn about the likelihood of *h* given *K* alone.

Note that robust confirmation implies non-negligible confirmation, which in turn implies incremental confirmation, but not vice versa, under the assumption that 'significantly more likely' implies at least 'non-negligibly' more likely. Thus, an assumption that a piece of evidence *e confirms* a hypothesis *h* can have weaker or stronger implications, depending on the kind of confirmation we have in mind. Later (in Chapter 6), when I present my solution to the Paradox of the Ravens, we shall see that coupling the assumptions of the paradox, together with the adequacy conditions for incremental, non-negligible, and significant confirmation, leads to consequences with increasing implausibility.

Acknowledging the connection between confirmation and what is learnable from evidence suggests how to deal with the Grueland example above. If the ciphertext coupled with the public key confirms whatever it is that the plaintext confirms, then for every feasible learning process that takes you from the plaintext to a more positive verdict about the likelihood of a hypothesis, there must be a feasible learning process that takes you from the ciphertext and the public key to the same verdict. But if one-way functions exist, this is false, at least for realistic learners.

The model of a learner we have in mind determines, of course, what counts as a feasible learning process. A godlike learner like Laplace's demon has unlimited computational resources. For such entities, whatever logically follows from the evidence (at least in the syntactical sense) is learnable from the evidence and thus confirmed by it. Thus, the ciphertext and the public key *do* confirm whatever the plaintext confirms for an entity like Laplace's demon. The solution I shall suggest to the Paradox of the Ravens won't work for Laplace's demon, but this is a feature, rather than a bug. Laplace's demon supposedly knows that all ravens are black, based only on seeing many non-ravens in some early moment of the universe. The paradox of the ravens is inherently connected to our limitations as learners. It shouldn't appear as a paradox when applied to gods and demons.

Chapter 4: Algorithmic Complexity and Epistemic Complexity

1. Introduction

This chapter introduces the concept of *algorithmic complexity* and discusses its relation to inductive learning. Unlike the *computational* complexity discussed earlier, algorithmic complexity is (at least on the face of it) not a property of computational processes, but rather a property of data, that is, structures that encode information, like binary strings (which can function as inputs or outputs of computational processes).

It was Solomonoff (1960) who first introduced the concept of algorithmic complexity, which was later independently (see Li & Vitanyi, 2019, §1.13) introduced by Kolmogorov (1965) and Chaitin (1966). It became very common to use the equivalent[1] term *Kolmogorov complexity*, but I will use the more indicative and less historically misleading term "algorithmic complexity". Interestingly, Solomonoff was interested in a general theory of inductive inference, and he thought that the concept of algorithmic complexity is what's needed to develop such a theory (see also his later 1996 paper). Algorithmic complexity turned out to be an interesting concept in its own right, but among philosophers, it is not widely considered to be a relevant concept for solving problems related to inductive learning. Two standard worries (see Neth, 2023) are that the concept of algorithmic complexity, as it is currently defined, is a relative, rather than an objective property, and that algorithmic complexity is uncomputable.

In this chapter, I shall discuss how data externalism may allow us to define a physically objective notion of algorithmic complexity and thus evade the first problem. This kind of reply is different from the one that is commonly given, which is based on the so-called Invariance theorem. I shall discuss the limits of the latter and explain where my reply (hopefully) does

[1] Sometimes, as in Li & Vitanyi (2019), the two terms are used to denote slightly different variants of the concept.

better. We shall see that data externalism can be used to extend the notion of algorithmic complexity to the physical phenomena themselves, and not only to structures that encode information about physical phenomena, like binary strings. I further develop a variant of algorithmic complexity, which I call *epistemic complexity*, and argue that it is more suitable for epistemology. Regarding the uncomputability of algorithmic complexity, I explain why computability, in itself, doesn't necessarily suffice for knowing the algorithmic complexity of a given string. However, the concept of algorithmic complexity, and especially epistemic complexity, can be useful for philosophy, nonetheless.

In the next section, I introduce the concept of algorithmic complexity and related notions like *algorithmic randomness* and *compressibility*. §3 introduces the Invariance theorem and the limits of its success in making algorithmic complexity objective. In §4, I explain how data externalism might succeed where the Invariance theorem failed. §5 introduces the notion of *resource-bounded algorithmic complexity*. In §6, I introduce the notion of *epistemically bounded algorithmic complexity*. In §7, I discuss the connection between predictability and compressibility.

Sections 2, 3, and 5 present mainly textbook material (with some emphasis on matters that are more relevant to the philosophical issues discussed in this thesis, especially in §3). Sections 4, 6, and 7 present novel contributions, as far as I know. When presenting textbook materials, the main source for this chapter is Li & Vitanyi's (2019, henceforth just Li & Vitanyi) comprehensive book on the topic of algorithmic complexity.

2. Algorithmic Complexity

Intuitively, data can be more or less complex. For convenience, let's think of data as coming in the form of binary strings.[2] Some strings are intuitively much simpler than others. Take a string of one billion bits, all of which are 1's: 1111111…1. This strikes us as a very simple string. Take another string of one billion bits: 1001110000111110… This string is slightly more complex, but only slightly. This is the string that starts with one '1', continues with two '0's, then three '1's, then four '0's, and so on until the billionth bit. In contrast, suppose we take a coin and toss it one billion times, writing 1 for heads and 0 for tails. Most likely, we will get a string that is very complex to describe. Our best way to describe it is likely to specify the digits of the string bit by bit. There will be no pattern or regularity that we can exploit to provide a shorter description of the string.

The algorithmic complexity of a string is a concept that tries to capture this kind of complexity. Informally, it is defined as the length of the shortest algorithm that generates the string. The 1111111…1 string is very simple because there are very short algorithms that generate it. For instance, "print '1' one billion times" is a very short algorithm (in English), containing only 27 characters, that generates it. In contrast, in the coin tossing case, the shortest algorithm that generates the string likely needs to specify the value of each of the one billion bits, so its length is at least one billion characters. Thus, the latter string is much more algorithmically complex than the former.

The phrase "length of an algorithm" is somewhat loose, as there are many ways to describe what we would refer to as essentially the same algorithm, and the length of each description can vary across (natural or programming) languages. More formally, algorithmic complexity is defined relative to a universal Turing machine. Let U be a universal Turing machine and let

[2] Nothing fundamental should change if our alphabet, or the number of "words", is larger than 2. I am working, though, under the assumption that it is finite.

x be a binary string. The algorithmic complexity of x relative to U, denoted by $C_U(x)$, is defined as the length of the shortest program p such that the output of U (i.e., what's on the tape of U when U halts), when given p as input, is x. Formally, $C_U(x) = \min\{|p|: U(p) = x\}$, where $|p|$ denotes the length of p in bits.

The program p itself is a binary string, so we can think of some binary strings as algorithms for producing other strings by a machine U. The machine U, relative to which we measure the algorithmic complexity of x, is called *the reference machine*. This dependence on a reference machine will be discussed in the next two sections.

Strings with complexity greater than or equal to their own length, like the coin tossing string above, are called *incompressible* or *algorithmically random*. Algorithmically simple strings are highly *compressible*, in the sense that if you want to store the string in memory, you don't have to store the entire string. You can just store a short program that generates it and save memory space. The cost is that to restore the full string, you will need to execute the program (that is, compute its output). Figuratively, we can say that you buy memory space with computation.

Mathematically, there must be at least one completely incompressible string. The reason comes from a counting argument (see Li & Vitanyi, §2.2). If we treat the empty string as a string with zero length, we see that there is exactly one string of length zero, two strings of length one, four strings of length two, and in general, 2^n strings of length n. Thus, for every number n, there are 2^n strings of length n, and $\sum_{k=0}^{n-1} 2^k$ strings of any length less than n (including the empty string). The sum of this geometric sequence is $2^n - 1$. Therefore, there are only $2^n - 1$ strings with length shorter than n, and 2^n strings of length n, so at least one string of length n cannot be the output of any shorter string, i.e., for every n, there is at least one string of length n that is completely incompressible.

Similar reasoning shows that at least half of the strings of length n are compressible by no more than a single bit if at all (because among the $2^n - 1$ strings of length shorter than n, 2^{n-1}

are of length $n-1$), at least half of the remaining half (i.e., a quarter) are compressible by no more than two bits (if at all), and in general, at least $\frac{1}{2^c}$ of the strings are compressible by no more than *c* bits (strings that are compressible by at most *c* bits are called *c-incompressible*). That is, the vast majority of strings cannot be compressed by more than several bits (if at all). Strings that are compressible by a negligible number of bits can be called *nearly random* or *almost random*. We see that the vast majority of strings are random or nearly random. Hereafter, I'll usually use the term "random" or "incompressible" to refer also to strings that are nearly random or nearly incompressible.[3]

According to our definition, if a string *x* is one billion bits long, that is, $|x|$=1,000,000,000, and the shortest program that generates *x* in *U* is *p*, when $|p|$=1,000, then $C_U(x) = 1{,}000$. We might want to compare strings of different lengths. Suppose a string *y* is of length one million bits, and a string *z* is of length one thousand bits, and the algorithmic complexity of both is $C_U(y) = C_U(z) = 1{,}000$. According to the definition above, *x, y,* and *z* have the same algorithmic complexity. However, we see that *x* is compressible to one millionth of its length, *y* is compressible to one thousandth of its length, and *z* is incompressible, that is, algorithmically random.

Clearly, there is a sense in which *x* is simpler than *y* and *y* is simpler than the (in this sense) maximally complex *z*. We can define the *normalised algorithmic complexity* of a string *w* relative to *U* as $\frac{C_U(w)}{|w|}$, that is, the algorithmic complexity of *w* according to *U* divided by *w*'s length. We will get that the normalised algorithmic complexity of the strings *x*, *y*, and *z* above is 0.000001, 0.001, and 1, respectively. In general, random strings will have normalised complexity close to one, and very simple strings will have normalised complexity close to zero.

[3] Some might see an analogy here with the concept of entropy. However, it is important to note that the two concepts, *entropy* and *algorithmic complexity*, are distinct. If a state of high entropy evolved from a state of low entropy, then supposedly, the low entropy state is relatively easy to describe, and the high entropy state can (in principle) be computed from the description of the low entropy state together with a description of the laws of nature. Thus, high entropy states can have low algorithmic complexity.

Another useful concept is *conditional algorithmic complexity* (see Li & Vitanyi, §2.1). Sometimes, a very complex string can be simple to generate from already existing knowledge. To see an extreme example, suppose our reference machine U already has a random string x in U's memory. A very short program instructing the machine U to output whatever is in its memory will generate x. The algorithmic complexity of x conditional on y, denoted by $C_U(x|y)$, is defined as the length of the shortest program p that makes U output x when U has access to the string y. "Access to y" is defined technically as the machine getting both p and y as inputs, usually as a concatenated string that includes both p and y, maybe with some separator between them. Formally, $C_U(x|y) = \min\{|p|: U(p,y) = x\}$, where (p,y) is a suitably defined input containing both p and y. We can treat non-conditional algorithmic complexity as a special case of conditional algorithmic complexity, defining the non-conditional algorithmic complexity of x as the complexity of x conditional on the empty string.

The complexity of x conditional on y can be very different from the non-conditional complexity of x. As we saw, for any string x, no matter how complex, there is a short program that generates x when it has access to x itself. That is, for every x, $C_U(x|x) \leq constant$, and the constant depends only on the complexity of the program that makes machine U print the string it was given access to. However, you cannot reduce complexity for free by using conditional complexity. For every x and y, it is always the case that $C_U(x) \leq C_U(x|y) + C_U(y)$ (see Li & Vitanyi, Corollary 2.8.1, p. 193).[4]

Thus, the discount you can get by generating x from y instead of from scratch cannot be greater than the cost of generating y from scratch. There are no shortcuts. The reason for this is straightforward: generating y from scratch and then using y to generate x is a way to generate x from scratch, so it cannot be simpler than the simplest way to generate x from scratch. In the

[4] I neglect some technical complications involved in how the machine knows that one string has ended and another string starts. For that, a machine either needs to use an unambiguous code (known as *prefix code*, see Li & Vitanyi Ch. 3), or needs to be supplied with information about the length of the strings, which for a string of length n requires no more than about $\log(n)$ bits. This is not crucial for what follows.

case of $C_U(x|x)$ we see that immediately, as having access to x means that x must have been generated somehow before U could use it for computation.

The algorithmic complexity of a string is, in general, uncomputable. The reason ultimately goes back to the halting problem. Some programs run forever, and there is no general way for us to compute whether a given program halts at some point. Thus, we can't in general be sure that a short program that is still running won't end up generating the string in question.

Another way to see why algorithmic complexity is uncomputable is that if algorithmic complexity *were* computable, we would get a case of the Richard-Berry paradox. Intuitively, "the lexicographically first one-billion-bit string with algorithmic complexity more than one million" is a very short description, and we know such strings exist, so there must be one that is lexicographically the first. If algorithmic complexity were computable, we should have been able to use this short description to generate that string. But this is a contradiction, because such a string would have complexity far less than one million. More accurately, for every universal machine U, we can find a constant c_U that depends only on U, such that if algorithmic complexity were computable, we would arrive at a contradiction with a program that finds the first string x such that $C_U(x) \geq c_U$ (see Li & Vitanyi, §2.7).[5] I shall return to this difficulty in §6 and §7.

3. The Invariance Theorem and Its Limits

A natural question is how algorithmic complexity is affected by the choice of a reference machine. Are some strings more algorithmically complex or simple relative to different

[5] A similar way to put it is that for any consistent formal system F that is rich enough to express algorithmic complexity, sentences of the form "x has algorithmic complexity greater than n" are not provable in F for a large enough n. This result is known as *Chaitin's Incompleteness Theorem*.

machines? The short answer is 'yes'; the longer answer is 'yes, but'; and the even longer answer is 'but still, yes'. This section elaborates on that.

At the most basic level, for any string x, the exact length of the shortest algorithm that generates x can change from one universal machine to another. Thus, the algorithmic complexity of strings depends on which machine is chosen as the reference machine. The troubles don't end here. For every string x, there is a universal machine (in fact, infinitely many of them) that is prebuilt to output x in response to getting the empty string as an input (see Li & Vitanyi, p. 111). Thus, for every string, there is a universal machine relative to which the complexity of the string is zero, i.e., the string is maximally simple. Similarly, strings that are very simple according to one machine can be very complex according to another.[6] This is why the short answer to the question at the beginning of this section is 'yes'. This might be a problem because if our choice of reference machine is subjective, conventional, or arbitrary, then the algorithmic complexity of a string is a subjective, conventional, or arbitrary matter.

However, there is a sense in which the judgment of two universal machines regarding any string cannot differ *too much*. This is the content of *the Invariance theorem*, which is the fundamental theorem of algorithmic complexity theory (see Li & Vitanyi, §2.1). The Invariance theorem tells us that for any pair of universal machines, U and U', there is a constant $C(U, U')$, which is a function of U and U' alone, such that for *every* string x, $|C_U(x) - C_{U'}(x)| \leq C(U, U')$. One way to put it is that every two machines agree about algorithmic complexity *up to a constant*.

The idea behind the theorem is that each universal Turing machine can simulate all others. Thus, for any given pair of machines, U and U', let $P_{U,U'}$ be the shortest program that simulates U' in U (or, figuratively speaking, takes you from U to U') and let $P_{U',U}$ be the shortest program

[6] Think of a machine that computes a bijection that matches each of the $2^n - 1$ strings of length less than n to each of the $2^n - 1$ strings of length n except for x and matches every other string to itself. The shortest input that generates x for this machine is x, which is of length n, so x is not compressible for this machine.

that simulates U in U' (or takes you from U' to U).[7] To generate x in U, we can always run $P_{U,U'}$ on U and then run the shortest program that generates x in U'. Thus, for every string x, $C_U(x) \leq |P_{U,U'}| + C_{U'}(x)$, where $|P_{U,U'}|$ denotes the length of $P_{U,U'}$. For analogous reasons, we get that for every string x, $C_{U'}(x) \leq |P_{U',U}| + C_U(x)$. Therefore, $|C_U(x) - C_{U'}(x)| \leq \max\{|P_{U,U'}|, |P_{U',U}|\}$ and $\max\{|P_{U,U'}|, |P_{U',U}|\}$ is a constant that depends only on U and U' and not on any string x. We can call $\max\{|P_{U,U'}|, |P_{U',U}|\}$ *the translation constant* between U and U'.

One thing we learn from the Invariance theorem is that, again, you can't reduce algorithmic complexity for free. Suppose you fix U as your reference machine, and suppose a string x is very complex relative to U. Suppose there is another machine U' relative to which x is very simple. We can define the shortest program that simulates U' on U, which we denoted earlier as $P_{U,U'}$, as the algorithmic complexity of U' according to U, that is, $C_U(U')$. $P_{U,U'}$, after all, can be seen as a description of U' in the language of U. Then we can rewrite the inequality $C_U(x) \leq |P_{U,U'}| + C_{U'}(x)$ as $C_U(x) \leq C_U(U') + C_{U'}(x)$.

In this form, we can immediately see that for any decrease in the algorithmic complexity of x you get by assessing it from U', you need to pay with the algorithmic complexity of the description of U' itself.[8] If x is very complex (according to our reference machine), but there is another machine according to which it is very simple, it implies that this other machine must be at least as complex as the reduction you achieve (cf. Li & Vitanyi, p. 112).[9] Thus, while the algorithmic complexity of a string changes from one machine to another, it can seem as though

[7] Formally, $P_{U,U'}$ is said to simulate U' on U if for every string x, we have $U((P_{U,U'}), x) = U'(x)$ where $((P_{U,U'}), x)$ is a concatenation of $P_{U,U'}$ and x with some separator between them. That is, you always get the same output whether you give x as an input to U' or give $P_{U,U'}$ and then x as inputs to U.

[8] For instance, if it's hardcoded on U' to generate exactly the bits of a random string x when receiving the empty string as an input, then describing the hardcoding must use the same number of bits as the length of x.

[9] The situation is somewhat analogous to thermodynamics. In thermodynamics, you can reduce the entropy of a system, but only at the cost of a greater or equal increase in entropy outside the system. See also fn. 23.

there is something objective in the combined complexity of a string and the machine that is used for measuring it.

Another thing we can learn from the Invariance theorem is that any two machines approximately agree about the complexity of strings that are *long enough*. Any two machines *U* and *U'* disagree about the algorithmic complexity of a string *x* only up to their translation constant, $C(U, U')$. The difference in the *normalised* complexity of a string *x* is $\left|\frac{C_U(x)}{|x|} - \frac{C_{U'}(x)}{|x|}\right| = \frac{1}{|x|}|C_U(x) - C_{U'}(x)| \leq \frac{1}{|x|}C(U_1, U_2)$. Thus, we get that $\lim_{|x|\to\infty} \frac{1}{|x|}C(U_1, U_2) = 0$, or in other words, for every $\varepsilon > 0$, there is some number *n*, such that for every string *x* that is longer than *n* bits, *U* and *U'* agree about *x*'s normalised complexity up to ε. Thus, while *U* and *U'* might not agree on an exact number, if a long enough string is very simple (complex) according to one machine, then it should be very simple (complex) according to the other.[10] In the case of infinite strings, *U* and *U'* agree completely. This is why the longer answer to the question whether algorithmic complexity depends on a choice of a reference machine is 'yes, but'.

It's time to explain why the even longer answer is 'but still, yes'. In essence, the Invariance theorem just doesn't cancel the fact that when we focus on given strings rather than comparisons between pairs of machines, every string can appear maximally simple or maximally complex. Some writers describe the Invariance theorem as saying that algorithmic complexity is *invariant up to a constant*. This is somewhat misleading, though, because as we saw, the translation "constant" in question is actually a *function* of the pair of machines that are compared. In general, different pairs of machines have different translation constants.

This is a crucial point. It's true that *for every two machines*, you can show that at the limit, when we consider longer and longer strings, they always approximately agree about the

[10] If there is an exact number that separates the "simple" strings from the "very simple" strings, we might always find strings that will fall on different sides of the border, but in this case the two machines will still agree about the strings being within ε distance to the borderline.

complexity of those strings. However, it is equally true that *for every finite string x*, we can find two machines (in fact, infinitely many) that would radically *disagree* about x's complexity (cf. Sterkenburg, 2018 §3.2.5). In other words, if we fix a pair of machines, we get agreement at the limit, but if we fix a string, we get as radical disagreements as we want among some possible pairs of machines.

In mathematical terms, we can say that while all pairs of machines converge in their normalised algorithmic complexity judgments at the limit $|x| \to \infty$, they don't converge *uniformly*. That is, while for every pair of machines there exists a length n from which they agree up to ε, there is no *single* value of n that applies to all pairs of machines. Rather, there are different values of n, which are a function of the pair of machines, and these possible values of n are unbounded.

Now, here is the question: is a given (finite) string x simple or complex? The question is about x, not about any machine. Say our reference machine U tells us x is very complex. Another machine, U', tells us that x is very simple. As I said above, this implies that according to U, U' must be very complex to describe, 'gruesome', one might be tempted to say. However, U' is complex, or gruesome, only from the perspective of U. U' doesn't need to take itself to be complex and in any case does not include the description of itself according to itself as part of the complexity of x. Thus, from the perspective of U', x is very simple, period. The Invariance theorem doesn't change this fact.

If the choice between taking U or taking U' as our reference machine is subjective, arbitrary, or conventional, nothing in the content of the Invariance theorem makes the complexity of a given string x objective, non-arbitrary, or non-conventional. Every finite string is simple from some perspectives and complex from others, and only an objective way to choose between perspectives can make the algorithmic complexity of strings an objective matter.

My use of the word 'gruesome' is not a coincidence.[11] There is a clear analogy between the situation here and the New Riddle. While for English speakers the predicate 'green' is simpler than the predicate 'grue', and the Green Hypothesis is simpler than the Grue Hypothesis, for a Grue-Bleen speaker, 'grue' is simpler than 'green' and the Grue Hypothesis is simpler than the Green Hypothesis. Moreover, for every moment in time *t*, let $grue_t$ be the variant of *grue* that coincides with *green* for objects examined before *t* and disagrees otherwise. For every $grue_t$-$bleen_t$ speaker, there is a moment in time (namely, *t*), such that after that time, English speakers and $grue_t$-$bleen_t$ speakers can start putting their competing hypotheses to the test and collect enough data to converge on their verdicts. However, for every moment in time *t*, there is a later moment *t'* (in fact, infinitely many of them), such that $grue_{t'}$-$bleen_{t'}$ speakers radically disagree in their current predictions with English speakers.

The New Riddle asks for an objective basis to discriminate between the Green Hypothesis and the Grue Hypothesis, or between English speakers and Grue-Bleen English (or Griblish) speakers, based on our evidence. Similarly, we need an objective basis to discriminate between "natural" and "gruesome" machines (or programming languages). In the next section, we shall see that essentially the same solution I offered to the New Riddle can be of service here.

4. Data Externalism and the Objectivity of Algorithmic Complexity

In Chapter 1, I argued that the thesis of data externalism suffices to break the symmetry between normal English speakers and Griblish speakers within each possible world. Data externalism, recall, is the thesis that for data to be about physical phenomena requires there being a causal chain between the state of the data and the state of the phenomena the data is

[11] And it is not a first, either. See for instance Neth (2023)

about. Recall that not every causal chain between the state of a thing and the data suffices for the data to be about that thing.[12] I assume only that the existence of such a causal chain is necessary. In this section, I argue that data externalism might provide the resources to deal with the analogous problem presented in the previous section.

Assuming data externalism, we shall see that the physics of a world restricts which computations can be performed at that world. When we focus on computations that are physically possible at a world, it is no longer clear that for every physically possible machine, there is another machine that is physically possible at that world, which radically disagrees with the former regarding the algorithmic complexity of binary strings. Additionally, we shall see that some machines that might appear to judge a specific string to be very simple should not be seen as computing the algorithmic complexity of a string, but rather its complexity conditional on other strings. Moreover, even if we can find two physically possible machines that radically disagree about the algorithmic complexity of a given string, I shall argue that it's not at all clear that there is no objective way to prefer the judgment of one over the other. This last claim is more speculative, and the argument is admittedly somewhat hand-wavy, but I believe the claim and the argument in its favour are correct in spirit if not in letter.

Hopefully, the discussion will let us specify an objective sense of algorithmic complexity. The kind of objectivity involved is physical, not mathematical. It does not encompass all possible worlds, but rather all the worlds within a class of worlds that share the same physics. As a bonus, we will get a natural way to extend the definition of algorithmic complexity to apply to physical objects, facts, and properties, or, in short, to physical phenomena, and not only to strings or data structures that are supposed to encode those physical phenomena.

The key move is that any physical computation involves encoding information. For convenience, let's think only about computational processes whose inputs and outputs come in

[12] Further specifications may be needed for a causal chain to be "of the right kind".

the form of binary strings. For a process to be considered computational, it must include a component that functions as the input string, and a component (call it 'the head') that can "read" the bits of the input string and, based on what it reads, "write" the bits of an output string.[13] What does it mean for the head to "read" the bits of an input string and "write" the bits of an output string? I contend that it must at least mean that the head encodes information about the bits of the input string, and that the bits of the output string encode information about the state of the head at the time of writing. For the writing to be based on what the head reads, the state of the head while writing the output bits must depend on the information about the input bits encoded by the head.

If a computational process is physical, the components of the process that function as the input and output strings must themselves be physical. For them to realise binary strings, they must be composed of a sequence (either in space or time) of physical bits, that is, a sequence of physical objects, each of which can be in one of two possible states. If, during the computational process, the head encodes information about the bits of the input string, then, according to data externalism, there must be a causal chain from the states of the bits of the input string that are being encoded to the state of the head while it encodes them. If the writing of the output bits is based on the reading of the input bits, there must be a causal chain from the state of the head while it's encoding the input bits to the state of the head while it's writing the output bits. If the bits of the output string encode the state of the head when the head wrote them, there must be a causal chain from the state of the head when it was writing those bits to the states of those bits after the head finished writing them.

The physics of each world restricts which kinds of causal connections (if any) are possible at that world. Thus, the physics of each world determines what can function as strings and as

[13] It doesn't matter for our current purposes whether the input string is a read-only or the head writes the output string on top of the input string.

heads, and whether and how such heads can compute an output x from an input p. There is no guarantee that every (mathematical) machine that runs computations and outputs x on an input p is physically possible, so for starters, it is not clear anymore that for every physically possible machine there exists another physically possible machine that radically disagrees with it regarding the algorithmic complexity of a given string. The mathematical possibility of such a machine does not suffice to establish its physical possibility.

A clarification is needed here. Of course, no physical computing machine is a universal Turing machine, as no physical computing machine has anything that can function as an indefinitely long tape. The resources provided by the physical universe (at least in the actual world, as we believe it to be) are limited. Algorithmic complexity was defined relative to a universal Turing machine, so technically, it is not defined relative to physical computing machines. However, some physical computing machines (like the laptop I'm using to write this chapter) are *universal in principle*, in the sense that they can compute anything a universal Turing machine can compute, provided that they are supplied with a sufficiently long tape (and the needed supply of spare parts for worn-out components). We can decide that the algorithmic complexity of a string relative to these physical machines that are universal in principle is identical to what a machine with the same configuration, but unlimited tape, would say. Alternatively, we can narrow the discussion to resource-bounded algorithmic complexity, which will be presented in the next section.

We can say that it is not obvious that, for every physical computing machine that is universal in principle, there exists another physical computing machine that is universal in principle, which disagrees with it radically. It might even happen to be the case that in our world, the translation constants between all pairs of physically possible machines are not too big, so the convergence of the judgments of different machines *is* uniform. We might never encounter radical disagreements between physically possible machines about reasonably long strings.

An obvious objection might come to mind: surely, if it's physically possible to output a string x, then it is physically possible to build a machine with the string x hardcoded to its memory, such that the machine outputs x for the empty string. Thus, for every possible physical output, there can be a machine that assigns complexity zero to it.

Data externalism can be of help here, too. Let's ask: what is the difference, according to data externalism, between conditional and unconditional algorithmic complexity? Recall that the complexity of x conditional on y, $C_U(x|y)$, is defined as the length of the shortest program p that lets the reference machine U output x for input p, when U *has access* to the string y. What do we mean by U having access to the string y? We mean that U can read the bits of y during the computation process. According to data externalism, what this means is that there is a causal connection between a physical copy of the string y and the state of the head during the computational process.

A machine that has a copy of x hardcoded to its memory, such that it is programmed to output x for the empty string, is a machine whose head causally interacts with a physical copy of x to produce x as an output. From the perspective of data externalism, what this machine does, in effect, is to output x conditional on x, rather than outputting x unconditionally. It might show, unsurprisingly, that the complexity of x conditional on x is low, but not that the unconditional complexity of x is low.

Mathematically, we can distinguish between a machine that is hardcoded to output x for the empty string and a machine that outputs x for the empty string *conditional on* x, but it is unclear whether there is a way to make sense of this distinction when we think about computational processes as physical. The point here is that data externalism blocks easy "cheats" for reducing complexity by using gruesome machines as reference. We should treat the causal interactions a machine has with a physical copy of a string *y*, even if this physical copy is part of the functioning of the machine itself, as something that comes with a cost. Either what is judged is the complexity conditional on *y*, or the bits of *y* need to count as part of the input.

This point generalises. We might worry that, still, there can be a very complex physical machine (from our point of view) that outputs some random string *x* for a very short input, without access to any physical copy of another string. Let's call this machine *G* (for 'gruesome'). It is far from trivial that *G* is physically possible, but even if it is, we should still ask how such a machine can come into existence.

The physics of our world might allow *G* to be assembled out of nowhere by a random fluke. While this might be physically possible, it is definitely a very remote physical possibility, and it's not clear that this kind of remote physical possibility is very relevant to epistemology. Compare: suppose a person comes into existence by a random fluke with a specific decryption key in their memory. Suppose even that there is a person who, by sheer luck, whenever they see a ciphertext, their brain randomly reassembles, now with the needed decryption key in their memory. While such a person would be able to learn from ciphertexts what no other person could learn, the implications of that for what humans can learn or ought to believe are minimal. As I said in Chapter 3, epistemology usually focuses more on what can be learnt methodically rather than on what can be learnt by flukes. In this regard, it might be more appropriate to talk not about physically possible machines, but rather about machines that are possible in nearby physical worlds. This restricts the set of machines we need to take into account even further.

If we do that, a more plausible option for machine *G* to come into existence is for agents with access to other universal machines to build *G* based on computations they performed using these other machines, which in turn required various inputs. If the only plausible way for *G* to come into existence involves computations performed by other machines, there is a sense in which *G*'s computations already include the cost of the previous computations that were required to build G. There's something somewhat arbitrary in not counting the necessary[14] computational cost of building G as part of the computational cost of getting a certain output from G.

This highlights another point, and this is where the argument becomes more speculative and hand-wavy. Given the physics of a world, some computing mechanisms can be more likely than others to come into existence without any prior computation (e.g., by a process of natural selection). For instance, suppose that the physics of a world is such that mechanisms that function like (say) AND, OR, and NOT gates are somewhat likely to emerge without any previous computation. Let's call these basic computing mechanisms that are likely to emerge without any prior computation *computational atoms*. Given the physics of a world, other, more complex computing mechanisms may be more likely than others to be assembled out of those computational atoms. Then other, even more complex computing mechanisms can be assembled from previous ones. From some stage, the computations performed by some computing mechanism might contribute to the assembling of the next stage.

If this is the case, then the physics of that world seems to induce a natural ordering of computation mechanisms by some suitable function[15] of their likelihood of coming into being and the computational cost associated with bringing them into being using other, already existing computing mechanisms. From some point in the ordering, the order might include computing mechanisms that are universal Turing machines (in principle). We can define algorithmic complexity relative to these physical universal machines (or sometimes maybe even relative to non-universal Turing machines that can still generate the required output).

[14] Where the scope of the necessity is the nearby physically possible worlds.

[15] I don't have a worked-out account of what such a function should be. My hunch is that the function should take all machines whose likelihood of coming into being spontaneously is above some threshold to be on a par, ignore all others, and assign to strings the minimal algorithmic complexity it receives according to any of those machines. The details go beyond the scope of the current discussion.

If two physical machines radically disagree about the complexity of a string, it seems natural to prefer the verdict of the machines that appear earliest in the ordering. Since the ordering is an objective physical feature of the world, the decision between the verdicts of two machines can be (physically) objective, rather than subjective, conventional, or arbitrary. Algorithmic complexity remains dependent on the choice of a reference machine, but the choice of a reference machine can be objective. The Invariance theorem is an important tool in algorithmic information theory, but we don't need it to make algorithmic complexity a physically objective matter.

If the computing mechanisms in our world are not assembled from radically different computational atoms, the translation constant between universal machines that come early in the ordering is likely not too big. Since we ourselves are products of the physical world, the computations performed by our bodies, or by the computation machines we have built, are probably not too late in the ordering. Thus, taking our current computers as reference machines need not be subjective or arbitrary. There are reasons, not decisive but still significant, absent counterarguments, to think that we will not get a very distorted picture by doing that.

As a bonus, data externalism, by restricting how physical objects can be encoded, lets us extend the notion of algorithmic complexity so that it applies not only to strings but to physical phenomena in general. Without assuming data externalism, it's tempting to think that all encoding schemes are on a par. That is, we can encode anything we want with any string we want by stipulation. For instance, it seems that we can just say "let the string '1' of length one bit encode the exact current microscopic state *m* of the 10^{24} gas molecules in this box". If all encoding schemes are on a par, then the algorithmic complexity of a string that encodes a physical phenomenon says nothing about the phenomenon, because it could have been encoded by any other string.

However, data externalism severely restricts which strings can encode information about which phenomena and how. For a string to encode information about a phenomenon, there must be a causal connection between the phenomenon and the bits of the string. As far as we know, there is no causal mechanism that directly changes the state of an object that functions as a physical bit only if the molecules in a box are exactly in a specific microscopic state m. If our world is approximately like our current physical theories tell us it is, the only (theoretical) way to encode full information about the molecules in the box involves distinct causal interactions with roughly each and every molecule, resulting in a string with at least 10^{24} bits.[16] Once we have this 10^{24}-bit string, we can encode this string using a single bit, by (say), creating a machine that checks whether its input is exactly this 10^{24}-bit string that directly encodes m, and outputs '1' only if that's the case. But then the one-bit string '1' encodes *m* only parasitically on the 10^{24}-bit string encoding *m*, and the string '1' encoding something about the 10^{24}-bit string.

Furthermore, for the one-bit string '1' to encode *m*, rather than encoding the fact that the machine got a specific input that *happens to coincide* with a string that encodes *m*, there must be a causal connection between the state *m* and the machine being configured as it is, and this causal connection must be mediated by a specific 10^{24}-bit string. That is, the string '1' can encode *m* only in virtue of being a part of a much larger mechanism that includes encoding the state *m* using the 10^{24}-bit string. A full description of how *m* is encoded must include the 10^{24}-bit string. There is no way to encode *m* without it. We can say that a string *directly encodes* a phenomenon if the causal process in virtue of which the string encodes the phenomenon does not involve the encoding of another string.

[16] The molecules can be in more than two states, so the string needs to have more than 10^{24} bits, but it's not important for what follows.

If data externalism restricts the ways physical phenomena can be encoded, then only some encoding schemes and not others are physically permissible. Furthermore, if the physics of a world is approximately like we believe our world's to be, then there are only finitely many ways to directly encode a phenomenon. If that is so, we can identify the algorithmic complexity of a phenomenon with the algorithmic complexity of the simplest string that directly encodes it. Since, given data externalism, the set of strings that can directly encode a phenomenon is an objective physical property that is associated with this phenomenon, so is its algorithmic complexity. Thus, algorithmic complexity can be not only a mathematical property of abstract objects like strings and Turing machines, but also a physical property of everything that exists in the physical world, or at least of everything about which information can be encoded.

5. Resource-Bounded Algorithmic Complexity[17]

The algorithmic complexity of a string x is defined as the length of the shortest program p that generates the string x as an output. In this definition, the resources the program p requires are unbounded. We don't care how much computational time and space the program p requires for generating x. Even if the shortest program that generates x runs in (say) exponential time, we would still take its length as the algorithmic complexity of x.

However, bounded learners like us are usually less interested in algorithms that are too demanding in computational resources. If all the short programs that generate x require computational time exponential in their length to run, and those programs are still not *very* short, then we will not be able to use them to generate x, because we'll die long before the end of the process. In practice, such an x is as incompressible as a random string.

[17] This section is based on Chapter 7 of Li & Vitanyi.

Here is an example: let F be a formula in propositional calculus with n propositional variables (think of n as reasonably big, say, 100 or 1000). Suppose that the length of F, in number of symbols or in binary representation, is not *much* greater than n (n^2 or n^{10} are still alright). Every formula with n variables is associated with a specific truth table with 2^n rows. Every such truth table can be represented by a binary string of length 2^n with the i_{th} bit being '1' if and only if the i_{th} row is T. Computing this string from F requires exponential time (even just writing the required string takes at least 2^n steps). However, the formula F itself is a description of the required string, as the string is (in principle) computable from it. Thus, we have a case of a string that is very compressible but the compression algorithm is infeasible.[18]

We might be more interested, then, in the shortest *feasible* algorithm that generates x. The *resource-bounded algorithmic complexity* of a string x, relative to a machine U, denoted by ${C_U}^{s,t}(x)$, is defined as the length of the shortest program p that generates x in U within a space bound s and a time bound t (which can be functions, e.g., specific polynomials, of $|p|$ or $|x|$). More formally, it is defined as min $\{|p|: U(p) = x \textit{ AND the computation of x from p by U requires less than s memory cells and less than t computational steps}\}$. If no such p exists, we say that ${C_U}^{s,t}(x) = \infty$. Note that unbounded algorithmic complexity can be seen as a special case of resource-bounded algorithmic complexity, with $s = t = \infty$.

If realistic learners can never use more than s_0 computational space and t_0 computational time, then for realistic learners, an algorithm that uses more than s_0 space or t_0 time is of no use. Thus, with the right bounds, the resource-bounded version of algorithmic complexity can be much more relevant for realistic learners than the unbounded one.

Note that resource-bounded algorithmic complexity is computable in principle (when $t \neq \infty$). For every string x, we can enumerate all strings up to a length $|x|$, feed them one by one to

[18] It's an interesting fact about the way we do logic that for formulas with several variables or more, we usually work with formulas that are much shorter (even in binary representation) than the length of their truth table. That is, there is some simplicity bias in the way we build formulas.

the reference machine U, and document the state of U up to t (or earlier if U halted). Then we consider all strings for which U halted on x (and didn't use more than s space), and take the length of the shortest one as the resource-bounded algorithmic complexity of x. Note that this does not create a contradiction, because while the algorithm that enumerates the output of U after t steps for all inputs of length up to $|x|$ might be short, it requires more than t steps.

Of course, such a computation requires a running time exponential in $|x|$, so we can never make such a computation (for sufficiently large $|x|$). This shows that the focus on whether algorithmic complexity is computable is misguided. Computability by itself doesn't give us a lot. The concept of algorithmic complexity is useful not because it tells us something about specific strings, but because of the general lessons we can learn from it.

6. Epistemic Complexity

Resource-bounded algorithmic complexity is still not fully suited to serve the purposes of epistemology. Indeed, if we are bounded in the time and space complexity of the algorithms we can execute, then we are more interested in variants of algorithmic complexity that respect these bounds. However, as we saw in chapter 2, the fact that an algorithm is simple to execute (i.e., doesn't require a lot of computational resources) doesn't imply that it is easy to *learn* the algorithm from the available data. For instance, if strong encryption is possible, there are decryption algorithms that are easy to execute once known, but hard to learn from examples. [19] A measure for the complexity of data that is relevant to epistemology should be sensitive to whether a program p that generates a string x is learnable, not only to the computational

[19] There have been interesting connections discovered recently between encryption and algorithmic complexity. See (Liu & Pass, 2020, Ilango et. al., 2021, Hirahara et. al., 2023, Hirahara & Nanashima, 2023, Cavalar et. al., 2025).

resources it requires. The aim of this section is to introduce such a measure, which I call *epistemically bounded algorithmic complexity*, or *epistemic complexity* for short.[20]

The discussion above suggests adding another parameter to the definition. Let's define the epistemic complexity of a string x, for a learner l, given prior data y, denoted by $C_U^{s,t,l}[x,y]$, as the length of the shortest program p such that $U(p) = x$, when (a) the computational complexity of p is space-bounded by s and time-bounded by t, and (b) p is learnable by a learner l when l is given y as input for l's learning process.[21] If no such p exists, we say that $C_U^{s,t,l}[x,y] = \infty$. As in the unbounded case, we can extend the definition to physical phenomena by assigning them the epistemic complexity of the epistemically simplest string that can encode them.

To see the difference between algorithmic complexity and epistemic complexity, think of the string composed of one billion '1's, 111111…1. Suppose someone takes this string and encrypts it using a simple encryption algorithm, *E*. *E* can be something like "encrypt in the RSA method using public key *k*". The resulting ciphertext, *x*, is very compressible in the unbounded sense. "Print '1' one billion times and then give it as input to *E*" is an example of a short algorithm that generates it.[22] However, suppose *l* is a realistic learner, and the only data available to *l* is *x* itself. If strong encryption is possible, then *l* cannot distinguish *x* from random strings, so *l* cannot learn any compression algorithm for *x* from *x* alone.[23]

To avoid confusion, note that the role of the prior data *y* in this definition is distinct from the case of conditional algorithmic complexity that we denoted as $C_U(x|y)$. The complexity of

[20] "Epistemic complexity" really is a shorthand. The term is not intended to suggest that this is the only kind of complexity relevant for epistemology. As we saw in previous chapters and shall see in the next section, other kinds of complexity are relevant, too.
[21] If l knows p innately, i.e., without learning p from data, we can treat p as learnable by l from any prior data (including the empty string).
[22] Modern encryption schemes will use some random bits in the process, and these bits will not be compressible, but the number of random bits required is relatively small.
[23] We have noted earlier (fn. 3) that high entropy does not imply high algorithmic complexity. It would be interesting to think about the relations between entropy and epistemic complexity for physical learners, which seem closer in meaning to one another.

x conditional on *y* is the length of the shortest program *p* that (when given) generates *x* using another (given) string *y*. The epistemic complexity of *x* given prior data *y* is the length of the shortest program *p* that a learner *l* can learn from evidence *y* such that the machine *U* can run *p* alone and output *x* (even if *U* no longer has access to *y*).

Note further that the complexity of *x* conditional on *x*, $C_U(x|x)$ can be very low while the epistemic complexity of *x* given prior data *x*, $C_U^{s,t,l}[x,x]$, can be very high. For instance, if *x* is incompressible, then the shortest program that generates *x*, a fortiori the shortest *learnable* program is at least as long as *x*. However, the very short program "print the string that is given" will print *x* conditional on being given *x*. Note further that the algorithm "print *x*" might not be learnable from *l*'s specific data, so the epistemic complexity of a string can be much greater than its length (and indeed, can be infinite if *l* cannot learn from *l*'s data any program that generates *x*).

We can define the *epistemic complexity of x conditional on y given prior data z*, denoted by $C_U^{s,t,l}[(x|y),z]$, as the length of the shortest program *p* that is learnable by *l* from *z,* such that $U(py) = x$. This conditional form of epistemic complexity can be useful because, many times, we learn a general algorithm that requires additional data to apply to specific cases. For instance, we can learn algorithms from weather data gathered in previous years, the output of which is the weather 10 days from execution, but these algorithms are conditional in the sense that, to produce the needed output, they need to have access to further data about the weather at the time of execution. Again, non-conditional epistemic complexity can be defined as epistemic complexity conditional on the empty string.

Disagreements between different machines, *U* and *U'*, can be decided in accordance with the considerations of the previous section. However, under the assumption that the learner *l* can be modeled as a universal machine that comes early in the natural ordering, it might be useful to think about the epistemic complexity when *U*=*l*, and the bounds *s* and *t* are the bounds that apply to l (we might in this case wish to understand the bounds s and t as applying to the resources required for the learning process and for executing p combined). This would give the epistemically bounded algorithmic complexity relative to the learner l, which amounts to the shortest program that l can learn from y and then execute to generate x. As a convention, we can omit the "U" if $U=l$.

The input y that the learner l gets for l's learning process can be anything, but it might be particularly interesting to think of y as a string that is composed of subsegments of x. For instance, x can be a string that encodes full information about some general phenomena X, say, the position and motion of the planets at any given moment in the last and next one billion years, and y can be information about particular instances, say, 1000 data points about the position of the planets at different moments in time. The program p is an algorithm that computes the position and motion of the planets at every moment. We can treat p as already including the needed data to predict, or as a conditional algorithm that, conditional on the right data, outputs the positions and motions of the planet for the last and next one billion years. A realistic learner might require enough data of the right kind to learn a short p. That is, the epistemic complexity of x in this case can be infinite without the right data. However, if p is short relative to x and is learnable from a suitable input y, then x is epistemically simple given the right data. Below are some further clarifications.

i. Relations to Other Forms of Algorithmic Complexity

Resource-bounded complexity can be seen as a special case of epistemic complexity, when the learner l is unbounded. Thus, there is a hierarchy in the strength of the definitions of unbounded algorithmic complexity, resource-bounded algorithmic complexity, and epistemic complexity. If there is no short program that generates x, then a fortiori there is no short program that requires bounded resources that generates x, and a fortiori there is no such program that, on top of that, is also learnable for some learner l. Thus, algorithmic randomness is a stronger notion than resource-bounded randomness, which, in turn, fixing the bounds, is stronger than epistemic randomness. In the other direction, if a string x is epistemically simple for l, i.e., there is a short learnable program that generates x within the given bounds, a fortiori there is a short program that generates x within the given bounds, and a fortiori, there is a short program that generates x.[24] Schematically:

Algorithmic Randomness ⇒ Bounded Randomness ⇒ Epistemic Randomness

Epistemic Simplicity ⇒ Bounded Simplicity ⇒ Algorithmic Simplicity

ii. *Epistemic Complexity is Not Invariant*

There is no version of the Invariance theorem for epistemic complexity. In the case of epistemic complexity, there are two distinct cases to consider. First, we can compare the epistemic complexity for the same learner according to two different machines, U and U', that is, $C_U^{s,t,l}[x,y]$ vs. $C_{U'}^{s,t,l}[x,y]$. Second, we can compare the epistemic complexity for two different learners, l and l', according to the same machine, that is, $C_U^{s,t,l}[x,y]$ vs. $C_U^{s,t,l'}[x,y]$. In both cases, there is no invariance, generally.

In the first case, let the shortest program learnable by l that generates x in U, given previous data y be p. Nothing guarantees that l can learn the program $P_{U,U'}$ that simulates U' in U from l's data. U' can be so complex from l's point of view that l cannot learn from y how to simulate it. Thus, it is not automatic that l can learn to generate x in U' by running $P_{U,U'}$ and then p. Furthermore, nothing guarantees that there is a different program p' that is learnable by l that

[24] The hierarchy of strength between unbounded and resource-bounded algorithmic complexity is strict, as we saw in §5. There are algorithms that can compute the resource-bounded complexity of strings, and "the lexicographically first string of length n that is boundedly random" is an example of a description of an algorithmically simple string that is boundedly random. In the case of resource-bounded complexity and epistemic complexity, a lot depends on the bounds on p and on the learner l, but if indeed not all of class $\mathcal{P}$ is learnable, then there are cases of strings that are algorithmically simple under polynomial bounds on the program p, but not epistemically simple.

generates x in U'. Thus, the difference between the epistemic complexity for l relative to U and U' can be infinite.

In the second case, there might be a short program p that generates x and is learnable from y by a very talented learner l, but is not learnable from y by the more limited l'. Furthermore, it might be that no other program that generates x is learnable by the limited learner l' from y. Thus, here too, the difference in the epistemic complexity for two learners can be infinite.

This should not come as a surprise and is not a bug.[25] Something that goes beyond the limits of one learner might still be within the ability of another learner. As we shall see in the next section, epistemic complexity is related to what a learner can predict, and it should be the case that what we and learners like Laplace's demon can predict from the same data is radically different. Moreover, it should be the case that what learners can predict is sensitive to the computing machines available to the learners. We can use a simple computing machine (from our point of view) to generate a prediction, but if a computing machine is alien to us and too complicated for us to learn how to use it, we will not be able to use it to generate a prediction. We want a concept of epistemic complexity that gives a place to qualitative differences in what different types of learners can learn.

iii. Computability and Learnability

Note further that, like resource-bounded algorithmic complexity, epistemic complexity, at least when the time bound on the learner and on machine U is finite, is computable. However, again, there is no reason to think that the epistemic complexity for l given l's data of a given string x is *learnable* by l given l's data. Intuitively, there are many learning processes that a learner can use, and it is not necessarily feasible for the learner to check them all to learn which

[25] In general, different computable approximations of algorithmic complexity do not satisfy the Invariance theorem (see Sterkenburg, 2018, Neth, 2023). Epistemic complexity for different learners can be seen as a special case of this phenomenon.

one is optimal. As I said above, the concept of epistemic complexity is useful not because it tells us something about specific strings, but because of the general lessons we can learn from it.

An example of a lesson we can already draw is that epistemically simple strings are probably rare even among the algorithmically simple ones, which are already a tiny minority among all strings. To see why, let k be a sufficient length of an encryption key to produce ciphertexts that a learner is unable to break. In general, k need not be too small but also not too big. There are roughly 2^k possible keys. Since there are simple encryption algorithms, for every epistemically simple string x of length n, there are roughly 2^k encrypted versions of x that are only slightly less algorithmically simple but are epistemically random.[26]

Additionally, consider the following conjecture: for a learner l with bounded computational time, algorithmically random strings are not learnable by l if their length is above a threshold, except maybe when the learner's data include enough *algorithmic information* about a random string to reduce its complexity below the threshold. That is, if the threshold is k and a random string x is of length $n>k$, then x is not learnable by l, except maybe when $C(x|y) < k$ when y is l's data (e.g., *l can* learn a random string x if l's data includes x, or all the bits of x except the last, or the XOR form[27] of x, etc.). The reason is that if l's data provides no algorithmic information about a random string (i.e., the string is not simpler conditional on the data), then there is no way for l to generate the string except by brute force. However, the learner's data don't prefer any random string (on which the data don't provide any information) over another,

[26] To be more precise, we can think only of strings of length n. If strong encryption is possible, there exist pseudorandom permutations (see Goldreich, 2001, §3.7), which take every string of length n to another string of length n, using an encryption key, such that inverting the permutation without access to the key is infeasible for realistic learners. Suppose the algorithm for the pseudorandom permutation requires p bits (which shouldn't be a lot) plus the length of the key, which we denote by k. For a long enough (though not too long) k, we get that for every string with epistemic complexity $c \ll n$, which implies algorithmic complexity of at most c, there are roughly 2^k encrypted strings that are epistemically random but with algorithmic complexity of at most $c + p + k$, which is not a lot greater than c.

[27] That is, another string y, which has 1 in the n'th bit if and only if x has 0 in the n'th bit.

so to produce a specific random string unknown to *l*, *l* would need to enumerate random strings one-by-one, and above some threshold, there are just too many of those for this task to be feasible for *l*.

We can strengthen this conjecture. The same reasoning applies to strings that are *epistemically* random relative to *l*. In fact, it applies to any string that is sufficiently epistemically complex. Furthermore, it doesn't suffice that *l*'s data simplify a given string in the algorithmically unbounded sense. To *epistemically* simplify an epistemically random string *x*, it's not enough that there will be an algorithm that simplifies *x* given the learner's data; this algorithm must also be learnable by the learner. Thus, we can make the conjecture that above a threshold, learners bounded in their computational time cannot learn an epistemically complex string, unless the learner's data sufficiently epistemically simplify it (or provide enough *epistemic information* about it).

If this conjecture is correct, then learners who, on top of being bounded in computational time, are also bounded in their memory, cannot store information about more than a tiny fraction of the strings that are epistemically random for them. Thus, for such learners, at any point in time, the vast majority of strings are not learnable given their data. To clarify, every random string that is short enough to be stored in the learner's memory can, in principle, be learnable *given the right data*, but no data the learner can store in their memory at a given time suffice for learning more than relatively few random strings. To learn more, the learner needs to forget their current data.[28]

What this means is that the epistemic complexity of most strings relative to a realistic learner is infinite. No matter what data a realistic learner has, most strings are not learnable for that learner, given the learner's data. While we usually can't learn the algorithmic or epistemic

[28] We may say that the domain of random strings that are still short enough to be stored in the learner's memory is only *negligibly learnable*, in the sense that while every member of the domain can be memorized by the learner given the right data, only a negligible fraction of them can be memorized given any possibly current data.

complexity of a *given* string, we do learn that, in general, most strings are not learnable. Learnability of a string is not the default. Second, we see the importance of having the right data to learn a given string, because some strings might not be learnable without the right data but can be epistemically simplified given the right data, that is, data from which the learner can learn the needed string.

iv. More Sophisticated Notions of Epistemic Complexity

For the purposes of actual learners, when they try to learn about general phenomena, they don't need to be able to generate an exact description of each and every instance of the general phenomenon. Suppose a string *x* encodes full information about some general phenomenon X. Actual learners usually don't need and don't aspire to find an algorithm that exactly generates *x*. They usually need an algorithm that, with high probability, gives something close to *x*. For instance, when predicting the position and motions of the planets, we tolerate some error and some risk that our model will be entirely wrong due to unexpected events (e.g., an unexpected collision of one of the planets with a huge celestial body).

There are ways to make the notion of epistemic complexity more suitable to accommodate these issues. For instance, following the notion of probably approximately correct (PAC) learnability, we can define the *PAC epistemic complexity* of *x* by: $C_U^{s,t,l,\delta,\varepsilon}[x,y] = \min\{$|p|: *p* is learnable by *l* from *y*, AND *p* requires less than *s* space and *t* time to run, AND with probability greater than (1-δ), $d(x, U(p)) < \varepsilon\}$, where $d(x, U(p))$ is the distance between *x* and *U(p)*, according to some suitably chosen metric, and δ, ε are chosen in some epistemically relevant way. However, choosing such a metric and defining these epistemically relevant ways will take us too far afield and will not contribute to the arguments to come, so for the purpose of this thesis, we will make do with the simpler definition.[29]

Further possible complication comes from the fact that actual learners are usually interested not in probably generating an approximate *full* description of a wide domain of phenomena X (that supposedly can be fully and accurately encoded by a string *x*). Rather, they are interested in the ability to (with high probability) generate an approximate description of any segment of *x* upon demand. Even less, they wish to be able to denote for each demanded segment what further data are needed to generate (with high probability) an approximation of the segment with known algorithms.

For instance, we don't really need to know the position and motion of the planets at every moment in the next and last one billion years. We want to be able to know when the next solar eclipse is, or the approximate position of the planets tomorrow night when we plan to observe them, and so on. By *being able to know*, I include knowing what data we need to gather that will let us answer these questions. This applies not only to less urgent matters like astronomy but also to various scenarios realistic learners encounter in their struggle to survive. For example, human learners don't need to know where each lion is at every moment. They need to have methods that with high probability notify them about the existence of lions nearby, or even methods that direct them where to look to check whether a lion is nearby.

There are some variants of algorithmic complexity, e.g., Loveland's (1969) *uniform complexity,* that might be more suitable for the task. The uniform complexity of a string *x* of length *n* is defined as the shortest program p such that the input (p,i) generates the ith bit of *x* for every $i \leq n$ (see also Li & Vitanyi, p. 130). Note that when a string is very compressible but just too long, for instance, 10^{100} '1's, a realistic learner cannot print the whole string (they

[29] If we understand learnability in terms of PAC-learnability, this is already included in the original definition. However, I haven't defined distance between strings, and such a definition is required to think about non-exact generation of strings.

would run out of ink, for starters) but can still learn how to generate any short enough segment of it. We might want to say that a realistic learner can still know how to generate the 10^{100} '1's string, and a notion like uniform complexity better captures that. To not overcomplicate the discussion, we'll make do with the simpler definition, but I shall mention this concept again in Chapter 6.

7. Predictability and Compressibility

By "prediction" I mean something that we say or think regarding the behaviour of something we have not yet observed, which is distinct from mere guessing.[30] Suppose we are going to toss a coin 100 times. We can guess the outcome of the coin tosses and might have some luck. With a tiny probability of $\frac{1}{2^{100}}$ we can even be extremely lucky and guess the entire sequence of coin tosses correctly. However, we wouldn't say in such a case that we have *predicted* the outcome of the coin tosses, in the sense I'm using the term. We were lucky to guess the outcome, but to count as a prediction, what we say or think about the behaviour of some range of unobserved phenomena, call it X, should be the result of us trying to learn about X from things we have already observed. For instance, we *predict*, rather than guess, that the sun will rise tomorrow because we reached this conclusion by learning from previous observations.

Note, though, that trying to learn doesn't imply succeeding. Scientists who use bad scientific theories, and even sincere astrologers, make predictions. 'Prediction' is not a success term, and one can be wrong in one's predictions. When a prediction turns out to be correct, we usually say it was a successful prediction. However, a prediction can be successful or

[30] Understood this way, retrodiction, i.e., what we think or say about the unobserved past that is distinct from mere guessing, also counts as a kind of prediction.

unsuccessful due to luck. A successful prediction about X doesn't imply that the predictor knew anything in advance about X. This is expected because not all attempts to learn are successful, and some unsuccessful learning attempts can still sometimes give good results by luck. We say that one has *predictive knowledge* (or *inductive knowledge*) about an unobserved phenomenon X if one successfully learns something about X from one's previous observations.[31]

I reserve the term *predictable* for things about which one can have predictive knowledge. If nothing can be learnt from observations about some unobserved X, then X is unpredictable, regardless of whether one *thinks* one has learnt something about X from observations, and of whether one was lucky enough to say something true about X. The exact outcome of the sequence of coin tosses above is, in this sense, unpredictable. One can try to predict the unpredictable, and even luckily succeed, but one cannot have predictive knowledge about the unpredictable, or methodically succeed in predicting it. Stated this way, *predictability* is relative to a learner and previous data. What is predictable for one learner *l*, given data *y*, might not be predictable for another learner *l'* or given other data *y'*.[32]

We shall see that predictability, understood this way, is highly connected to epistemic compressibility. Before explaining how, let me first clarify that the notions of predictability and compressibility, even epistemic compressibility, are distinct. Predictability is an epistemic notion. It is about what can be known about something without observing it. Compressibility is an algorithmic-information-theoretic notion. It is about the structure of strings or other means to encode data, or, as we saw in §6, about the structure of the means to encode information about physical phenomena. Even epistemic compressibility, which is a somewhat epistemic

[31] One need not learn that something is the case to have predictive knowledge. One's learning that something is likely can also count as predictive knowledge.

[32] Of course, we are free to reserve the word 'predictable' to apply to any phenomena X for which there exists some learner and some data such that this learner given that data can have predictive knowledge about X. I'm not sure though how useful this weak and general sense of 'predictable' is, so in this thesis I will treat the word 'predictable' as implicitly or explicitly referring to a particular learner and particular data.

notion as it is about what is learnable by various kinds of learners given various kinds of data, is distinct from predictability.

If a coin is fair, there is no way to know the exact outcome of one million coin tosses except by observing the outcome, so the exact sequence is not predictable.[33] However, if the coin happens to land heads one million times, the resulting sequence will be very compressible and even epistemically compressible for learners like us. We can easily find a way to compress the sequence when it is given to us.

On the other hand, if we know that in the other room there is a computer that outputs '1' if the number of heads in the sequence is odd and '0' if it's even, but we have not observed its screen yet, we can learn what digit is presented on the screen without observing it, by counting the number of heads. Thus, the digit on the screen is predictable, but you can't compress a single bit,[34] so it is not compressible.[35] Thus, compressibility, whether unbounded or epistemic, does not imply predictability, and predictability does not imply compressibility, whether unbounded or epistemic.

However, if learning is computational, then predicting X by l amounts to running an algorithm that non-accidentally outputs a string x that encodes X. What this shows, immediately, is that if the epistemic complexity of X for l given l's data is infinite, then X is not predictable for l given l's data. Since the vast majority of strings are not predictable for l, X is predictable only if it is encodable by a string from a very tiny set of strings with finite epistemic complexity.

[33] What about clairvoyants? Presumably a clairvoyant can know the sequence in advance and even tell us about it. I'm not convinced that clairvoyants are relevant for two reasons: first, in a natural understanding, clairvoyants "see" the future. They have a "sixth sense". This makes their knowledge observational, rather than predictive. Second, there is probably a reason why clairvoyants don't exist and without further details on how their clairvoyance works it's hard to know exactly what to say about it.

[34] Or at least, you can compress at most one of the binary digits to the empty string.

[35] If we repeat the experiment many times (say k times) we can encode the results using k 1,000,001-bit strings that encode the outcome of the coin tosses + the digit on the screen. These k 1,000,001-bit strings together *are* at least somewhat compressible for a large enough k. Together they include $k \cdot 1{,}000{,}000 + k$ bits, but describing them requires at most $k \cdot 1{,}000{,}000 + constant$, where the constant is the number of bits required to represent the rule that the 1,000,001th bit encodes the parity of the number of 1's in the former 1,000,000.

We can see that predictability is associated with two separate costs: first, there is the cost associated with learning an algorithm that predicts X. Let's call it *the cost of learning how to predict X*. Second, there is the cost associated with using the learnt algorithm to predict X. Let's call it *the cost of predicting X*.

Both these costs can be further divided. In the case of the cost of learning how to predict X, learning processes need data and acquiring data incur costs, both in terms of what's needed to get access to the data (call it *the data collection cost*) and in terms of storing the collected data in the learner's memory for as long as needed (call it *the storage cost*). Then the learning process itself requires both computational space (*the space cost*) and computational time (*the time cost*).

In the case of the cost of predicting X, let's denote the learnt prediction algorithm by *p*. There is a storage cost for storing *p* in the learner's memory, and if *p* predicts X only conditional on further specified data, *d*, then acquiring *d* is associated with collection and storage costs. Here, too, executing *p* (maybe with access to additional data *d*), requires computational resources, so it comes with space and time costs. Note that when the learner that learns the algorithm *p* is the same as the machine that later executes *p*, in a sense the storage cost and space cost are not distinct, as they both make use of the learner's memory, but it is useful to distinguish between them.

I can't think of a general reason why there should be some dependence between the cost of learning how to predict X and the cost of predicting X. It can be the case that X is very easy to predict once you've learnt how to predict X but learning how to predict X is hard. Sometimes it might be easy to learn an algorithm that predicts X, but the algorithm can be hard to execute. It's not clear that the cost of learning how to predict X (when we don't know anyone who has learnt it and can tell us about it) can in general be known in advance rather than learnt retrospectively after the learner has learnt how to predict X. As a result, I don't have a lot to say about it.

Things are clearer, however, when we focus on the cost of predicting X, and specifically for our discussion, on the *storage cost* of predicting X. The storage cost of predicting X using prediction algorithm p and further data d is $|p|+|d|$ (in case no further data is required we can take $|d|$ to be zero). The *minimal* storage cost of predicting X by l is the length of the shortest *learnable* pair p and d, which is exactly the epistemic complexity of X for l. We can think of the ratio $\frac{|p|+|d|}{X}$ as the storage cost-per-bit of predicting x using p and d. Thus, the minimal storage cost-per-bit of predicting X is exactly its normalised epistemic complexity.

One consequence is that if X *is* predictable, the more epistemically complex X is, the more data are needed for l to predict X, or the more epistemically complex[36] the algorithm that is used to generate x from d (or both). Furthermore, we get that the phenomena that are very cheap or easy to predict (per bit) in terms of storage cost are the ones that are very epistemically compressible. Non-accidental epistemic compressibility and *ease of predictability* (in terms of the storage needed) are essentially the same thing.

It is important how easy it is to make predictions (in terms of storage). A scientific theory is considered more successful the more it can predict a very wide domain of phenomena based on a simple algorithm and relatively few measurements. That is, for a scientific theory to be successful, it must be about an easily predictable domain. If the world contains wide and easily predictable domains of phenomena, it means that the world is very epistemically compressible.

Our most successful scientific theories can be seen as massive compression devices. Suppose, for convenience, that the world is like classical physics told us it is. An unbounded learner like Laplace's demon can compress the full information about the entire world into full

[36] If it is learnable from l's data that p can be compressed by a shorter algorithm p', then the pair (p',d) will be shorter than (p,d), so it would be p' rather than p that determines the epistemic complexity of x.

data about one instant in time and a very simple algorithm (Newton's three laws, or Euler-Lagrange equation, together with specific force laws like Newton's law of gravity and Maxwell's equations). More limited learners can use the theory to massively compress information about practically isolated systems like the solar system. Quantum physics somewhat complicates the picture, as in some cases it gives only statistical regularities, but statistical regularities can also be used for compression.

The fact that our science is so successful (or at least it has been successful so far) shows that our world (or at least the observed parts of our world) can be encoded in a way that is very epistemically simple for us, given our data. There's something striking about this fact, given how rare epistemically simple strings are. What makes a world encodable in an epistemically simple way for learners that encode data about it is a question that might teach us a lot about Hume's problem of induction and other matters. However, this thesis is not about Hume's problem, so we shall leave it here as a question.

Part 3: Ravens

Introduction: Two Paradoxes of Confirmation

It is common to present Hume's Old Riddle and Goodman's New Riddle as the two problems of Induction. However, it is equally common to present Goodman's New Riddle together with Hempel's (1945) Paradox of the Ravens as the two paradoxes of confirmation (see, for instance, the entry by Crupi, 2025). Goodman's riddle and Hempel's paradox seem to lead to the unwelcome result that any or almost any evidence confirms almost any hypothesis. Below, I present Hempel's paradox and then discuss the similarities and differences between Hempel's paradox and Goodman's New Riddle.

1. Hempel's Paradox of the Ravens

Take the following seemingly plausible assumptions. First, special cases of a general hypothesis confirm the hypothesis. For example, a black raven is a special case of the hypothesis that all ravens are black, so if we see an object which is a black raven, this fact confirms the hypothesis that all ravens are black. This assumption is referred to as *Nicod's Condition.* Second, if evidence E confirms a hypothesis H, and H is logically equivalent to another[1] hypothesis H', then E confirms H' too, and to the same degree. This assumption is referred to as the *Equivalence Condition*. From these seemingly plausible assumptions, we can derive the *Paradoxical Conclusion* that our seeing, say, a white shoe confirms the hypothesis that all ravens are black.

[1] I individuate hypotheses as linguistic entities, such that H and H' can be seen as different hypotheses even though they are logically equivalent. Alternatively, we can also say that H and H' are two guises of the same hypothesis and take EC to state that if evidence E confirms a hypothesis H under one guise, then it confirms H to the same extent under every guise.

The derivation of the Paradoxical Conclusion is quite simple. Take the hypothesis H' stating that all non-black things are non-ravens. A white shoe is a special case of H', so according to Nicod's Condition, seeing a white shoe confirms H'. But H' is logically equivalent to H, the hypothesis that all ravens are black. Thus, according to the Equivalence Condition, seeing a white shoe also confirms H.

This seems paradoxical because white shoes (and brown tables, and green apples, and so on) seem completely irrelevant to the colour of ravens and learning about the colour of shoes doesn't seem to provide any information about the colour of ravens. As Goodman (1953/1983, p. 70) quipped, this can seem to give rise to a new field of indoor ornithology. We are left with the dilemma of whether to accept this seemingly paradoxical conclusion or reject one of the premises. If we choose to accept the conclusion, we had better also be able to explain why it seems paradoxical at first sight.

Like the predicate *grue*, the colour of ravens is, of course, just an example. The paradox seemingly applies to many other properties, F and G. We should be able to explain why it so often seems reasonable to infer the hypothesis that all Fs are Gs from seeing Fs that are Gs but not from seeing non-Gs that are non-Fs.[2]

2. Haven't Bayesians Solved It Already?

There is, I think, a widely (though not unanimously) held view among philosophers that Hempel's Paradox of the Ravens is no more than a historical anecdote since the paradox has already been solved by the Bayesian approach. We can find this view expressed in some encyclopedic entries on confirmation (Huber, 2024) and promoted in slides shown to

[2] Thus, idiosyncratic properties of the ravens example, e.g., that there are crows who are similar to ravens, or that many people mostly hear stories about ravens and never see the actual bird need not distract us.

undergraduate students in epistemology or philosophy of science classes. Before discussing any novel solution, it's worth explaining why I don't accept Bayesian solutions to the paradox. This is what I do in Chapter 5, which will serve as a prelude to Chapter 6, where I present my solution to the paradox.

3. Grue and the Ravens

Some thinkers, most notably Quine (1969), saw the Paradox of the Ravens and the New Riddle as essentially the same problem, and consequently as calling for the same solution. According to Quine, both the New Riddle and the Paradox of the Ravens stem from failing to respect the notion of *natural kinds*. A disjunctive predicate like *grue* is not a natural kind, and similarly, a negation of a natural kind, like *non-raven,* need not be a natural kind. Only natural-kind terms can be used for good inductive inferences.

Seen from this specific angle, I can see why some philosophers take the two paradoxes as the same problem. But on a similar note, I can see how people can take a cylinder and a ball to have the same shape when they look at them from a very specific angle. My view is that the Paradox of the Ravens and the New Riddle pose distinct challenges and deserve distinct solutions. As in the case of my solution to the New Riddle, one can try to cast my solution to the Paradox of the Ravens in terms of natural kinds, and in a loose sense, from a specific angle, there will be some commonalities with my solution to the New Riddle. However, as we shall see, my approach to the New Riddle is very different from my approach to the Paradox of the Ravens. The former focuses on encodability, the latter, building on the last three chapters, focuses on complexity.

At first glance, we might think that, based on what I have said about the New Riddle, the Paradox of the Ravens is a non-issue. If my solution to the New Riddle is correct, then it can be the case that the examples of emeralds we have observed, which were both green and grue, confirm the hypothesis that all emeralds are green but do not confirm the hypothesis that all emeralds are grue. Thus, it is not the case that examples of objects that are both A and B automatically confirm the hypothesis that all the *F*s are *G*s. In other words, Nicod's Condition is false, and we are not compelled to accept the Paradoxical Conclusion. Technically, this is correct, but ending the discussion here will make us miss the lesson of the Paradox of the Ravens. Tackling the paradox directly leads to further insights regarding learning and confirmation.

Let's look at the key difference between the New Riddle and the Paradox of the Ravens. The New riddle asks why a single set of examples (the emeralds we have observed) does not confirm distinct hypotheses (both natural and gruesome hypotheses) to which it stands in seemingly symmetric relations. The Paradox of the Ravens, on the other hand, asks why distinct sets of examples (sets containing black ravens and sets containing non-black non-ravens) do not both confirm a single hypothesis (that all ravens are black) to which they stand in seemingly symmetric relations. If the question that the New Riddle raises is what is wrong with gruesome hypotheses, we can say that the question the Paradox of the Ravens raises is what is wrong with *gruesome evidence*, or *gruesome examples*.

Furthermore, unlike the case of *grue*, there doesn't seem to be anything too peculiar about predicates like *non-black* or *non-raven*. Negation is just a simple logical operation. If we understand what 'black' or 'raven' mean, we quite immediately understand what 'non-black' or 'non-raven' mean. If we can detect whether an object is a raven or not, we can immediately detect whether it is a non-raven or not. This amounts to the same thing. That is, even if we have a way to discriminate against gruesome predicates, it is still a riddle why we can discriminate between the evidential import of positive examples and negative examples of non-gruesome predicates. Indeed, positive examples of the predicate *raven* are negative examples of the predicate *non-raven* and vice versa.

4. Symmetry, Again

As in the case of the New Riddle, we have a seeming symmetry. This time, it is not a symmetry between hypotheses but a symmetry between sets of examples. The hypothesis that all ravens are black seems to be symmetrically confirmed by examples of (black) ravens and examples of (non-black) non-ravens. As in the case of the New Riddle, my aim is to break the symmetry. I shall argue that the limitations of realistic learners are what break the symmetry. What might be symmetrical for supernatural learners will turn out to be asymmetrical for realistic ones.

Here, again, I do not aim to show that we *can* learn from examples of black ravens that all ravens are black. This requires dealing with Hume's old riddle. Rather, I aim to show that *if* we can learn from examples of ravens that ravens are black, there should be no remaining riddle as to why we can't learn that ravens are black from examples of non-ravens. For that, it suffices to show why the apparent symmetry breaks. It will turn out that the basic asymmetry is not between *the concepts raven* and *non-raven*, but rather between ravens and non-ravens *as objects*. Specifically, the asymmetry stems from the kind of information examples of ravens and examples of non-ravens provide to realistic learners, and what realistic learners can learn from these examples about both ravens and non-ravens (as such).

Chapter 5: Have Bayesians Solved the Paradox of the Ravens?

1. Introduction

By 'the Bayesians', I shall refer for now to those who choose to accept that both black ravens *and* non-black non-ravens confirm the hypothesis that all ravens are black, and who therefore accept the seemingly paradoxical conclusion. However, they emphasise that confirmation is a matter of degree, and from the fact that both black ravens and non-black non-ravens confirm the hypothesis that all ravens are black, it doesn't follow that a non-black non-raven confirms the hypothesis that all ravens are black *to the same degree* as a black raven does. In fact, with further assumptions about our prior probabilities of sampling objects from our universe, we can derive the conclusion that the degree of confirmation provided by a non-black non-raven is negligible compared to the degree of confirmation provided by a black raven.[1] The conclusion seemed paradoxical to us because it did not include further information that the degree of confirmation provided by a non-black non-raven is negligible. People take non-black non-ravens to be irrelevant to the hypothesis that all ravens are black because they confuse a minute degree of confirmation with zero confirmation. It doesn't change a lot in practice.

Thus, in what follows, to qualify as a 'Bayesian' solution to the paradox, the two desiderata below need to follow from it:

D_1: Both sampling a black raven and sampling a non-black non-raven confirm the hypothesis that all ravens are black.

[1] Examples of Bayesian solution to the paradox can be found as early as Hosiasson-Lindenbaum (1940), Alexander (1958), Mackie (1963), and others. More recent Bayesian treatments of the paradox include Horwich (1982) and Howson & Urbach (1993). Vranas (2004) presents a version of a Bayesian solution that is sometimes referred to in the literature as *the Standard* or *Canonical* Bayesian Solution. Fitelson & Hawthorn (2010a, 2010b, and 2010c, which is an addendum to 2010b) discuss the paradox and the Bayesian approach to it in detail. Rinard (2014) criticizes the canonical solution and promotes adopting what she takes to be an alternative Bayesian solution (but see fn. 24).

D_2: The degree of confirmation provided to the hypothesis that all ravens are black by sampling a non-black non-raven is negligible compared to the degree of confirmation provided by sampling a black raven.

Some might wonder whether this is too restrictive. It should suffice for a solution to be qualified as 'Bayesian' if it uses statistical and probabilistic considerations to solve the paradox, whether they satisfy D_1 and D_2 above or not. While indeed the most famous Bayesian solutions satisfy D_1 and D_2, maybe other solutions that can be qualified as 'Bayesian' don't. I will say something about this issue in the conclusion.

I hope to show that any solution that satisfies D_1 and D_2 suffers from a fatal flaw. The central idea of this chapter is a rather simple one. A satisfying solution to the Paradox of the Ravens need not only show that the support provided to the hypothesis that all ravens are black by a *single* non-black non-raven is negligible compared to the support provided by a *single* black raven. It also needs to show that the support provided by *all* the non-black non-ravens we see is negligible compared to the support provided by *all* the black ravens we see. Otherwise, even if the support provided by a single non-black non-raven is negligible compared to the support provided by a single black raven, it might be that we see many more non-black non-ravens than black ravens and the cumulative support provided by all the non-black non-ravens we see can be non-negligible compared to the cumulative support provided by all the black ravens we see. This way, the prospects of indoor ornithology are on the rise, and we are left again in a paradoxical position.

On most Bayesian solutions, *what makes it the case* that a non-black non-raven contributes negligibly compared to a black raven is exactly the (obviously true) assumption that it is overwhelmingly more likely when we sample an object at random to sample a non-black object than to sample a raven. As we shall see in the next section, this assumption implies that it is also overwhelmingly more likely to sample a non-black non-raven than to sample a black raven. Thus, the attention to the cumulative support provided by all non-black non-ravens we see and all black ravens we see is a natural worry.

If the cumulative support provided by all non-black non-ravens we see is non-negligible, then the paradox retains its full force. We can still do indoor ornithology and learn about the colour of ravens without ever seeing any ravens. To get a feel for the kind of results that will follow, suppose that the cumulative support provided by all the non-black non-ravens I see is equal to the cumulative support provided by all the black ravens I see. Suppose further that my late grandmother, who had similar priors as I have, saw roughly similar things in her life as I did, except that she lived all her life never seeing any raven (or getting any testimony about their colour, for that matter). It follows that my grandmother, who was more than twice my age when she passed away, had stronger evidence than I have for the hypothesis that all ravens are black, even though she had never seen any ravens. She probably saw more than double the amount of non-black non-ravens that I saw at the time of her death, and they more than compensate for all the ravens I have seen, and she hadn't. If the cumulative contribution of all the non-black non-ravens I see is (say) half of that of the black ravens, we only need to go back in time to a moment when my grandmother's age tripled mine (I was old enough to know the colour of ravens by then).

Thus, what seems to matter more is not whether a single non-black non-raven provides a non-negligible degree of confirmation, but rather whether the *cumulative* support that all the non-black non-ravens we saw give to the hypothesis that all ravens are black is still negligible compared to the support provided by all the black ravens we saw. The paradox stemmed from the paradoxical flavour of the conclusion that we can learn the colour of ravens by looking only at non-black non-ravens. The Bayesians allegedly solved the paradox by showing that under some plausible assumptions, a single non-black non-raven contributes only negligibly. But it is barely a consolation if, from these very assumptions, it follows that the tons of non-black non-ravens we see will together let us gain substantive confirmation for the claim that ravens are black, without us ever seeing any ravens.

In my view, the need to show that the cumulative support of all the non-black non-ravens is negligible in normal circumstances is part of the difficulty posed by the original paradox, surprisingly unnoticed as it is. Hence the chapter's title and my choice of words in it. Some might contend that this is not part of the original puzzle but rather a separate revenge puzzle. I don't want to pay too much attention to this matter because nothing crucial hangs on that. It's open for the readers to defend the Bayesians' honour and insist that the Bayesians *did* solve *the* Paradox of the Ravens, even if I'm right to say that they lack the means to solve an equally disturbing revenge puzzle that stems naturally from the first. The bottom line will remain the same.

To sum up, I take the Paradox of the Ravens to pose another desideratum that must follow from every plausible solution to it:

> *D_3: The totality of non-black non-ravens we see contributes negligibly to the confirmation of the hypothesis that all ravens are black compared to the totality of black ravens we see.*

To solve the paradox, one needs to derive *both* D_2 and D_3 from plausible assumptions. The Bayesians want to establish D_1 and D_2, and they (like everyone else) usually ignore D_3. I aim to show that, except under some extremely narrow (and usually implausible) assumptions, D_3 is incompatible with D_1 and D_2. Therefore, to solve the Paradox of the Ravens, that is, to establish D_2 and D_3, we need to reject D_1, that is, we need it to be the case that a non-black non-raven does *not* positively support the hypothesis that all ravens are black.

Showing that inevitably requires some undergraduate-level calculus. In a sense, there is nothing particularly important in the specific mathematical details. The philosophical value of the mathematical details comes from the conclusion they support. For this reason I decided to cut some of the details out of the thesis. The full details can be found in the mathematical appendix in (Karmon, 2026).

The rest of the chapter is structured as follows. In the next section, I present the most famous version of a Bayesian solution, sometimes referred to in the literature as the *standard*, or the *canonical*, Bayesian solution. This version is adapted from Vranas (2004) and is the one that, to my impression, is commonly shown to undergraduates in their introductory classes.[2] In §3, I show why the assumptions in Vranas' version imply that (except for extreme conditions) the cumulative contribution of all the non-black non-ravens is not negligible compared to that of the black ravens. In §4, I explain why that problem is not specific to Vranas' version. The desiderata D_1-D_3 can hold together only under implausible conditions. §5 concludes.

2. The Standard Bayesian Solution

The standard Bayesian solution aims to establish D_1 and D_2 from our priors regarding the probabilities of sampling black ravens, non-black ravens, black non-ravens, and non-black non-ravens and from whether and how these probabilities change if the hypothesis that all ravens are black is true. It is common to model the situation as follows: we sample objects from the universe, like drawing balls from an urn. Since we are going to discuss multiple draws below, we will assume we replace the ball we draw back into the urn.[3] According to the objects we sample (the balls we draw), we try to estimate what the rest of the objects in the universe are (what all the balls in the urn are like). Let's denote the hypothesis that all ravens are black by H, and denote the proposition that the object in question that we have sampled is a black raven,

[2] Vranas himself doesn't necessarily adopt it as a solution. His paper is a critical but inconclusive evaluation of whether the solution is successful.

[3] Modelling the situation with replacement simplifies the calculation, but is not crucial. In section (ii) of the appendix in (Karmon, 2026), I show that under plausible assumptions we get the same results for models without replacement.

or non-black raven, or black non-raven, or non-black non-raven, by *br,* or *nbr*, or *bnr*, or *nbnr*, accordingly.

We can get all the relevant information regarding the prior probability distribution by specifying the probability of sampling an object from each category conditional on *H* and conditional on *~H,* and by specifying the prior probability of *H*. That is, we need to specify the values of all the probabilities in the two tables below:

With probability *P(H):*

	r	*~r*
b	*P(br|H)*	*P(bnr|H)*
~b	*P(nbr|H)*	*P(nbnr|H)*

Table 3: Prior probabilities conditional on H

With probability *P(~H):*

	r	*~r*
b	*P(br|~H)*	*P(bnr|~H)*
~b	*P(bnr|~H)*	*P(nbnr|~H)*

Table 4: Prior probabilities conditional on ~H

That is, to fully specify our priors, we need the value of the following 10 variables: *P(br* | *H), P(nbr* | *H), P(bnr* | *H), P(nbnr* | *H), P(br* | *~H), P(nbr* | *~H), P(bnr* | *~H), P(nbnr* | *~H), P(H),* and *P(~H).*

Not all ten variables are independent. First, of course, $P(\sim H) = 1 - P(H)$. Moreover, $P(br|H) + P(nbr|H) + P(bnr|H) + P(nbnr|H)$ should sum up to 1 as together the four variables cover the whole universe (conditional on *H*). Ditto for $P(br|\sim H) + P(nbr|\sim H) + P(bnr|\sim H) + P(nbnr|\sim H)$. That is, one in each of these sets of four variables is dependent on the other three. Since *P(bnr|H)* and *P(bnr|~H)* will not play any role in the calculations that follow, we will treat them as the dependent variables. Furthermore, due to the specific meaning of *H* (i.e., all ravens are black), *P(nbr|H)* must be zero. Thus, we are left with six independent variables:

P(br|H), P(nbnr|H), P(br|~H), P(nbr|~H), P(nbnr|~H), and *P(H).*

By ‘independent’, of course, I don’t mean that they can get any value. We are talking about probabilities, so each must range between 0 and 1, and as I said, the probability that an object we sampled is from the whole universe needs to be 1. To avoid triviality and divisions by zero, let’s adopt the common practice and assume that all these six variables are *strictly* between 0 and 1. The Bayesians aim to solve the paradox for all plausible cases anyway, so surely there is no harm in assuming that.

Note that the probability of sampling a black raven, say, regardless of whether H is true is (according to the law of total probability) simply $P(br)=P(H)P(br|H)+P(\sim H)P(br|\sim H)$, and ditto for sampling a non-black raven, black non-raven, or non-black non-raven. Thus, whenever we see a probability of sampling one of the four categories unconditional on H or $\sim H$, we can treat it as an abbreviation of an expression that uses only the six variables above.

We are now ready to present the standard Bayesian solution. It is common to present it as follows. Our background knowledge, it is claimed, seems to agree with the following three assumptions (*nb* stands for the proposition that the object in question we have sampled is non-black, and *r* stands for it being a raven):[4]

A_1) $P(nb) \gg P(r)$

A_2) $P(r|H) = P(r)$

A_3) $P(nb|H) = P(nb)$

A_1 states that the probability of sampling a non-black object at random is overwhelmingly higher than the probability of sampling a raven. A_2 states that the probability of sampling a raven is independent of whether the hypothesis that all ravens are black is true, that is, the probability of sampling a raven conditional on the hypothesis that all ravens are black is simply

[4] Usually, the assumptions are presented with the probabilities conditionalized on our background knowledge K. E.g., A_1 is presented as $P(nb|K) \gg P(r|K)$, and ditto for A_2 and A_3. This is because, as Good (1967) has famously shown, according to some background knowledge black ravens can *disconfirm H,* so it’s a good practice to always conditionalize on our background knowledge K. However, since K will not contribute anything to the calculations below, I chose to omit it.

the probability of sampling a raven in general. A_3 states that the probability of sampling a non-black object is also independent of the hypothesis that all ravens are black. Note that it's a direct result of A_2 and A_3 that $P(r|H) = P(r|{\sim}H)$ and $P(nb|H) = P(nb|{\sim}H)$.[5]

These three assumptions suffice to satisfy D_1 and D_2. There are various ways to show that. It will be useful for what follows to present the case in terms of the six independent variables above, so let's see what each of A_1-A_3 above tells us about them. Let's denote the ratio $\frac{P(nb)}{P(r)}$ by m. A_1 tells us that m is a huge number. Note that since both nb and r are probabilistically independent of H (as we learn from A_2 and A_3), m is independent of whether H is true or not, that is, $m = \frac{P(nb)}{P(r)} = \frac{P(nb|H)}{P(r|H)} = \frac{P(nb|{\sim}H)}{P(r|{\sim}H)}$. It follows[6] that:

$$(2.1)\ P(nbnr|H) = mP(br|H)$$

And therefore:

$$(2.2)\ P(nbnr|H) \gg P(br|H)$$

Moreover, it follows[7] from A_2 and A_3 that:

$$(2.3)\ P(nbnr|{\sim}H) = mP(br|{\sim}H) + (m-1)P(nbr|{\sim}H)$$

Therefore, $P(nbnr|{\sim}H)$ is at least m times greater than $P(br|{\sim}H)$ and at least (m-1) times greater than $P(nbr|{\sim}H)$. If $P(br|{\sim}H)$ and $P(nbr|{\sim}H)$ are not too small compared to one another, then $P(nbnr|{\sim}H)$ is even more than m times greater than both $P(br|{\sim}H)$ and $P(nbr|{\sim}H)$. That is, we can derive from A_1 both:

$$(2.4)\ P(nbnr|{\sim}H) \gg P(br|{\sim}H)$$

$$(2.5)\ P(nbnr|{\sim}H) \gg P(nbr|{\sim}H)$$

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[5] It follows from the law of total probability. For instance, in the case of A_2:
$P(r) = P(H)P(r|H) + P({\sim}H)P(r|{\sim}H) = P(H)P(r) + P({\sim}H)P(r|{\sim}H)$.
Therefore, $\big(1-P(H)\big)P(r) = P({\sim}H)P(r) = P({\sim}H)P(r|{\sim}H)$.
Therefore, $P(r) = P(r|{\sim}H)$.

[6] Since $P(nbr|H) = 0$, we have $P(nbnr|H) = P(nb|H) = P(nb) = mP(r) = mP(r|H) = mP(br|H)$

[7] $P(nbnr|{\sim}H) + P(nbr|{\sim}H) = P(nb|{\sim}H) = P(nb) = mP(r) = mP(r|{\sim}H) = mP(br|{\sim}H) + mP(nbr|{\sim}H)$.
Thus, $P(nbnr|{\sim}H) = mP(br|{\sim}H) + (m-1)P(nbr|{\sim}H)$.

The $\gg$ sign is interpreted as at most slightly weaker than the $\gg$ sign in A_1 and in many (perhaps most) cases as even stronger.

From A_2 we infer that $P(r|H)=P(r|{\sim}H)$ and therefore we can derive[8] that:

$$(2.6)\ P(br|H) = P(br|{\sim}H) + P(nbr|{\sim}H)$$

From A_3 we know that $P(nb|H)=P(nb|{\sim}H)$ and therefore we can similarly derive[9] that:

$$(2.7)\ P(nbnr|H) = P(nbnr|{\sim}H) + P(nbr|{\sim}H)$$

Now, there is no single agreed-upon way to measure the degree of confirmation a piece of evidence e confers on a hypothesis h. In the literature (Fitelson and Hawthorne, 2010), there are three major relevance measures:

The Difference: $d(h, e) = P(h|e) - P(h)$

The Log-Ratio: $r(h, e) = \log\left(\frac{P(h|e)}{P(h)}\right)$

The Log-Likelihood Ratio: $l(h, e) = \log\left(\frac{P(e|h)}{P(e|{\sim}h)}\right)$

Among these three, the third one, the log-likelihood ratio, is the most suitable for quantitative comparisons, and there are independent reasons to prefer it (Fitelson, 2001), so I'll use this measure.[10]

Measured by the log-likelihood ratio, the degrees of confirmation provided to the hypothesis that all ravens are black by a single black raven and by a single non-black non-raven are:[11]

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[8] $P(r|H) = P(br|H) + P(nbr|H) = p(br|H) + 0 = P(br|H)$. Now, since $P(r|H) = P(r|{\sim}H)$ we get that: $P(br|H) = P(r|{\sim}H) = P(br|{\sim}H) + P(nbr|{\sim}H)$.

[9] $P(nb|H) = P(nbnr|H) + P(nbr|H) = P(nbnr|H) + 0 = P(nbnr|H)$. Since $P(nb|H) = P(nb|{\sim}H)$, we get: $P(nbnr|H) = P(nb|{\sim}H) = P(nbnr|{\sim}H) + P(nbr|{\sim}H)$.

[10] Nothing crucial seems to change in the case of the other measures. The same results we get using the log-likelihood ratio seem to hold using the other measures as well. The math in these cases is much more complicated so it's harder to prove the results with the same rigor. Suffice it to say that if for some reason one is reluctant to use the log-likelihood ratio as a measure, one carries the burden of proving that the other measures don't lead to the same bottom line.

[11] According to (2.6) and 2.7),

$l_{br} = \log\left(\frac{P(br|H)}{P(br|{\sim}H)}\right) = \log\left(\frac{P(br|{\sim}H)+P(nbr|{\sim}H)}{P(br|{\sim}H)}\right) = \log\left(1 + \frac{P(nbr|{\sim}H)}{P(br|{\sim}H)}\right)$

$l_{nbnr} = \log\left(\frac{P(nbnr|H)}{P(nbnr|{\sim}H)}\right) = \log\left(\frac{P(nbnr|{\sim}H)+P(nbr|{\sim}H)}{P(nbnr|{\sim}H)}\right) = \log\left(1 + \frac{P(nbr|{\sim}H)}{P(nbnr|{\sim}H)}\right)$

$$(2.8)\ l_{br} = \log\left(1 + \frac{P(nbr|{\sim}H)}{P(br|{\sim}H)}\right)$$

$$(2.9)\ l_{nbnr} = \log\left(1 + \frac{P(nbr|{\sim}H)}{P(nbnr|{\sim}H)}\right)$$

Since all the terms above are positive, we see immediately that the confirmation provided by sampling a black raven and by sampling a non-black non-raven are both positive and therefore both the sampling of a black raven and the sampling of a non-black non-raven confirm the hypothesis that all ravens are black. Thus, D_1 is satisfied. Furthermore, since $P(nbnr|{\sim}H) \gg P(br|{\sim}H)$ we get that $\frac{P(nbr|{\sim}H)}{P(br|{\sim}H)} \gg \frac{P(nbr|{\sim}H)}{P(nbnr|{\sim}H)}$. Therefore, the contribution of a non-black non-raven is negligible compared to that of a black raven, and D_2 is also satisfied. However, we shall now see that D_3, in general, cannot be satisfied, except under extreme and implausible assumptions regarding our priors.

3. The Non-Black Non-Ravens Strike Back

To satisfy D_3, we need to compare the support provided to H by all the black ravens we see and all the non-black non-ravens we see. Since we model each sampling of a black raven or a non-black non-raven as independent and the log-likelihood ratio is additive, we can compare instead the support H receives from seeing a single black raven, and the support H receives from seeing the number of non-black non-ravens we see per black raven. What is this number?

The expected value of the number of non-black non-ravens we see for each black raven is given by $\frac{P(nbnr)}{P(br)}$, which under A_1-A_3 can be rewritten as $\frac{P(nbnr|{\sim}H)+P(H)P(nbr|{\sim}H)}{P(br|{\sim}H)+P(H)P(nbr|{\sim}H)}$. One can argue that $\frac{P(nbnr)}{P(br)}$ is not the right value to use. We can narrow the discussion to the confirmation provided by all the non-black non-ravens only when it is in fact the case that all ravens are black. In such a scenario, the expected value of the number of non-black non-ravens we see for each black raven is given by $\frac{P(nbnr|H)}{P(br|H)}$ that under A_1-A_3 can be rewritten as $\frac{P(nbnr|\sim H)+P(nbr|\sim H)}{P(br|\sim H)+P(nbr|\sim H)}$, which is smaller than $\frac{P(nbnr)}{P(br)}$. Taking $\frac{P(nbnr|H)}{P(br|H)}$ as the expected number simplifies the calculations, and the resulting expected cumulative support of all the non-black non-ravens we see per raven is smaller than if we had taken $\frac{P(nbnr)}{P(br)}$ instead as the relevant magnitude. Since I want to be generous towards the Bayesian solution and simplify the calculation, I'll indeed use $\frac{P(nbnr|H)}{P(br|H)}$ as the relevant magnitude (let's call it choice 1). I'll sometimes say (in the footnotes, without the full calculation) what is added by taking $\frac{P(nbnr)}{P(br)}$ instead (let's call it choice 2).

Section (i) of the appendix in (Karmon, 2026) shows that under the assumption that each sampling of a non-black non-raven is independent,[12] the contribution of this number of non-black non-ravens can be approximated by:[13]

$$(3.1)\ l_{nbnr's\ per\ br} \approx \frac{1}{\frac{P(br|\sim H)}{P(nbr|\sim H)}+1}$$

The ratio $\frac{P(br|\sim H)}{P(nbr|\sim H)}$ appeared (in its inverse form) in l_{br} as well (see 2.8). This ratio tells us what we take in advance the proportion between black ravens and non-black ravens to be, in case not all ravens are black. For instance, if we think in advance that if not all ravens are black, then half of them are, then the value of this ratio is 1. If we think in advance that if not all ravens are black, then 10% of them are, then the value of this ratio is $\frac{1}{9}$. If we think in advance that if not all ravens are black, then either (with a probability of 50%) half of them are, or (with

[12] Section (ii) of the appendix in (Karmon, 2026) explains why the assumption of independence is not crucial.

[13] If we take choice 2, we get instead $l_{nbnr's\ per\ br} \approx \frac{1}{\frac{P(br|\sim H)}{P(nbr|\sim H)}+P(H)}$.

a probability of 50%) none of them are, then the value of this ratio is $\frac{0.5\cdot 0.5+0.5\cdot 0}{0.5\cdot 0.5+0.5\cdot 1}=\frac{1}{3}$. In general, I think, we expect this ratio not to be extremely big or extremely small.[14] At the very least, it seems *legitimate* to take it in advance to be moderate.

For simplicity, let's denote this ratio by *x,* that is, $\frac{P(br|\sim H)}{P(nbr|\sim H)}=x$. We can then rewrite l_{br} and $l_{nbnr's\ per\ br}$ as $l_{br}=\log\left(1+\frac{1}{x}\right)$ and $l_{nbnr's\ per\ br}=\frac{1}{1+x}$.[15] To satisfy D_3, we need $\frac{1}{1+x}$ to be *negligible* compared to $\log\left(1+\frac{1}{x}\right)$. One way to formalise this requirement is with the inequality below:

$$(3.2)\ \log\left(1+\frac{1}{x}\right)>K\left(\frac{1}{1+x}\right)$$

Here K is a big *negligibility constant*. For instance, if to be negligible the value of $l_{nbnr's\ per\ br}$ needs to be (say) less than 1% of the value of l_{br}, then $K=99$. I tend to think that *K* should be significantly greater than 99, but we don't need to decide upon it now to proceed. (3.2) is equivalent[16] to:

$$(3.3)\ \log\left(\left(1+\frac{1}{x}\right)^{1+x}\right)>K$$

Exponentiating both sides of the inequality yields:[17]

$$(3.4)\ \left(1+\frac{1}{x}\right)^{1+x}>e^{K}$$

The function $f(x)=\left(1+\frac{1}{x}\right)^{1+x}$ has been studied in great detail. For our purpose, though, we need only note two things. First, *f(x)* is descending for every $x>0$. Second, (provided that we

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[14] We can think that it's a possibility that if not all ravens are black then none of them are. However, as long as we think that there is a non-negligible probability that if not all ravens are black then still a significant portion of them are, we expect $\frac{P(br|\sim H)}{P(nbr|\sim H)}$ not to be extremely small.

[15] If we take choice 2, we can rewrite: $l_{nbnr's\ per\ br}=\frac{1}{P(H)+x}$

[16] Because $(1+\mathrm{x})\log\left(1+\frac{1}{x}\right)=\log\left(\left(1+\frac{1}{x}\right)^{1+x}\right)$.

[17] If we take choice 2, we get instead that $\left(1+\frac{1}{x}\right)^{P(H)+x}>e^{k}$.

take K to be greater than 2, say),[18] the inequality holds for $x = \frac{1}{e^k}$, but fails to hold for $x = \frac{2}{e^k}$. That is, for the inequality in (3.4) to hold, x must be smaller than a number that lies somewhere between $\frac{1}{e^k}$ and $\frac{2}{e^k}$. Therefore, to satisfy D_3, the ratio $\frac{P(br|{\sim}H)}{P(nbr|{\sim}H)}$ must be smaller than $\frac{2}{e^K}$.

Since K is supposed to be a pretty big number, e^K should be gigantic.[19] This means that for the standard Bayesian solution to satisfy D_3, we must think in advance that if not all ravens are black, and we sample a raven, then the probability that the raven we sample *is* black should be extremely tiny. To get a feel of how tiny, look at table 3 below. The table states how small x should be for some chosen values of K, that is, how likely it is that a given raven we sample is black, conditional on not all ravens being black. To help the reader get a grasp of the magnitudes, in each row, I gave an example of a roughly equiprobable event we are more familiar with.

Chosen value of K	The non-black non-ravens contribute at most…	x should be at least smaller than…	Suppose that ~H and that we have sampled a raven. This raven being black must be less likely than…
19	5%	$\frac{2}{e^{19}} \approx 10^{-8}$	Winning the lottery[20]
99	1%	$\frac{2}{e^{99}} \approx 10^{-44}$	Winning the lottery six times in a row
999	0.1%	$\frac{2}{e^{999}} \approx 10^{-434}$	Randomly sampling the same specific elementary particle from the whole universe five times in a row

Table 5: How small x should be for some chosen values of K

[18] Or not smaller than about 1.92 to be exact.

[19] Note that we get from (3.4) that while the cumulative support of the non-black non-ravens is not negligible compared to the cumulative support of the black ravens it still must be smaller. To see that, simply take K to be 1. However, if we had taken choice 2, we could have easily come up with examples of priors according to which the cumulative support of all the non-black non-ravens is greater than that of the black ravens. E.g., take x to be 1 and P(H) to be 1/3.

[20] The chance of winning the lottery changes from country to country. I take it to be in the order of magnitude of 10^{-7}.

Since ravens are discrete objects and there aren't that many of them, we get that for the standard Bayesian solution to satisfy D_3, it must be the case that we think in advance that if not *all* ravens are black, then it's overwhelmingly likely that none of them are.[21] However, there is no reason to think that if not all ravens are black, then none of them are.

Furthermore, remember that the colour of ravens is just an example. The paradox seemingly applies to many other properties F and G, and there is even less reason to assume in advance that for all these properties F and G, either all the Fs are Gs or, in probability close to certainty, none or only an unimaginably tiny minority of them are. For example, it seems unreasonable to think we are committed not only to thinking in advance that if not all ravens are black then none of them are but also to think in advance that if not all parrots are green then none of them are, if not all chairs have three legs then none of them do, if not all oranges are orange then none of them are, etc. I conclude that the standard Bayesian solution cannot plausibly satisfy all three desiderata, and if I'm right in saying that both D_2 and D_3 are crucial to answering the paradox, it follows that the standard Bayesian solution fails to address the paradox of the ravens.[22]

4. The Prospects of Any Alternative Bayesian Solution

At this point, the reader might suspect that the challenge I presented applies only to a specific variant of the Bayesian solution, namely, the one presented by Vranas, and while indeed this is probably the most famous version of a Bayesian solution, it needn't be the only

[21] Taking very small values for K like 19 allows one very special raven to be black even if not all of them are, but taking K to have such a small value seems implausible (at least to me).

[22] It is known at least since (Horwich, 1982) that the standard Bayesian solution has the consequence that seeing black non-ravens *disconfirms* H (see Rinard, 2014). I believe people tend ignore it because the contribution of a single black non-raven is negligible. The cumulative negative contribution of all the black non-ravens we see is non-negligible, though.

one. What if instead of assumptions A_1-A_3, we adopt different assumptions? Can't we then save a Bayesian approach to the paradox of the ravens? My answer is that the prospects for that are not promising, at least when "Bayesian approach" is understood as the strategy of establishing both D_1 and D_2 (that is, acknowledging that both seeing a black raven and seeing a non-black non-raven confirm H but a black raven provides stronger support). The (somewhat complicated) mathematical details appear in section (iii) of the appendix in (Karmon, 2026). Here, I present the general outline of the argument.

The idea of the argument is to examine what follows from adopting the three desiderata D_1-D_3, without making any assumptions regarding our background knowledge. That is, recall, we examine what follows from assuming that both a black raven and a non-black non-raven support the hypothesis that all ravens are black (D_1), but the contribution of both a single non-black non-raven compared to a single black raven (D_2), and the aggregate contribution of all the non-black non-ravens we are expected to see compared to the aggregate contribution of all the black ravens we are expected to see (D_3) are negligible. We derive that for these three desiderata to hold together, at least one of three conditions must hold, but it is implausible to impose any of these conditions on our priors.

The first condition that allows D_1-D_3 to hold together states that the ratio $\frac{P(br|H)}{P(br|\sim H)}$ is extremely big. That is, it is overwhelmingly more likely to sample a black raven if all ravens are black than to sample a black raven if not all of them are, or practically, that if not all ravens are black, then none of them are. Again, there is no reason that our priors regarding ravens must be tuned this way, all the more so our priors regarding any properties F and G for which the paradox applies. Note that this is a generalisation of the result we got in the previous section. Earlier, we got from A_1-A_3 that to satisfy D_1-D_3, the ratio $\frac{P(br|\sim H)}{P(nbr|\sim H)}$ needs to be extremely small, or equivalently, that $\frac{P(nbr|\sim H)}{P(br|\sim H)}$ needs to be extremely big. It follows from A_1-A_3 that $\frac{P(br|H)}{P(br|\sim H)} > \frac{P(nbr|\sim H)}{P(br|\sim H)}$ so if $\frac{P(nbr|\sim H)}{P(br|\sim H)}$ is extremely big, then so is $\frac{P(br|H)}{P(br|\sim H)}$.

The second condition that allows D_1-D_3 to hold together states that $P(br|H) \geq P(nbnr|H)$, i.e., that if all ravens are black, then sampling a raven is at least as likely as sampling a non-black object. An implausible condition indeed. This condition comes from the fact that, in the argument presented in the appendix of (Karmon, 2026), I assume for convenience that $\frac{P(nbnr|H)}{P(br|H)} > 1$, so negating this assumption seemingly lets us evade the conclusion of the argument presented there. In fact, the argument can work with even smaller values than 1,[23] but it is not crucial. A solution that requires our priors to be such that $P(br|H) \geq P(nbnr|H)$ is a non-starter.

The third condition that allows D_1-D_3 to hold is somewhat more complicated. It states, roughly, that for the ratios $l = \frac{P(br|H)}{P(br|\sim H)}, m = \frac{P(nbnr|H)}{P(br|H)}$ and $n = \frac{P(nbnr|\sim H)}{P(br|\sim H)}$, the following bounds on n must hold:

$$(4.1)\ ml - \frac{l}{K}\log(l) < n < ml$$

When K is the negligibility constant.

While not easy to see at first glance, this condition is completely implausible. When you look more closely at the inequality in (4.1), you see that the length of the interval of permissible values for n is dependent only on the values of l and K and independent of the value of m. Suppose that (say) $l = 10, K = 100$. It follows that the length of the interval of permissible values for n is about 0.23. If $m = 1000$, that is, if conditional on all ravens being black, it is 1000 times more likely to sample a non-black object than to sample a raven, then n must be in

[23] Technically, $\frac{P(nbnr|H)}{P(br|H)}$ just needs to be big compared to $\frac{1}{K}$.

the interval between 10,000 and 10,000 minus 0.23. If $m = 1\ billion$ then n must lie in the interval between 10 billion and 10 billion minus 0.23. This is too exact to be taken seriously. The fact that what remains invariant for various values of l, m, and n is the absolute length of the interval rather than its proportional length makes it even more bizarre. D_1-D_3 severely restrict the set of permissible priors, and in a way that is seemingly hopeless to motivate. At the very least, I can't see any way to motivate it. Anyone who argues for a solution that maintains D_1-D_3 needs to provide us with an argument for why our priors must be so fine-tuned.

I conclude that it is not the specific three assumptions A_1-A_2 of the standard Bayesian solution that get the Bayesians into trouble. It is the very strategy of taking a non-black non-raven to support the hypothesis that all ravens are black, but only negligibly, that does.

5. Conclusion

The standard Bayesian solution fails because according to it, the aggregate contribution of all the non-black non-ravens we expect to see must be non-negligible unless we severely and implausibly restrict the set of permissible priors. However, it is not only the standard Bayesian solution that fails. Any solution that wishes to maintain all of the desiderata D_1-D_3 needs to severely restrict the set of permissible priors in an unmotivated way. Since D_2&D_3 are desiderata for any solution to the paradox of the ravens, it follows that for a solution to work, D_1 needs to be false. A non-black non-raven contributes zero or negatively to the confirmation of the hypothesis that all ravens are black. Since there is no reason to think that a non-black non-raven should *dis*confirm this hypothesis, the conclusion is that the contribution of a non-black non-raven to the hypothesis needs to be zero. For some reason, that will become clearer in the next chapter, non-black non-ravens (as such) are irrelevant to the confirmation of the hypothesis that all ravens are black.

It is time to discuss the accusation that I have been too restrictive with my use of the term 'the Bayesians'. Offering a Bayesian solution to the Paradox of the Ravens, one might protest, does not commit one to adopt D_1. Rather, it just commits one to solving the paradox using considerations regarding our prior probabilities for sampling black ravens and non-black non-ravens in every scenario.

However, we just saw that any plausible solution to the paradox must take the support provided by a non-black non-raven to be zero, and only a very specific set of possible prior probability distributions (namely, those in which $P(nbnr|H) = P(nbnr|{\sim}H)$) give that. There doesn't seem to be any reason our priors must be like that, a fortiori why they should be like that for every F and G the paradox applies to. There must be something else in play here beyond our prior statistical beliefs that makes it the case that non-black non-ravens are irrelevant (and that non-*F*s non-*G*s are irrelevant for *F*s and *G*s the paradox applies to). The paradox will not be solved by statistical considerations alone.[24] If what I said here is correct, the conclusion seems to be that Bayesian considerations alone will not solve the paradox. We need a different strategy.

[24] Rinard (2014) discusses why our priors are and should be such that a non-black non-raven provides zero support to *H*. Although she calls her paper "A New Bayesian Solution to the Paradox of the Ravens", she actually demonstrates my point. There is nothing particularly Bayesian in her solution. What does the work in her solution are metaphysical considerations regarding the hierarchy of natural kinds, rather than statistical considerations. I will not go into the details of her solution here.

Chapter 6: Learning from Examples and the Paradox of the Ravens

1. Introduction

We have seen that the Bayesian approach falls short of solving the Paradox of the Ravens. This chapter offers my solution to the paradox, which is based on considerations of learnability and complexity, that were discussed in Part 2 of this thesis. If confirmation is essentially about what can be learnt from evidence, then the paradox is essentially about what can be learnt from various kinds of examples. Focusing on Hempel's specific case, it is about what can be learnt from examples of ravens vs. what can be learnt from examples of non-ravens.[1]

Intuitively, we think there should be an asymmetry in what can be learnt from examples of ravens and examples of non-ravens regarding the colour of ravens, and thus there should be an asymmetry in the evidential import of seeing ravens and seeing non-ravens. I will try to make this intuition more precise. My analysis will show that indeed, there is an asymmetry in the evidential import provided by examples of ravens and examples of non-ravens. The asymmetry lies in the difference between ravens and non-ravens themselves, as objects, rather than between the concepts *raven* and *non-raven*. There is nothing in the concept *non-raven* that makes it problematic or "unprojectable" (Quine, 1969) in any way.

In the next section, we shall see a toy example of a learning algorithm that naturally discriminates between the evidential import of positive and negative examples of certain kinds of concepts, called *conjunctive* or *disjunctive* concepts. This serves as a proof of concept against assuming automatically that positive examples and negative examples need to be symmetrical for any learner in the evidential import they provide. In §3, I argue, more generally, that in some cases, there *must* be an asymmetry in the evidential import of positive

[1] And examples of black vs. non-black objects. For reasons that will become clearer later, my solution focuses on ravens and non-ravens.

and negative examples for realistic learners, and specifically, that there must be an asymmetry between examples of ravens and examples of non-ravens. In §4, I explain how this helps to solve the paradox.

In §5, I explain what went wrong with the premises that led to the Paradox of the Ravens, namely, Nicod's Condition and the Equivalence Condition. Nicod's Condition and the Equivalence Condition are false because they completely ignore considerations of complexity. We shall see that Nicod's Condition and the Equivalence Condition imply extremely strong (and implausible) consequences regarding the possibility of strong encryption and the computational capabilities of realistic learners. Nicod's Condition and the Equivalence Condition get their intuitive appeal by considering how they apply to examples that are not very complex. However, amending the Equivalence Condition, and especially Nicod's Condition, and restricting their scope only to simple examples does not preserve the paradox.

2. A Toy Example - The Elimination Algorithm

Meet *the Eliminator*. The Eliminator's task is to determine, for some concept *C*, whether an object is *C* or non-*C*, based on the properties the object possesses. Suppose there are *N* properties in the property space, call them $P_1, P_2, \dots, P_N$. Let us assume that for each object and each property, the Eliminator can know whether an object possesses the property when the object is presented to the Eliminator. The Eliminator gets examples of objects together with a label stating for each object whether it is a *C* or a non-*C*.

The Eliminator is a learner equipped with a simple learning algorithm called *the elimination algorithm* (see Valiant, 2013, chapter 5). Here is how it works: the Eliminator starts with a list *L,* which, before seeing any example, contains 2*N* items – the *N* properties and their negations:

$P_1, P_2, \ldots, P_N$ and $\tilde{P}_1, \tilde{P}_2, \ldots, \tilde{P}_N$.[2] Then, for each new labelled object o, the Eliminator acts according to the following rule:

> *If o* ***is*** *C, delete all the items in L that o* ***does not*** *have. If o* ***is not*** *C, delete all the items in L that o* ***does*** *have.*

Upon seeing the first object, then, the list L will necessarily contain N items, and the more objects the Eliminator sees, L's length decreases or remains the same.

To see the algorithm in action, suppose for a moment that we have only five properties in our property space. The Eliminator starts with the list $L_0 = P_1, P_2, P_3, P_4, P_5, \tilde{P}_1, \tilde{P}_2, \tilde{P}_3, \tilde{P}_4, \tilde{P}_5$. Let the first object, o_1, be a *positive* example with the properties $P_1, \tilde{P}_2, \tilde{P}_3, P_4, \tilde{P}_5$. After seeing o_1, we delete the properties o_1 *lacks*, and the updated list is $L_1 = P_1, \tilde{P}_2, \tilde{P}_3, P_4, \tilde{P}_5$. If the next object o_2 is a *negative* example with the properties $\tilde{P}_1, P_2, \tilde{P}_3, P_4, \tilde{P}_5$ then we delete the properties o_2 *has*, and the list is updated to $L_2 = P_1, \tilde{P}_2$, and so on.

So far, I have described only how a list is being produced. However, creating such lists can be very useful for learning specific types of concepts, which are, in effect, quite like lists of properties. A concept C is called a *conjunctive concept* (relative to a property space) if being a C is equivalent to satisfying a conjunction of a list of properties and their negations from the property space. For example, if for every object o, o is C if and only if $P_{13}(o) \wedge P_{30}(o) \wedge \tilde{P}_{31}(o) \wedge P_{58}(o) \wedge \tilde{P}_{92}(o)$, then C is a conjunctive concept. Conjunctive concepts, therefore, are lists of separately necessary and jointly sufficient conditions – the kind of concepts philosophers tend to like. For instance, if *man* and *married* are taken to be basic properties, then *bachelor* is a conjunctive concept, where satisfying each of the conjuncts – *man* and *married*, is a necessary condition for an object to be an instance of *bachelor*.

Similarly, a concept C is called a *disjunctive concept* if being a C is equivalent to satisfying a disjunction of a list of properties and their negations from the property space. Thus,

[2] Nothing crucial changes if we assume objects can lack both a property and its negation.

disjunctive concepts are lists of separately sufficient and jointly necessary conditions. For instance, *non-bachelor* is an example of a disjunctive concept, because to be a non-bachelor just is to be non-man *or* non-married.

It is provable (ibid., p. 70) that conjunctive and disjunctive concepts are PAC-learnable by the elimination algorithm (for the concept of PAC-learnability, see Chapter 3). That is, if *N* is the number of properties in the property space (and thus there are 2^N possible combinations), for every $1 > \varepsilon, \delta > 0$, a learner using the elimination algorithm can, with probability $(1 - \delta)$, learn a list that for a randomly sampled object would lead to an error in less than ε of the cases, when the required computational time and number of examples are at most polynomial in $\frac{1}{\varepsilon}, \frac{1}{\delta}$ and *N*.

There is a twist, though. For the elimination algorithm to be useful for (PAC-)learning *conjunctive* concepts, the Eliminator needs to be shown only *positive* examples. To learn *disjunctive* concepts, the Eliminator needs to be shown only *negative* examples. To get an intuition why, recall that every property (or its negation) that a *positive* example of a *conjunctive* concept *C fails* to have is not *necessary* for being *C*, so we want to delete it from the list of necessary conditions for being *C*. Similarly, every property (or its negation) that a *negative* example of a *disjunctive* concept *C' fails* to have is not *sufficient* for being *C'*, so we want to delete it from the list of sufficient conditions for being *C'*.[3]

Now, suppose, only for the sake of the example, that the concept *raven* is a conjunctive concept. We can think of it as a finite list of separately necessary and jointly sufficient conditions an object must satisfy to qualify as a raven. For instance, the list can be something

[3] We see that it is not sufficient for the Eliminator to only build a list. The Eliminator then needs to know whether the list should be used as a list of necessary conditions or as a list of sufficient conditions. Furthermore, one might feel uncomfortable from restricting the set of examples to only positive or only negative examples. We can consider, if we want, the Eliminator 2.0, that runs two elimination sub-algorithms in parallel, one algorithm that reacts only to positive examples and does nothing for negative examples and another that does the opposite. Each sub-algorithm tells you whether to treat the resulting list as containing necessary or sufficient conditions. The classification is eventually done according to the algorithm that was more successful during the learning process.

like "have two legs AND have two wings AND be black AND have a beak AND have two eyes above the beak …". The eliminator, using the elimination algorithm, can thus learn the concept raven, and specifically, *that being black is a necessary condition for being a raven*, from a reasonable number of positive examples of ravens (in principle, even two carefully selected examples that agree on all the necessary conditions for being a raven and disagree on all the rest would suffice).

What about the concept *non-raven*? As we saw in the case of *bachelor* and its inverse concept *non-bachelor*, if *raven* is a conjunctive concept, then its inverse *non-raven* is a disjunctive concept (not having two legs OR not having two wings OR…). Since *non-raven* is a disjunctive concept, the Eliminator can learn the concept *non-raven*, and specifically that being non-black is a *sufficient* condition for being non-raven, from a reasonable number of negative examples of non-ravens. However, and this is the key point, negative examples of non-ravens are, well, ravens. That is, both the concepts *raven* and *non-raven* and the specific claims that all ravens are black and all non-blacks are non-ravens can be learnt by the Eliminator from examples of ravens but cannot be learnt by the Eliminator from examples of non-ravens.

We see that a concept and its inverse might have different structures (relative to the property space) and therefore need not be learnable from examples in a symmetrical way. Second, and more importantly, we see that there need not be a symmetry in the *evidential import* provided to a learner by seeing objects from one set and seeing objects from its complementary set. In our toy example, ravens are crucial for learning both the concept *raven* and the concept *non-raven*, and specifically for learning that ravens are black and non-blacks are non-ravens, while examples of non-ravens contribute nothing to the learning process.

3. Generalising: Structure and Lack of Structure

This was a toy example. Actual learners usually have much more complicated learning algorithms. In the case of natural learners (as opposed to artificial ones), most of their learning algorithms are still unknown. Furthermore, the concept *raven*, like most of the concepts we use, is likely not a conjunctive concept,[4] and therefore, *non-raven* is likely not a disjunctive concept. We do not think that being black is necessary for being a raven.[5] However, the point that a concept and its inverse need not have the same structure and therefore need not be equally learnable for every learner from positive and negative examples still holds.

Even if *raven* is not a conjunctive concept, assuming that learning is a computational process and that data externalism is true, there is an asymmetry in what can be learnt by realistic learners from examples of ravens and examples of non-ravens. Without loss of generality, we can think of a learner l (not to be confused with the list L from the last section) as a learning machine with reasonable computational resources that is trained on picture files depicting ravens or non-ravens.[6] For simplicity, let's say that every picture that contains at least one raven is a picture depicting ravens, and every picture that does not is a picture depicting non-ravens. Suppose each picture is encoded as a binary string of size s (in bits), and that s is pretty big. For most phone cameras today, s is in the order of magnitude of tens of millions, but the argument still stands for much smaller lengths. This means that the set of all possible pictures is 2^s, an inconceivably huge number.

Let us examine, first, what it would take for l to gain the ability to classify pictures as depicting ravens or non-ravens with low enough false positive and false negative rates.

[4] Since Wittgenstein's (1953/2009) and the work of Putnam (1973) and Kripke (1980) most of our concepts are not believed to be conjunctive.
[5] When we say that all ravens are black, we do not mean that very strictly. We know that there are albino ravens and it does not seem to affect the way we think about the paradox.
[6] Nothing substantive changes in the argument if we think of humans learning from visual and other kinds of data, provided that humans encode information about the environment and run computations on their data.

Mathematically, unbounded learners can just enumerate all the 2^s possible examples a priori and simply memorise one by one, which count as depicting ravens and which do not.[7] However, we assume that *l* doesn't have the amount of time and memory required for that. *l* must see a relatively small number of examples and learn from them how to classify whether a picture depicts ravens, regardless of whether it was one of the examples provided to *l* or not. For that, *l* needs to learn some characteristics that are shared by all or most of the strings of pictures depicting ravens or the strings depicting non-ravens.

Absent any restrictions, membership in any arbitrary set among the power set of the 2^s possible strings, i.e., membership in any arbitrary collection of strings, can be such a common characteristic. That is, every first-order predicate, no matter how gerrymandered or 'gruesome' can serve as such a shared characteristic. However, and here is a key point, to learn such shared characteristics, storing these characteristics must require no more storage than *l* has, and learning them from *l*'s examples must take no more time than *l* has.

The limitations of realistic learners dramatically restrict which shared characteristics can be learnt by them. Since there are 2^s possible strings, we can order them somehow (lexicographically, say), and represent each predicate with a binary string of length 2^s, that has '1' in the i_{th} bit if and only if the i_{th} string is in the extension of the predicate. To classify whether a string is in the extension of the predicate, we need a classification algorithm that given the i_{th} string gives us the i_{th} bit of that binary representation of the predicate, but only a tiny fraction of the strings of length 2^s have short classification algorithms. For a string to have a short classification algorithm, it needs to have low *uniform complexity* (introduced in §4.6.iv), and as in the case of the standard notion of algorithmic complexity, and for the same reasons, most strings don't have low uniform complexity. Thus, qua arbitrary collections of strings or

[7] Perhaps, though, they must be fed with the labels from the outside, to know what the word "(non-)raven" applies to.

possible pictures, most predicates are not realistically learnable, so most predicates cannot serve as shared characteristics of ravens or non-ravens for the learning process of realistic learners.

I suspect, though, that most predicates are not learnt as arbitrary collections of strings or possible pictures. Rather, they are learnt *from* examples that satisfy them. In our case, a predicate can be learnt from the specific array of bits, or pixels, in pictures that are given as inputs. A bunch of pixels in one picture can share characteristics of the kind that can be stored in a realistic learner's memory with a bunch of pixels of another picture. For that, the binary representations of these shared characteristics need to be very short or very compressible. Let's use the term *patterns* for such shared characteristics that are also short or compressible. If some pictures share patterns, then there should be a short classification algorithm that can check whether these patterns can be found in a given picture.

Note that if every string can encode every picture, then all pictures share some learnable patterns with one another under *some* encoding scheme. However, if data externalism is correct, then any given picture can be encoded only by some strings and not others. This is where data externalism becomes important. If not everything can encode everything, then it's not automatic that all sets of possible pictures share patterns, a fortiori learnable patterns. In fact, the opposite is the case.

Most of the 2^s possible pictures are algorithmically random (or nearly random) arrays of bits. To us, they will look like the screen of a broken old TV, as in Figure 6.[8] There is no common pattern among all random arrays of pixels. Importantly for us, random arrays of pixels do not depict ravens, that is, they are examples of non-ravens. Therefore, there is no common pattern shared by all or most of the pictures of non-ravens.

[8] In fact, the picture in Figure 6 is not random but pseudo-random. That is, we believe that there is no feasible computational process that can distinguish it from a random picture. It is an encrypted version of the picture in Figure 7.

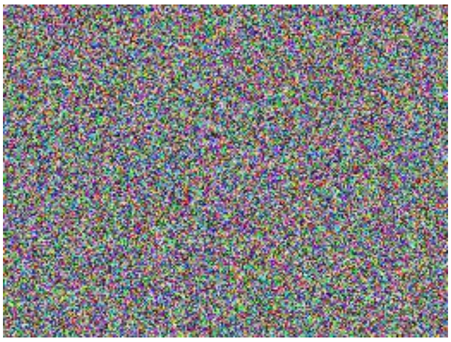

Figure 6: A random array of pixels

Pictures of ravens, on the other hand, are highly non-random. There are recognisable patterns that are shared by pictures of ravens like the one in Figure 7 (eyes, wings, legs, black feathers, etc.). Let's use the term *structure* for a package of patterns that are usually bundled in our examples. We can use *C structure* to refer to a structure that is associated with a concept *C*, and specifically, we can then refer to the package of patterns that characterise pictures of ravens as *the raven structure*. Pictures of ravens, or at least most of them, share the raven structure, or significant parts of it. What pictures of non-ravens have in common is not any pattern or structure, but rather that they *lack* the raven structure.

Figure 7: A non-random array of pixels. Source: Common Raven - Wikipedia. Picture by Fernando Losada Rodríguez.

Thus, there is an asymmetry between pictures of ravens (that share the raven structure) and pictures of non-ravens (that share the lack of the raven structure). For *l* to learn how to classify (non-)ravens, *l* needs to learn a classification algorithm that determines whether a picture has the raven structure (or significant parts of it) or lacks it. The question now is what role examples of ravens and examples of non-ravens can play in learning an algorithm for classifying objects as ravens or non-ravens. To learn the algorithm from examples of ravens, *l* needs to be able to learn to detect the raven structure or its lack from examples that have it. To learn the algorithm from examples of non-ravens, *l* needs to learn to detect the raven structure or its lack from examples that lack it.

In the former case, there is nothing particularly problematic. *l* can use some pattern recognition algorithm, and if learning the raven structure using this algorithm is not too demanding, then *l* can learn the raven structure in a reasonable time from a reasonable number of examples. However, in the latter case, *l* has two options. Either *l* learns that *l*'s examples share a lack of a specific, not previously learnt structure, or *l* learns all possible structures except for the raven structure and classifies any picture it detects to contain a previously unlearnt structure as depicting a raven. The latter option is likely infeasible because there are too many structures for a realistic learner to learn all of them except for one. Furthermore, likely not all structures are instantiated in *l*'s data, which contain only a reasonably small number of examples. Thus, the latter option, like the former one, requires *l* to learn structures from examples that lack them.

Can realistic learners learn structures from examples that lack them? Intuitively, it seems that, in general, it should be much harder to learn a lack of structure from examples that lack it than to learn a structure from examples that have it. If it were easy, in general, to do the former, then *l* could, in general, learn any *possible* lack of structure from practically any picture, including completely random ones. Since not lacking a structure is equivalent to having a structure, *l* could, in general, learn any structure from practically any picture. Moreover, since generating random pictures is an easy task, *l* could just generate some random pictures a priori, and, in general, learn any structure from them, and thus all structures would be learnable a priori, and seeing examples would become inessential for learning. In contrast, learning a structure from examples that have it only requires the learner to detect an existing pattern *actually* presented to the learner.

There are strong reasons to think the intuition above is correct. If, in general, learning a structure from examples that have it is comparable in hardness to learning a structure from examples that lack it, then strong encryption is impossible. One of the requirements for strong encryption is that encrypted data are indistinguishable from random data for learners with only reasonable computation time available (see §2.3). Suppose there is a strong encryption function E. Now, take a set S of all the pictures that share a specific common structure (e.g., they can be all the pictures whose pixels are of the exact same colour as the ones in Figure 8) and encrypt them using E. The encrypted pictures (the ciphertexts) also share a structure. They are all members of the image of S under E. If strong encryption is possible, then without access to the key, *l* cannot learn from the encrypted pictures that the encrypted pictures share a structure.

Figure 8: Completely monochromatic pictures

Suppose, though, that *l* can learn this shared structure from pictures that lack it. Then *l* can use that to distinguish between random pictures and the encrypted pictures. First, *l* can notice that *l* can learn the required structure from the random pictures but not from the encrypted ones. Second, *l* can, after learning the structures from the random pictures, use that to distinguish between the random pictures and the encrypted ones without access to the key, contrary to our assumption.[9]

[9] One may suggest that maybe structures that are not realistically learnable from examples that have them (like the one shared by the encrypted pictures) should also not be realistically learnable from examples that lack them. However, other structures should be easily learnable both from examples that have them and examples that lack them. It would be interesting to see the details of such a proposal and motivations for it. In any case, any structure

Therefore, if strong encryption is possible, then learning the lack of structures from examples that lack them should be inherently harder than learning structures from examples that have them. Thus, learning the raven structure from examples of non-ravens should be inherently harder than learning the raven structure from examples of ravens. This is a common phenomenon. If being a *C* is instantiating a certain *C* structure that is feasibly learnable from examples, then being a non-*C* is *failing* to instantiate this *C* structure. If the *C* structure is not learnable a priori, then in most cases the only way for *l* to learn both the concepts *C* and *non-C*, or to classify whether an object is *C* or non-*C*, is from positive examples of *C*. That is, many times there is an asymmetry in the evidential import of positive examples and negative examples of a concept, because many times only examples of one type contain information that can be feasibly used to learn both a concept and its inverse.[10]

4. Implications for the Paradox

So far, I have talked only about learning to classify whether an object is a positive or a negative case of a concept (e.g., whether a picture depicts ravens or not). The Paradox of the Ravens is about learning the colour of ravens, not about learning the concept *raven* or learning how to classify whether something is a raven. However, the relevance to the paradox is straightforward. First, if being black is part of the set of patterns that compose the raven

that is learnable from examples that lack it is learnable a priori, so a consequence of any proposal of this sort is that everything that is learnable from examples is also learnable a priori, which is implausible.

[10] To be more accurate, carefully selected examples of non-ravens can lead to learning the raven structure. For instance, examples of a lion with a raven head, a horse with raven wings, a fish with raven legs etc. might suffice for a learner to construct a chimera structure which is identical to the raven structure. More crudely, a raven head by itself is not a raven and ditto for other proper parts of ravens. The pieces of a raven jigsaw puzzle, each of which does not by itself depict a raven, even give instructions of how to form a complete raven from them. Supposedly, then, if one can learn the raven structure by observing pictures of ravens, then one can learn the raven structure by observing the jigsaw puzzle pieces of all these pictures, provided that putting the pieces together is not too computationally demanding. What we should say is that *l* cannot learn the raven structure from a random sample of examples of non-ravens, or even a non-random sample that is not carefully selected to allow *l* to easily learn the raven structure from them.

structure, then learning the raven structure is thereby learning that all ravens are black or at least that being black is part of what characterises a typical raven. If we can learn the raven structure only from examples of ravens, then initially we learn that ravens are (at least typically) black only from examples of ravens.

Another option[11] is that being black is not part of the raven structure, and what we learn when we learn that ravens are black is that the raven structure is usually associated with another property of being black, which is not part of the structure. Here, too, there is no reason to think that there is any symmetry in what *l* can learn about this from positive and negative examples of ravens. In the case of positive examples, *l* needs to associate a specific instantiated structure with a specific instantiated property. In the case of negative examples, *l* needs to associate the lack of a specific property, among all properties the example lacks, with the lack of a specific structure, among all structures the example lacks. Absent any cheats, the latter is probably infeasible for realistic learners, because there are too many properties and structures any given example lacks.[12]

As we saw, there are some fundamental asymmetries between learning from examples that have a structure and learning from examples that lack it, and we can't just assume without argument that after a structure is learnt the asymmetries evaporate. In any case, learning from lack of structures requires different learning processes from learning from examples that instantiate structures. Thus, at the very least, there is no mystery in *specific* realistic learners

[11] Which in the case of ravens, specifically, is somewhat implausible, though might be the case in other examples.
[12] Developmental psychology provides some idea of how the difference between structure and lack of structure plays a role in human learning as well as empirical support that it does. The leading developmental psychologist Susan Carey discusses this topic in her (2009) book (especially in ch. 13). Empirical studies reveal the important role that prototypes play in categorisation decision. According to Carey, "[t]he existence of prototypicality structure and its importance in the process of categorization are absolutely beyond doubt." (pp. 496-497). She argues, however, that "[m]any complex concepts do not even have prototypes or places in theories. We do not categorize the entities that fall under many complex concepts by matching current exemplars with stored representations of exemplars of those entities (for we often have no stored representations of exemplars). An example is *not a tiger*." (p. 511). If we can develop a prototype for *raven* but not for *non-raven* (partly due to reasons explained in this chapter), and if learning about either ravens or non-ravens (as such) is done mainly by enriching the prototype structure we have for ravens, then it is clear why examples of ravens can be more relevant than examples of non-ravens for enriching this prototype.

with their specific learning algorithms learning that ravens are black (at least mainly) from examples of ravens rather than examples of non-ravens.

One might insist that for learners like us, who *already* learnt the raven structure and can easily detect whether an object is a raven or not, seeing a non-black non-raven can still provide confirmation for the hypothesis that all ravens are black (or that ravens are typically black). The intuition behind this kind of objection, I suspect, stems from a kind of Bayesian picture. This picture introduces three major simplifications that distort the analysis of the situation we encounter in the paradox.

The first simplification is that like in the urn model, the concepts *raven* and *non-raven* are treated as basic properties, or merely as labels, without any attention to the more basic properties that might be associated with these labels and are relevant to the learning process. For this reason, being a raven or a non-raven is usually modelled as balls drawn from an urn having different colours, and that is it, as if the information you get from an example of a raven and an example of a non-raven is only the label, and is thus entirely symmetrical. The second simplification is that the learner is taken to have prior probabilities about the quantities of ravens and non-ravens in the universe (and black objects, non-black objects, etc.). The third simplification is the assumption that learners learn, or at least should learn, by conditionalisation on prior probabilities.

I agree that in a situation where all three simplifications are correct, seeing a non-black non-raven can confirm the hypothesis that all ravens are black. However, in such situations we get the absurd results of the previous chapter, so clearly this is not our usual situation. In our usual situation, *raven* is a complex concept with a specific structure that is relevant for learning. Furthermore, we don't need to have an idea about the quantity of ravens, non-ravens, non-black objects, etc. in our universe, and any vague idea we might have regarding those quantities doesn't seem to play a role in learning anything about ravens. Given the last point, we can't really be learning about the colour of ravens by conditionalisation because we have nothing to conditionalise on.

One might still contend that we *should* form priors and conditionalise on them. I argue against this Bayesian picture in the next chapter, but before that, we have still some work left in dealing with the Paradox of the Ravens. The paradox was presented as a valid argument. I have rejected its paradoxical conclusion. It's time to revisit the premises that led to the paradox, namely, Nicod's Condition and the Equivalence Condition, and explain what went wrong with (at least one of) them and why they initially seem appealing.

5. Revisiting the Premises of the Paradox

Towards the end of Chapter 3, I presented three adequacy conditions, one for incremental confirmation, one for the stronger notion of non-negligible confirmation, and one for the even stronger notion of robust confirmation. We shall see that Nicod's Condition and the Equivalence Condition, coupled with these adequacy conditions, lead to unexpectedly strong consequences. The stronger the notion of confirmation, the stronger and more implausible the resulting consequences. We shall then discuss whether Nicod's Condition and the Equivalence Condition can be amended in a way that saves the paradox. The answer will turn out to be that the Equivalence Condition can be easily amended, but there is no way to amend Nicod's Condition in a manner that remains compelling and still preserves the paradox.

Nicod's Condition, recall, says that examples of objects that are both F and G confirm the hypothesis that all Fs are Gs. The Equivalence Condition says that if evidence e confirms hypothesis h and h is equivalent to h' then e confirms h' too and to the same degree. It is not explicitly stated what the strength of the notion of confirmation in Nicod's Condition and the Equivalence Condition should be. Interpreting Nicod's Condition and the Equivalence

Condition, somewhat implausibly, as applying merely to incremental confirmation for realistic learners (negligible as it might be) already implies some non-trivial consequences in cryptography. Interpreting Nicod's Condition and the Equivalence Condition as applying to non-negligible confirmation for realistic learners implies that strong encryption is impossible. In Chapter 3, I explained why if all of class $\mathcal{P}$ is PAC-learnable then strong encryption is impossible. Interpreting Nicod's Condition and the Equivalence Condition as applying to robust confirmation for realistic learners (as is most plausible) can be used to derive ridiculously strong results about learnability of the class $\mathcal{NP}$.[13]

The problem starts already with Nicod's Condition. Suppose we intercept some enemy ciphertexts, all of which are from the same source and all decryptable by a key *k*. According to Nicod's Condition, this confirms the hypothesis that all enemy communications from that source can be decrypted using the key *k*.[14] Even if by confirmation we mean only incremental confirmation, this already implies that we should be (maybe marginally) more confident that *k* is the right key.

You might think that this is trivial because by seeing even one ciphertext we can check keys one-by-one, and every time we rule out an incorrect key the confidence that *k* is the key increases (marginally). However, while this might apply to the first ciphertext you get, Nicod's Condition implies that each new ciphertext provides incremental confirmation that the key is *k*, and it's not clear why this should be the case. If confirmation is interpreted as at least non-negligible confirmation, this already implies that each new ciphertext provides non-negligible confirmation that the key is *k*, which even for only the first ciphertext already implies that strong encryption is impossible.

[13] The class $\mathcal{NP}$ was introduced briefly in Chapter 2, fn. 17.

[14] I understand *k* as a rigid designator, denoting a specific number, rather than as a description, like "the correct key". It's trivial and can be learnt deductively that the correct key is the correct key.

To see the problem with the Equivalence Condition, suppose the encrypted enemy messages were encrypted with an encryption scheme that uses a key pair, k_{public} and $k_{private}$. Suppose we intercept each ciphertext together with k_{public}. According to Nicod's Condition, this confirms the hypothesis that the enemy messages were encrypted using k_{public}. Since the hypothesis that the enemy messages were encrypted by k_{public} is equivalent[15] to the hypothesis that the enemy messages can be decrypted by $k_{private}$, according to the Equivalence Condition our evidence equally confirms both, so the same problems from the previous paragraph arise. In this case, since supposedly we have very strong confirmation that the enemy messages were encrypted using k_{public}, we have not only non-negligible but also very strong confirmation that the enemy messages can be decrypted by $k_{private}$.

Suppose, however, as is more plausible, that the kind of confirmation that Nicod's Condition and the Equivalence Condition are about is *robust* confirmation. Consider the following problem, familiar to every philosophy student from their introductory logic classes, called *the Boolean Satisfiability Problem*, or simply *SAT*: given a formula in propositional calculus, is the formula satisfiable (i.e., is there an assignment of truth values to the propositional variables that makes the formula true)? You have probably solved instances of this problem for formulas containing 2 or 3 propositional variables, or maybe even 4 if your logic teacher was particularly mean.[16] It is a whole other matter to solve this problem when there are many propositional variables (e.g., 1,000) because as far as we know, the (worst-case) complexity of the problem grows exponentially with the number of propositional variables.

SAT is the first $\mathcal{NP}$ problem that was proved to be *$\mathcal{NP}$-complete*. That is, every instance of an $\mathcal{NP}$ problem can be translated into an instance of SAT, the solution to the instance of

[15] The hypotheses are mathematically equivalent. Some would want to distinguish between logical and mathematical equivalence, but in such a case, note that the Paradox of the Ravens can be formulated with the Equivalence Condition applying to mathematical equivalence.
[16] Usually, we needed to state exactly which rows satisfy the formula and which don't, which is a harder task.

SAT can then be translated back into a solution to the original problem, and the translations back and forth require at most polynomial time (in the length of the input). Thus, every $\mathcal{NP}$ problem is at most as hard as SAT up to a polynomial factor, so any progress we can make in solving SAT is a progress we make in solving all $\mathcal{NP}$ problems, which is a huge matter.

Now, here is a (parodic) "proof" that you can, in general, learn whether any formula is satisfiable in polynomial time. Take a formula with n variables, denoted by F. Suppose we agreed on the order by which we write rows in a truth table. Then for every formula there is a corresponding truth table. Let's refer to the truth table of F by T_F. T_F has 2^n rows.[17] Let's refer to the set of all natural numbers up to 2^n by A. Let 2^A be the power set of A. The members of 2^A are all the sets of natural numbers up to 2^n. Among all members of 2^A, one member is a set of natural numbers that happens to correspond exactly to the rows in T_F that satisfy F. Let's denote this set by S_F (if F is unsatisfiable then $S_F = \emptyset$).[18]

Now randomly pick numbers between 1 and 2^n, and do it $P(n)$ times where $P(n)$ is some polynomial of n (that can be of low degree with small coefficients). For each number you pick, you can check whether the corresponding row in T_F satisfies F. This takes only polynomial time. There are two options. Either one of the rows you checked satisfies F, and in such a case, you have solved the problem for F, or none of the rows you checked satisfy F. In the latter case, you have examples of numbers, all of which don't correspond to rows that satisfy F, and all of which are not members of S_F. According to Nicod's condition, the above confirms the hypothesis that all and only[19] the numbers that don't correspond to rows that satisfy F are numbers that are not members of S_F. According to the Equivalence Condition, this confirms

[17] Though recall from Chapter 4 that if encoding F requires significantly less than 2^n bits then a string encoding the truth table of F is highly compressible, and F itself can be used for a compression algorithm.

[18] I take S_F to be a rigid designator of a specific set of natural numbers.

[19] Nicod's Condition is not sensitive to the order of the predicates.

the hypothesis that the numbers that correspond to the rows that satisfy F are all and only the members of S_F. The confirmation we get for this hypothesis should be robust.

There is some flexibility in what we can mean by 'robust confirmation', but let's say that *e* robustly confirms *h* for you if you can learn from *e* that *h* should be assigned a high probability to be true. If $S_F \neq \emptyset$ you get a set of numbers that with high probability are the numbers of the rows that satisfy F. You can then even try some of these rows and with high probability you should find an assignment that satisfies F. If $S_F = \emptyset$ then you can learn that with high probability, F is unsatisfiable. So solutions to instances of SAT, and thus to $\mathcal{NP}$ problems in general, can be found with high probability in polynomial time. So easy!

The example is somewhat grotesque, but its grotesquery comes from the formulation of Nicod's Condition, and helps to uncover its deep problem. You might object to the ease with which I let the learner pick the set S_F from all the possible members of 2^A, and rightly so, but it's Nicod's Condition that allows us to do so. In a sense, this is exactly what went wrong with the Paradox of the Ravens in the first place.

Choosing specific sets among all the subsets of objects in the universe and using them to form hypotheses is not a trivial task! It is a computational process, and only some sets of objects, a tiny minority, actually, are such that they can be feasibly selected based on the information provided by the examples we examine. The paradox uses the fact that we, who learnt what ravens are through other means, already made the process needed to be able to select the set of all ravens as one of the lucky privileged sets among the countless many others, and reason accordingly. This hides from us how computationally non-trivial it is to select the set of ravens among the power set of all the objects in the universe, and how special our evidence should be for us to be able to do that.

Because Nicod's Condition and the Equivalence Condition completely neglect all that, the results we get from them are ridiculously strong. In their current form, they cannot be true. At the very least, the fate of such strong results regarding computational complexity theory will not be decided based on vague philosophical intuitions regarding Nicod's Condition and the Equivalence Condition. Can we amend Nicod's Condition and the Equivalence Condition in a way that saves the paradox? And how can we explain their initial appeal?

What went wrong with the Equivalence Condition and Nicod's Condition is that in their original formulation they completely neglect considerations of complexity. If we want to amend this flaw, we likely need to restrict Nicod's Condition and the Equivalence Condition to apply only to cases that are not too complex. Let's examine each one in turn.

In the case of the Equivalence Condition, we should probably say that if evidence *e* confirms hypothesis *h*, and it's within (or maybe safely within) the remaining computational resources of *l* to learn that *h* is equivalent to *h*', then *e* confirms *h'* to the same degree for *l*. This in effect says merely that deduction is a way of learning. The intuitive appeal of this is clear, and I suspect the original Equivalence Condition seems initially appealing by the fact that the cases we have in mind are usually cases where seeing the equivalence is easy.

Note further that the only role the Equivalence Condition plays in formulating the paradox is to transfer confirmation by contraposition. Since it's learnable by undergraduate students that conditionals are equivalent to their contrapositions, we can use the amended version of the Equivalence Condition to play the role the original Equivalence Condition played in deriving the paradoxical conclusion.

Things are different regarding Nicod's Condition. Nicod's condition states that observing objects that are both *F* and *G* confirms the hypothesis that all *F*s are *G*s. A reasonable suggestion might be to amend it slightly by switching the order of "objects" and "that" to get the following condition:

Amended Nicod's Condition: observing that objects are both F and G confirms the hypothesis that all Fs are Gs.

In the case of the encrypted messages, the messages are decryptable by the key *k*, but we do not *observe that* they are decryptable by the key *k*.

By this point, however, we should be very suspicious of what *observing that* means. How does one *observe that* an object is *F*? A reasonable answer is that to observe that an object *o* is *F* is for one to learn that *o* is *F* directly through observation, rather than by other means, such as deduction. Without providing an answer to what "directly through observation" means (which is a very difficult question), it seems to me clear enough that realistic learners don't learn that an object is a non-raven directly through observation, but rather by checking whether a previously learnt structure is instantiated in the object they observed. This seems to be the case not only for the specific example of ravens. Thus, interpreting the amended condition this way doesn't give rise to the paradox.

Another plausible answer is that to observe that an object *o* is *F* is simply for one to learn that *o* is *F*. We can learn about observed objects that they are non-ravens. If we interpret the amended condition this way, however, we cannot ignore the question of *how* we can learn about observed objects that they are non-ravens. The way we learn that an object *o* is both *F* and *G* can affect whether we can learn from this fact that it is more likely that all *F*s are *G*s.[20]

It might be slightly compelling (though not very compelling, given how little we know about learning) that if an object *o* has two learnable structures, the *F*-structure and the *G*-structure, then we can learn from observing *o* that it is more likely that all *F*s are *G*s. I suspect the intuitive appeal of Nicod's Condition stems from considering mainly such cases. But this doesn't give rise to the paradox because non-ravens don't share a structure. The statement that if we learn that an object *o* lacks the previously learnt *F* structure and *G* structure, then we can learn from it that it is more likely that all objects that lack the *F* structure lack the *G* structure,

[20] For example, if a learner learns that *o* is both *F* and *G* by observing that *o* is *F* and then, using the learner's previous knowledge that all *Fs* are *G*s, learns that *o* is also *G*, this doesn't seem to provide further confirmation to the hypothesis that all *Fs* are *G*s

seems much less compelling. In any case, the latter is not symmetrical to the former in terms of what they imply about learnability, so if one accepts the former, one is not compelled to accept the latter in any way.

Can Nicod's Condition be amended in another way that is compelling and preserves the paradox? I suspect not, and at this point the burden of proof is on those who argue otherwise. For an amended Nicod's Condition to be even slightly compelling, it must not apply to cases that are too complex, but any plausible way of restricting Nicod's Condition seems to rule out learning the properties of ravens and other natural properties from the lack of properties in negative examples.

Part 4: Idealisation in Epistemology

Chapter 7: Which Ideal Epistemology?

1. Introduction

In political philosophy, philosophers distinguish between *ideal theory* and *non-ideal theory*. The distinction originated, to the best of my knowledge, in Rawls' *A Theory of Justice* (1971/1999). A normative theory in political philosophy provides norms that prescribe how agents should behave in the political sphere. However, as a matter of fact, not everybody complies with the norms. Are we to theorise about the norms that apply in case everybody complies with the norms, or about the norms given that some fail to comply? If we are doing the former, according to Rawls, then we are doing ideal theory. If we are doing the latter, we are doing non-ideal theory. Today, the terms 'ideal theory' and 'non-ideal theory' are used more liberally to refer to whether a normative theory idealises its subjects and abstracts away imperfections and limitations the subjects might have.

The distinction between ideal and non-ideal theory has entered the epistemological discourse (see Pasnau, 2013; Dogramaci, 2015; Greco, 2023; McKenna, 2023). In a recent paper (Carr, 2022) that already sparked some discussion (Thorstad, 2024; Hughes, 2024), titled "Why Ideal Epistemology?", Jennifer Carr makes a case in favour of the indispensability of ideal epistemology and its theoretical priority over non-ideal epistemology. While not committing to a specific definition of ideal epistemology, Carr characterises ideal epistemology as "concerned with questions about what perfectly rational, cognitively idealised, computationally unlimited believers would believe" (Carr, 2022, p. 1132). She observes that:

> "Often this involves presupposing or defending epistemic norms that, arguably, no actual humans can satisfy: norms mandating logical omniscience; consistency and closure of binary beliefs; infinitely precise credences; credences satisfying the Kolmogorov probability axioms; updating by conditionalization; immediate updating

(rather than temporally extended reasoning); closure of doxastic attitudes under Boolean operations; ... and so on" (ibid.).

Carr argues that ideal epistemology is at least in one respect superior to non-ideal epistemology. Only ideal epistemology is normatively robust in the sense that it is not conventional or context-dependent. Thus, only ideal epistemology "carves nature at the normative joints" (ibid. p. 1134), and only by doing ideal epistemology can we get a clear grasp of the epistemic norms.

Carr's use of the term 'ideal epistemology' is somewhat ambiguous. On the one hand, the term 'ideal epistemology' can be understood broadly as any epistemological theory that involves some idealisation. On the other hand, the term 'ideal epistemology' can be understood more narrowly as epistemology of those ideal unbounded believers, or, as I call them henceforth, *ideal unbounded learners*, roughly characterised by Carr. Carr doesn't seem to distinguish between the two, and this can induce confusion, which ultimately, I argue, infects Carr's argument.

Note that not only normative imperfections are abstracted away when idealising. In science, ideal models are used to simplify matters and gain a better understanding of the distinct components of a system. Such idealisations, which Carr refers to as *descriptive idealisations* (as opposed to *normative idealisations*), are also common in epistemology. I discuss this distinction at length in the next section.

Let me start by first clarifying what I do *not* argue in this chapter. I'm *not* going to argue against the need for any idealisation in epistemology. On the contrary, the model of realistic learners, presented in chapter 3, is an idealised one. Almost any theorising involves *some* idealisation. As Robin McKenna says, "It's hard to make sense of 'fully non-ideal' theory… Idealisation and abstraction is a necessary part of theorizing" (2023, p. 20). Particular cases are distinct from one another, and a general theory usually needs to abstract away some features of particular cases to separate the signal from the noise, so to speak.

However, idealisation is not always a binary or one-dimensional matter. In some cases, epistemology included, there are many ways to idealise, and idealisation can come with a variety of strengths. What I *do* argue in this chapter is that Carr's argument in favour of "ideal epistemology" applies not only to her narrow conception of ideal epistemology as epistemology of ideal unbounded learners but also to many other idealised epistemological models. This fact is overlooked by Carr due to a gap in her argument. At the same time, there are good reasons to think that for most of what interests us in epistemology, the model of ideal unbounded learners is not a good idealisation, because it over-idealises. Other models, like the one of ideal realistic learners, are better suited for our needs and have the same advantages Carr argued that only the former model has.

To be more precise, let's refer to Carr's narrow use of 'ideal epistemology' as *epistemology of ideal unbounded learners*, and let's save the term 'ideal epistemology' to refer to any epistemology that involves some idealisation in the process of determining the epistemically normative judgments, as in Carr's broader use of the term. Using this terminology, we can summarise my argument in this chapter as follows: First, Carr might be right that only ideal epistemology is normatively robust, but she's wrong in saying that only epistemology of ideal unbounded learners is normatively robust. Second, doing epistemology of ideal unbounded learners is probably not the right way to go, and anyway, not the *only* way to go.

In the next section, I discuss some issues related to idealisation in general. In §3, I argue that the model of ideal unbounded learners is not a good idealisation for studying the norms that apply to realistic learners like ourselves. Sections 4-7 present Carr's argument and its problems and discuss (in §7) what kind of 'can' is implied by the epistemic 'ought'. In §8, I explain how the preceding discussion helps us understand what went wrong with Bayesian solutions to the Paradox of the Ravens.

2. Idealisation

The claim that idealisation is not always binary or one-dimensional is not new. David Enoch (2018, p. 12) argues similarly in the context of political philosophy. I hope to convince the reader that Enoch's point generalises to epistemology as well, and given that, the question of what we are trying to idealise and why becomes crucial. To illustrate his point, Enoch gives the example of a theory of punishment:

> "[T]he theory of criminal punishment is, of course, a part of non-ideal theory in the most general sense, because under the assumption of full compliance with political justice and perhaps with morality as well, no one should ever be punished (because no one ever commits a crime). But we can still usefully and importantly distinguish between ideal and non-ideal theory of criminal punishment" (ibid.).

Imperfection is a constitutive component not only of a theory of punishment, but for other domains of inquiry as well, including epistemology. Epistemology, the study of how one comes to know, or justifiably believe, or bring oneself to adopt virtues that promote learning, is inherently about imperfect subjects. From an epistemological point of view, an absolutely perfect subject knows everything, but the epistemology of epistemically perfect subjects is as vacuous as a theory of punishment for perfectly moral agents.[1]

Thus, epistemology is inherently about *learners* – subjects that are imperfect in being at least partly ignorant but can improve their epistemic standing by learning. Any ideal epistemology is about partly non-ideal learners. On the other hand, as we said above, some idealisation is almost always required for theorising. The question we should ask, then, is not

[1] Carr specifically addresses this point (2022, pp. 1137-1138), from the perspective of why ideal epistemology doesn't treat omniscience as a norm for rationality or non-omniscience as irrational. She argues, first, that epistemic internalists can hold that rational status supervenes on the learner's internal state, so the level of the learner's knowledge is irrelevant for rationality. Second, she argues that it's not clear that an omniscience norm can have a coherent interpretation. I don't find her responses very satisfying from her perspective. However, since it's not going to play a role in my argument and I agree with her bottom line, I won't discuss her responses here.

about whether we should do the epistemology of ideal or non-ideal learners, but rather which idealisations are motivated or permissible for which epistemological purpose and why.

This can vary. Some idealisations can be permissible in one case when theorising about one matter, but impermissible in another case, when theorising about another matter. When studying objects in free fall near Earth, it is sometimes permissible to neglect air resistance, or in other words, to make an idealising assumption that there is no air resistance. When studying aerodynamics, neglecting air resistance defeats the whole purpose (as Kant's dove may find out).[2] Air resistance is part of the essence, or the core, of the questions we are trying to answer in aerodynamics. Ditto for neglecting friction when designing a car's braking system.

Similarly, in epistemology, some idealisations can be permissible in some cases, when dealing with some questions, and impermissible or even defeat the purpose in other cases, when dealing with other questions. In any case, any idealisation should be motivated. It needs to be clear what we gain from the idealisation and what we lose, and what the implications of the idealised case are to non-idealised cases.

In this regard, it is helpful to introduce Carr's distinction between *descriptive idealisation* and *normative idealisation* (Carr, 2022, p. 1134). Epistemology is *descriptively ideal* if it involves simplifying assumptions. These simplifying assumptions are meant to "abstract away from complicating extraneous factors" (ibid.), just as in physics we talk about idealised point particles, frictionless planes, perfectly rigid bodies, completely closed systems, and so on. Carr doesn't give a lot of details regarding how epistemology is descriptively ideal. She mentions talking about learners that are computationally unbounded, or capable of instantaneous belief revisions, as useful idealisations in the descriptive sense (which doesn't prevent them from being important idealisations in the normative sense as well). Presumably, according to Carr, not only is it simpler to neglect computational bounds and non-instantaneous thinking

[2] I thank David Enoch for the example.

processes when doing epistemology, but also, we are allowed to treat these as extraneous factors.

A lot has been written about idealisation in science.[3] Here is not the place for a comprehensive discussion of the topic. For what follows, it is useful to note two important features that are shared by most accounts of idealisation in science.

The first feature is what I call *separability*. The non-ideal system should be separable into its ideal components, which survive the idealisation, and its non-ideal components, which are abstracted away. The non-ideal components either don't make a difference and can therefore be discarded (this is how they are treated in Strevens' (2011) Kairetic account of idealisation or in what Weisberg (2007) calls *minimal idealisations*) or can be added later to make corrections to the predictions of the idealised model. We can do Newtonian mechanics in a frictionless plane and then add the friction later as a further component.[4] This process of adding back non-ideal components and making our model less ideal is called *de-idealisation* in the literature.[5]

The separability of the ideal and non-ideal components of a system, and sometimes, when needed, the *additivity* of the non-ideal component back to the model to correct for non-ideal disturbances, does two things: it significantly simplifies the calculation with no loss of predictive powers, and it provides a better understanding. It helps us explain the phenomena by understanding the separate contribution of each factor, and perhaps by isolating the factors that make the difference for the explanandum. Maybe more importantly to our discussion,[6] when the non-ideal components can be separated from the ideal ones (and perhaps reintroduced

[3] McMullin (1985), Sklar (2000), Weisberg (2007), Strevens (2011, Ch. 8), and Norton (2012) are some notable contributions.
[4] See Galileo's remark in McMullin (1985, p. 267) and McMullin's discussion of Bohr's model of the atom. See also Weisberg's discussion of Galilean idealisations.
[5] De-idealisation by no means needs to be a simple matter (Knuuttila & Morgan, 2019), and the possibility of de-idealising models in science has been attacked recently (Rice, 2019), though even Rice admits that de-idealisability is a feature of most accounts of idealisation. Some writers recently defended de-idealisation in economics (Peruzzi and Cevolani, 2022) and political science (Wajzer, 2024) too.
[6] I thank David Enoch for helping me put things this way.

later), it means that doing ideal theory is likely to be *helpful* to doing non-ideal theory. Doing ideal theory, and then maybe making adjustments for the non-ideal case by reintroducing the non-ideal components, is likely to be easier than doing non-ideal theory directly. When there is no separability, this is unlikely to be the case.

The second feature is *approximability*. The divergence between the behaviour of the target system and the behaviour of the ideal model, if such divergence exists, should depend on the strength or intensity of the non-ideal components. The more we weaken the effects that are abstracted away in the idealisation, the closer the actual system should resemble the idealised one. For instance, all else remaining equal, the lower the friction, and the closer the actual surface is to a plane, the closer the movement of a body should be to its hypothetical movement in an idealised frictionless plane. The idealisation would have been unsuccessful if, no matter how small the friction were, the actual bodies, or a significant portion of them, would move very differently from a body in an idealised frictionless plane.[7]

Normative idealisation, on the other hand, is something different. Carr's definition here is unhelpful, due to her ambiguity between ideal epistemology in the broader sense and epistemology of ideal unbounded learners in the narrower sense. She says that ideal epistemology is normatively ideal as it "describes norms of 'ideal', in the sense of perfect or maximal, rationality" (ibid.). This is too narrow, as we shall see.

[7] One might wonder whether these are the only motivations for building idealised models in science. Sometimes, it is claimed, models are not used for simplifying actual systems but rather to "exaggerate or isolate some feature of reality" (Gibbard & Varian, p. 673). Gibbard & Varian call such models "caricatures". Such models can highlight the distinct effect of a specific feature that can be usually masked by the effects of other features or can show that some observed effect is a robust outcome of some very minimal features (see also fn. 12). Greco (2023) argues that models in philosophy are usually more like caricatures than simplifications. I agree that at least sometimes this is the case (Williamson 2013 is an example). It is not one of the motivations proposed by Carr for idealisation, however, and it's hard to say how this kind of motivation can rescue Carr's argument. Presumably, according to Carr, ideal epistemology aims to find or explain the norms that guide our learning processes. It's clear why idealisations that simplify matters in a controllable way can help the epistemologist in theorising. It is less clear why isolating or exaggerating some feature of reality can help us with finding or explaining norms, unless doing so is for some reason of some normative significance. However, this latter case is already covered in the discussion of normative idealisation below.

Let's characterise normative idealisation more broadly as idealisation that is made to uncover or constitute norms or normative judgments. Consider Rawls' veil of ignorance. Rawls thinks neither that actual people can be separated into their non-ideal parts and their ideal parts that are behind a veil of ignorance, nor that actual people can approximate idealised people behind the veil of ignorance. Rather, the point of this idealisation is that the idealised scenario of the veil of ignorance uncovers, or constitutes, what it means for a society to be just. A decision made behind a veil of ignorance is fair *in virtue* of its being made behind a veil of ignorance.

Similarly, in epistemology, we can idealise because we think that non-idealised cases hide the core normative epistemic principles of the case. Various actual contingent facts about the learner, like how much sleep they had last night, might seem extraneous to the question of what the learner ought to believe based on the learner's evidence. Moreover, since lack of sleep introduces mainly errors, which, from an epistemic point of view, are bad, to discover what the learner ought to believe, we should think of a learner who doesn't make errors due to lack of sleep. Additionally, we might think that specific idealised processes are what constitute a norm. For instance, we might think that a belief or credence is rational *in virtue* of being formed in the right way. Idealisation is then motivated for making epistemic judgments because only by examining the ideal situation can we discover the epistemic norms and understand what grounds them.

This can naturally lead to the thought that the epistemic state a learner *ought* to be in is the epistemic state a properly idealised learner in such a case *would* be in. Counterfactuals of this kind notoriously lead to trouble (see Shope, 1978). To use Williamson's example (2000, p. 209), it might be rational to be confident that there are no ideal unbounded learners even if an ideal unbounded learner with access to our evidence would not think that there are no ideal unbounded learners. One can perhaps evade these worries by talking not about the epistemic state an ideal learner would be in, but about the epistemic state an ideal *advisor* would recommend you to be in based on your evidence. The key point is that a fully idealised learner should serve as a guide for finding the epistemic state we ought to be in.

The idea behind ideal theories, in epistemology as well as in political philosophy, is natural. You need to know where you're heading to know what your next step should be, so only by examining the fully ideal can you know how we can better approximate the ideal. The ideal case is simpler, so we can more easily start by theorising about it, and then add some corrections due to non-ideal complications, prescribing the non-ideal learners to approximate the ideal unbounded ones as much as they can, given their imperfections (cf. Enoch, 2018). This seems to be Carr's view (2022, §6.2) regarding epistemology of ideal unbounded learners.

Natural as the thought might be, it is wrong. First, while the fully ideal case can be simple, or even the simplest, it doesn't follow that less ideal cases must be too complicated to theorise about. For instance, the model of ideal realistic learners is less ideal than the model of ideal unbounded learners, but is still simple enough.

Second, this thought exemplifies what Estlund (2020, Ch. 4), in the context of political philosophy, called *the Fallacy of Approximation*. From the fact that in an ideal case one needs to do φ, it doesn't follow that in a non-ideal case one ought to approximate φ. If ideally, you need to take three different medications, but you only have access to two of them, it doesn't follow that you should take the two. The two taken without the third might cause more harm than good. Likewise, in epistemology, ideal unbounded learners are logically omniscient. However, for an actual bounded learner, trying to approximate ideal learners by endlessly spending time deducing propositions that follow from what they know seems more irrational than rational. This should stand as a warning against inattentively falling back on the natural thought that approximating the ideal is always better.[8]

When idealising in epistemology, then, there are various questions we should ask. We need to ask first which questions we are trying to answer. Second, we need to ask what we hope to achieve with each specific idealisation. Is the aim to simplify, as in descriptive idealisation? If so, is it a good simplification that provides a better understanding of the various relevant factors in the actual case? Are we sure that we are not abstracting away some features that are part of the core of the question we are trying to answer? Is the aim to discover normative truths that are hidden behind the intricate details of the actual cases, as in normative idealisation? If so, can we explain why each idealisation is normatively relevant? How can we infer from the ideal case to the non-ideal case, given that approximating the ideal is not always better in the non-ideal case? Keeping these questions in mind, let's examine whether epistemology of ideal unbounded learners involves a good idealisation.

3. Idealisation of What?

If epistemology is inherently about learners, then the norms discovered or accounted for in epistemology are the norms guiding learning attempts.[9] There can be various kinds of learners, though, and in principle, different learners can be guided by different norms. Whether all learners are guided by the same norms or not, the following seems to be clear: the epistemic

[8] Julia Staffel (2019) is a thorough discussion of how, in a Bayesian framework, we can make sense of the idea that a non-ideal learner should approximate the ideal. However, she also acknowledges that if there are constraints on the level of approximation the learner can achieve regarding one epistemic norm, then sometimes the learner would be best not to fully fulfill the other norms. See especially Ch. 6, §3.2.

[9] Of course, by "the norms guiding learning attempts" I mean norms that state what makes learning processes good ones, e.g., norms regarding what inferences to do and when. I do not mean norms about what enables (say) biological learners to be in a state where they are more likely to undergo good learning processes, like norms that guide humans to sleep well and have a nutritious breakfast).

norms *we* try to discover or explain[10] are first and foremost the norms that guide *our* attempts at learning (even if it might turn out that we can never fulfil them perfectly).[11] We are interested in norms for learners *like us*. Learners like us can be idealised in various ways, but if finding or explaining the epistemic norms for learners like us is what we're after, any idealisation is motivated only to the extent that it helps us find these norms.

Thus, the rationale for idealising learners as computationally unbounded in the way Carr suggests, at least so far as we want to discover the norms that apply to us, should be that making such idealisations helps us discover the norms for learning that apply to us. It can help by simplifying the discovery by getting rid of complicating extraneous factors that are irrelevant to the norms of learning, or by providing a setting where epistemic norms are constituted, such that what happens in this setting necessarily accords with the norms due to the features of the setting.

Does Carr's model of an ideal unbounded learner have these desirable features? I shall use this section to argue to the contrary. I first argue that the model is not a good descriptive idealisation. Then I argue that it is not a good normative idealisation, either. The norms it suggests might *coincide* with the norms for learning when all bounds are removed, but they are not the general norms for learning, whatever they may be.

[10] Some might say that epistemology is more about explaining norms or systematizing them, than about finding them. Most epistemologists, it seems, agree on the first order judgments regarding whether a specific method is a good way of reasoning, but they disagree on the best way to explain these judgments in a systematic way. While this might be true about the state of epistemology today, let me just say that agreement among professional epistemologists in 2026 is hardly an agreement among humanity at large in 2026, or an agreement among epistemologists throughout history. When people like Bayes and Laplace made their discoveries, they were (at least in part) doing epistemology, even though they discovered hitherto unknown norms. When a layman tries to convince another layman that perception is a good source of knowledge, but scripture and wishful thinking are not, they are engaging in epistemological discourse even if professional epistemologists don't argue about this anymore. If first-order disagreements between professional epistemologists about which methods are good for learning are rare, it mainly shows that the field has probably made some progress over the years.

[11] Carr seems to fully agree with that. She remarks: "Ideal epistemologists are often asked to defend their research programme. We're often asked: why spell out norms that only apply to fictional creatures in distant possible worlds? The question rests on a false presupposition. The norms described by ideal epistemology are meant to apply to actual humans; we just fail to satisfy them." (p. 1132).

We have mentioned that descriptive idealisations in science have the features of separability and approximability. Idealising away computational limitations clearly doesn't have them. We cannot separate an actual learning process into an ideal component, which is how a computationally unbounded learner would learn, and a non-ideal component, which is the computational bounds, add them together, and then get the actual learning process or even something that vaguely resembles it.

Think of encryption. An unbounded learner can try all the keys one-by-one and break the encryption instantly. Add bounds on computational resources to that, and you get a bounded learner who slowly tries every key one by one until the learner finds the right one. We get a learner that, besides being very stupid, also doesn't, in general, resemble what an actual learner would do in such a case. That's because non-ideal learners (good ones, at least) fit their learning strategy to their computational resources. Computational bounds are not an independent factor that can be separated from what makes a learning process of a bounded learner a good one.[12]

As for approximability, there are important respects in which weakening the bounds on bounded learners doesn't make bounded learners approximate unbounded ones. Take logical omniscience, for instance. It's true that for each formula that follows from a learner's body of knowledge, there exists a level of computational resources such that if the learner had this level of computational resources, the learner would be able to derive the formula. However, at the same time, it's also true that for any finite level of computational resources, there are infinitely many formulas that a learner with that level of computational resources cannot learn. That is, no matter how high the level of computational resources a learner has, if the resources are

[12] This does not imply that idealising away computational limitations is *always* futile. Some of our epistemic imperfections can stem from reasons other than computational bounds, e.g., imperfect perceptual discrimination. Using a model of a learner with limited perceptual discrimination but unlimited computational resources can enable us to isolate effects that are purely due to our imperfect perceptual discrimination from effects that are due (at least partly) to other imperfections as well (see Williamson, 2017). When we have more than one kind of imperfection, idealising away all but one imperfection can make us better understand the unique contribution of each imperfection. One can similarly abstract away perceptual imperfections when trying to isolate the effects of only computational limitations. These are examples of "caricature models" discussed in fn. 7.

bounded, the learner is infinitely distant from logical omniscience.[13] Therefore, there are qualitative differences between bounded learners and unbounded ones. Bounded learners, talented as they may be, have troubles that unbounded learners do not. As a result, it might be rational for bounded learners, even talented ones, to adopt different learning strategies from those of unbounded learners.[14]

The fact that the model of ideal unbounded learners doesn't have the features of separability and approximability already indicates that this idealisation, at least as a descriptive idealisation, is not a good one. I want to say something stronger, though. This idealisation (as a descriptive one, at least) is misguided to begin with, because fitting learning strategies to computational limitations is no less essential part of how bounded learners can learn about the world than air resistance is essential to aerodynamics (indeed, in my model of a realistic learner, the available computational resources are part of what *defines* a learning process).

Under the assumption that learning is computational, the level of computational resources determines which learning strategies are likely to succeed in the available time and which aren't, and consequently, which learning strategies are more rational than others. Deciding between learning strategies is hard and is a major part of what makes a good learning process. An unbounded learner doesn't face such difficult decisions. Learning how to make such decisions from an unbounded learner is like learning how to make decisions regarding what to inquire about from an omniscient being. The rational way to learn is defined partly *in terms* of the available computational resources,[15] in quite the same way as what is rational to inquire about is partly defined in terms of what you don't already know.

[13] In mathematical terms, we can say what is already a recurring theme in this thesis: while an infinite sequence of learners, ordered by increasing level of computational resources, converges to an unbounded learner, the convergence is not uniform.

[14] Note that I don't say that there are no *other* respects in which learners with better computational resources *do* approximate ideal unbounded learners. For instance, more talented bounded learners might know things that less talented learners do not, but ideal unbounded learners do. The point is rather that in *some* important respects there are qualitative differences between unbounded learners and bounded ones, however talented the latter might be.

[15] This does not exclude the possibility that *some* features of learning strategies are desirable irrespective of computational resources, or across a wide range of levels of computational resources.

Abstracting away these decisions between learning strategies removes a crucial component of any learning process of bounded learners, which is an essential part of what constitutes the normative evaluation of that learning process. This sheds light on why the bounds on the learning process are not a separable component that can be abstracted away. The rational way to learn is determined partly by the bounds. Different bounds might call for entirely different learning strategies, so a change in the bounds of a learner changes what's rational for the learner.

Consider the following example. Playing chess can be seen as a learning process. You have your previous chess knowledge and the current position on the board as inputs, and you try to compute the best move you can come up with (that is, what you are trying to learn is what are better or worse moves, or what are some possible consequences of some moves, in a *specific* situation of a chess board). However, in most cases, people play chess with a time limit. When you play 5-minute Chess, you use different learning strategies from when you play 2.5-hour Chess, and you are likely to come up with worse moves. Does this mean that you are less ideal, epistemically, when you play 5-minute Chess than when you play 2.5-hour Chess? This strikes me as implausible. The time limit is an inherent feature of the game. Strictly speaking, 5-minute Chess and 2.5-hour Chess are not the same game (in the game-theoretic sense of the word), given that they have different winning conditions. Using learning strategies that fit 2.5-hour chess in 5-minute chess would result in worse game-playing, not better. It will likely make you lose in just a little more than 5 minutes. Trying to learn as if you're unbounded, or even as if you're much less bounded than you actually are, is not, to Quine's words, "pathetic but praiseworthy". Only pathetic.[16]

[16] To avoid a possible misunderstanding, I don't mean that when you are training on how to become a better 5-minute chess player you cannot or should not play under different conditions or take more time to understand some chess positions. Of course, taking hours to understand a specific position not in the middle of a 5-minute game can make you better at detecting good moves quickly in other positions, and thus make you a better 5-minute chess player. The point is that trying to learn what the best move is in a *specific* position in the middle of a 5-minute game *as if* you are unbounded is pathetic.

Of course, God, who knows in advance every possible position in a Chess game and can make moves instantaneously, would play the same in both cases.[17] I agree that in some sense, the moves God would make define the ideal *moves* in both 5-minute Chess and 2.5-hour Chess. However, *given* computational imperfections and time limits, what's *closer* to the ideal is not determined only by the accuracy of the moves but also by the time it takes the player to make them. If a player runs out of time, the position on the board is irrelevant.[18] A player can try to approximate God also by making the moves as close to instantaneously as possible. Being more or less ideal, for bounded learners, is essentially about making better *trade-offs*. Acknowledging that it would have been nicer if no trade-offs needed to be made teaches you little about how to make trade-offs when you have to make them. I conclude that epistemology of unbounded learners is not a good *descriptive* idealisation of bounded learners.

It is not a good *normative* idealisation either. Idealisation, recall, is a theorising tool. We should ask, then, what a theorist can learn by examining ideal unbounded learners that she cannot (easily) learn otherwise? If the theorist herself is a bounded learner, as all actual epistemologists are, then she can't, by definition, use the idealisation to expand her knowledge beyond what is knowable by a bounded learner. Everything the theorist learns that an ideal unbounded learner would learn had they possessed the theorist's evidence is something the theorist can learn by herself from her evidence. What can be learnt by theorising about ideal unbounded learners, then, concerns less what these learners know, and more *how* they learn. However, as we have seen above, how an unbounded learner learns has at best limited implications for how bounded learners should learn, *given their bounds*.

[17] I assume here that God plays the whole game rather than joining in the middle. In the latter case, given that God may start playing in a non-ideal position, the time limit and the identity of the opponent may play a role in determining the optimal moves.
[18] That's a little inaccurate. If one player runs out of time, but the opponent cannot possibly win, e.g., the opponent has only their king left, then the game ends in draw.

To be more concrete, suppose we learn that ideal unbounded learners update only by conditionalisation, as Carr suggests. Realistic learners might also try to update only by conditionalisation, but given their bounds, realistic learners cannot anticipate every piece of evidence they might encounter, so they don't have conditional probabilities on unanticipated pieces of evidence. Learners can only conditionalise on existing priors. Realistic learners will likely gain by conditionalization only a tiny portion of the knowledge and credences an ideal unbounded learner would get from the same evidence by conditionalisation. Is there a guarantee that other learning processes, e.g., some pattern recognition algorithms, won't provide realistic learners with more, or better, knowledge or credences? I would like to see the argument for that. Moreover, upon seeing unanticipated evidence, it might be rational for the learner to revise their prior conditional probabilities, rather than disregarding the unanticipated evidence and conditionalising on the pieces of evidence the learner did anticipate. Unbounded learners don't need to bother themselves with such earthly matters.[19]

Given bounds, the best way to learn is the one that offers the best trade-off given those bounds. Of course, if there are no bounds, there is no need for any trade-off, so the best way to learn is by not making any trade-offs and learning as an unbounded learner does. Thus, the norms for learning when all bounds are removed collapse to the norms of ideal unbounded learners. Ideal unbounded learners are a special degenerate case of learners in general. But if trade-offs given bounds are an essential feature of the norms of learning, restricting ourselves

[19] Can't we still draw some general morals from the fact that ideal unbounded learners update by conditionalization? E.g., can't it reveal that, or explain why, surprising evidence gives stronger support to hypotheses that predict them than less surprising evidence? First, it can. There is nothing that excludes the *possibility* that epistemology of ideal unbounded learners provides *some* insights that are relevant for bounded learners, too. This is especially the case if we can show that the same moral can be drawn in the less ideal case, and that there is nothing in the fully ideal case that interferes with the reasoning in the less ideal case. In this specific example, though, we need to distinguish between surprise due to evidence with low prior probability and surprise from unanticipated evidence. Ideal unbounded learners can have only the former kind of surprising evidence. When bounded learners get surprising evidence of the former kind, exactly the same considerations apply as in the fully ideal case, so the fully ideal case doesn't reveal anything new. However, in the case of realistic learners, we see that the heuristic of taking surprising evidence as stronger is limited, because there can be cases where unanticipated evidence can defeat the import of the low-prior-probability evidence.

to a degenerate case just *hides* part of the normative picture from us, rather than revealing it. Thus, the model of ideal unbounded learners is also not a good normative idealisation.

Before proceeding, let me state clearly what I *don't* argue. I don't argue that there is no sense of 'ought' in which we all ought to be fully ideal. This is not unique to the epistemic 'ought'. In the moral case, there is a sense in which we all 'ought' to fly at Superman's speed and help everyone in trouble all over the world. We ought to do so in the sense that if we were to do it, the world would be better. Similarly, we ought to be ideal unbounded learners in the sense that if we were, we would be epistemically better off. However, this 'ought' is conditional (on having better abilities), not categorical. What we ought to do, categorically, is what is reasonably expected from us *given* our abilities.

I also don't argue that doing epistemology of ideal unbounded learners is a waste of time or pointless. There is nothing that is in principle wrong with doing ethics of superheroes. It might even be of interest to Marvel or DC fans. More broadly, there is nothing wrong with asking what duties we would have had if we could travel at infinite speed. However, there is a sense in which such superluminal ethics is of much less general interest to physical creatures. Similarly, in epistemology, idealising computational limitations away can lead to a normative theory of super-learners, and this theory can be of interest to super-learners' fans (and we philosophers are called 'philosophers' for tending to be fans of impressive reasoning abilities). However, there is a clear sense in which such a theory is of much less interest to human learners in general.

Let me emphasise again that I don't even say that nothing can be learnt about bounded learners by doing epistemology of ideal unbounded learners. In some carefully selected scenarios (e.g., betting), where the full specification of the scenario is simple, and there is no unanticipated kind of future evidence, we may very well try to approximate the behaviour of ideal unbounded learners. Compare: in a situation where we can easily help a person standing one meter from us, we may very well approximate what Superman would have done in such a case. In a simple chess position, when it is our move and there's a simple mate in one, we may try to approximate God's behaviour in such a scenario, that is, making the right move immediately.

Epistemology of ideal unbounded learners can teach us something about bounded learners, exactly in those situations where the bounds don't play an important role. This is somewhat similar to having a model in physics of bodies that behave according to Newton's laws, except that they have the supernatural ability to teleport randomly at times. These supernatural bodies are not a good idealisation of physical bodies, but in those cases where these supernatural bodies happen not to teleport randomly, we can learn from this model about actual physical bodies.

I believe the argument so far shows that doing ideal epistemology in the way Carr suggests, that is, epistemology of ideal unbounded learners, is not the way to go. However, Carr presented an argument for why only the epistemology of ideal unbounded learners is normatively robust, and this chapter has not yet dealt with Carr's argument. According to Carr, any drop from ideal unbounded learners is bound to make the theory conventional and context-sensitive. Carr is wrong about that, or so I argue.

The next section briefly presents Kratzer semantics for 'ought', which Carr uses as a general framework for her argument. I then explain why Carr's arguments for the non-robustness of the epistemic ordering (§5) and circumstances that are taken as fixed (§6) fail. In §7, I offer a way to understand what kind of 'can' is implied by the 'ought' of epistemology of less-than-perfect learners.

4. Kratzer Semantics for 'Ought'

Kratzer semantics (Kratzer, 1981; 1991) is a general framework for analysing the meaning of graded or comparative modality terms (e.g., 'good possibility' or 'better possibility'), in which absolute terms like 'necessary' and 'ought' are only special cases. This fact will turn out to be relevant in the next section, but for now, let's focus only on Kratzer Semantics for 'ought'. Kratzer semantics includes two parameters, the *modal base* and the *ordering source*. According to Carr, the choice of an epistemic ordering source and a modal base is conventional or context-dependent. This, in turn, renders the epistemology of less perfect learners normatively non-robust.

The modal base is a descriptive parameter. It determines which circumstances are taken as fixed. Formally, it's a set of propositions that must all be true at a world for this world to be accessible from the context of the utterance. The ordering source is the normative parameter. It ranks worlds in terms of ideality. Formally, an ordering source is a set of propositions A. Worlds are ranked based on the following preorder:[20]

$$\forall w, w' \in W, w \leq_A w' \ iff \ \{p \in A: p \ is \ true \ in \ w'\} \subseteq \{p \in A: p \ is \ true \ in \ w\}$$

In words: a world w is at least as close to the A-ideal as a world w' if all the propositions in A that are true at w' are also true at w.

What one 'ought' to do is what one does in the most ideal possible worlds. More formally, given a modal base P and an ordering source A, one ought to φ if there is a P-world v in which one φ's and for every P-world u that is at least as ideal as v, one φ's at u. If the modal base determines what one *can* do, then Kratzer semantics validates the principle that 'ought' implies 'can'. What one ought to do is what one does in the best worlds that are considered possible.

[20] In common terminology, this is a preorder rather than a partial order because it is not antisymmetric. Two distinct worlds w and w' can have the same set of propositions from A true in them, and then both $w \leq_A w'$ and $w' \leq_A w$.

5. Carr's Argument for the Non-Robustness of the Ordering

Carr argues that epistemology of any learner except for ideal unbounded ones is conventional, because there is more than one reasonable epistemic ordering source, and choosing between competing orderings is somewhat arbitrary and a matter of convention. Epistemology of ideal unbounded learners is not conventional, because all reasonable epistemic orderings rank the epistemic situation of ideal unbounded learners as the most ideal. However, in the case of less ideal learners, different choices of orderings lead to different epistemic evaluations, and thus evaluations of less ideal scenarios are conventional and not normatively robust.

Carr provides three distinct (though related) arguments for why more than one reasonable ordering is admissible, and thus why epistemology can't say anything robust in situations where different orderings clash. Rather than focusing on the details of Carr's arguments, I want to focus on a more fundamental problem they all share. What they all have in common is that they show specific cases where reasonable orderings may differ, together with an explanation of why these cases of disagreement are abundant. However, even if Carr is correct, and there are equally reasonable competing orderings which disagree in *some* (maybe many) cases, it doesn't follow that there isn't a consensus between reasonable orderings about a wide range of other cases. From the fact that we can only decide between *some* cases by convention, it doesn't follow that every epistemic comparison of non-perfect situations is conventional.

Granted, in many domains, it is not always easy to agree on the ranking of all non-ideal cases. Who is a better composer – Mozart or Bach? Who is a better football player – Messi or Maradona? It seems that reasonable people may differ. It doesn't follow that there is no fact of the matter here. Maybe we are just facing difficulties in detecting the one correct ordering, or maybe we all refer to the same ordering but are having trouble determining which candidate scores better according to that specific ordering. However, for the sake of the argument, let's suppose that indeed there is no fact of the matter here. Different reasonable orderings provide different verdicts, and which ordering we choose is a matter of convention.

Even in this case, it strikes me as implausible that any reasonable ordering would disagree regarding whether Mozart is a better composer than I or whether Messi is a better football player than I. The superiority of Mozart and Messi over me seems as objective and robust as any normative evaluation can be. Messi's and Mozart's superiority is invariant across all reasonable orderings, since an ordering that says otherwise is unreasonable.[21] This has direct normative implications. If you face a choice between going to a concert presenting Mozart's pieces or a concert presenting my pieces, then by all means (when everything else is equal), you really *ought* to choose Mozart. If you need to choose which football player to pick for your team, Messi or me, then by all means, pick Messi.

The analogy to epistemology is clear. Learners might adopt different ways of learning, and in some cases, we might find it hard to decide which is the best way to learn. All the more so, coming to know which way of learning is the better one might exceed our epistemic capabilities, so we just can't learn which one is better (even if one of them in fact is). Even if we think that in such cases choosing between various ways of learning that seem equally good is a matter of convention, it might still be the case that in any reasonable ordering, all those ways of learning are preferred over wishful thinking, random guessing, and all other methods that are clearly bad.

This highlights an important fact about bounded learners. Bounded learners often face specific choices between concrete ways of learning. Bounded as they are, they are usually not occupied with the full ordering of all ways of learning, but rather with how to choose between concrete ways of learning available to them at that moment. If epistemology can robustly help

[21] Comparing musical talent to football talent might be even harder, but it still strikes me as obvious that Messi is a much better football player than I'm a composer and Mozart is a much better composer than I'm a football player.

in many of these choices, it's not clear that we need, or should expect, much more. Even if epistemology can only tell us robustly that some non-ideal ways of learning are for sure better than some others, so we should not use the latter ones, it has done some service.

It's not clear that epistemology cannot even come up with *general principles* for choosing between available options in cases where the principles apply. Compare to the following plausible principle in decision theory:

> DOMINANT-STRATEGY: Whenever you face a choice between distinct strategies A and B, and A is dominant over B, you *ought* to choose strategy A.

DOMINANT-STRATEGY doesn't apply to every choice between strategies, but when it applies, it supposedly applies to all agents and all strategies, regardless of how ideal they are. Whether true or false, DOMINANT-STRATEGY seems to present itself as objective and normatively robust as any normative principle can be. There might similarly be general and normatively robust principles for choosing between non-ideal epistemic methods. Nothing in what Carr said indicates otherwise.[22]

6. Carr's First Argument for Non-Robustness of the Fixed Circumstances

Carr provides two arguments for the non-robustness of the modal base, which determines what circumstances are taken as fixed. I shall discuss each of them separately in this section and the next. According to Kratzer semantics, you ought to do the best you *can* do, or, equivalently, what you do in the best possible worlds. However, what you 'can' and 'cannot'

[22] Even if, implausibly, there is no consensus about anything between reasonable epistemic orderings, and the epistemic landscape is a complicated structure that just cannot be naturally ordered, it doesn't make epistemology of anything but ideal unbounded learners non-robust. Some structures, like the complex plane, are not naturally ordered. Some, like mountains, are not naturally ordered but still have one privileged point at the top. It doesn't mean we can't say anything non-conventional or not context dependent about them. However, it might mean that concepts that essentially refer to ordering, like (arguably) 'ought', are not fit to describe such structures.

do depends on the context. In one sense, to use Lewis's (1976) famous example, Lewis can speak Finnish. He has the right physiology for that. In another sense, Lewis can't speak Finnish. He has never learnt this language. Which sense of 'can' we mean depends on which circumstances are taken to be fixed, which in turn depends on the context.

Thus, according to Carr, if what one 'ought' to do depends on which circumstances are taken as fixed, what one 'ought' to do also depends on the context. This applies to the epistemic 'ought' as well. In one context (say), some computational bounds are taken as fixed, in another, they are not. Thus, what one 'ought' to do, epistemically, depends on the context. In practice, when doing non-ideal epistemology, we tend to take *some* actual limitations, like computational power and processing speed, as part of the fixed circumstances, and others, like disposition to delusional reasoning, as not. Carr is asking: "Why do some cognitive limitations *lower the bar for rationality* while others *constitute irrationality*?" (Carr, p. 1152). She answers that there is no clear boundary between limitations of the former kind and the latter, so determining where to draw the line is conventional. The only judgment that is not conventional or context-dependent, according to Carr, is what one ought to do when no circumstances are held fixed, and what one epistemically ought to do when no circumstances are held fixed is what an ideal unbounded learner does.

Carr's argument suffers from a major gap here. Indeed, if our use of the epistemic 'ought' at one time holds some circumstances fixed that are not held fixed at another time, then our epistemic judgments are context-sensitive. However, this shows only that for our epistemic judgment to be context-invariant, we need to hold fixed only those circumstances that are held fixed in every relevant context. Carr implicitly assumes that no such circumstances exist, but this is unwarranted. There is a very long way to go from noticing that some bound that applies to one person in one context doesn't apply to another person or in another context, to thinking that all bounds whatsoever depend on context.

Epistemology can discuss the norms for specific natural classes of non-ideal learners (the concept of a class of learners was introduced in Chapter 3). Those classes of learners can share some bounds regardless of context. For instance, suppose we are interested in what physical learners epistemically ought to do. It seems reasonable, then, to take only physically possible circumstances into account. Thus, we can take the set of physically possible circumstances as fixed. If this constraint is in place whenever we talk about 'ought' for physically possible learners, then an 'ought' that always takes all and only the physically possible circumstances as fixed is as context-invariant as an 'ought' that takes no circumstances as fixed. In all actual contexts, and even in all physically possible contexts, the phrase 'the set of physically possible worlds' denotes the same set. What physical learners ought to do, though, is distinct from what ideal unbounded learners ought to do, because ideal unbounded learners can do things that physical learners cannot.

We humans have a parochial interest in what we humans epistemically ought to do. We can fix the circumstances such that the only accessible worlds are those where the learner's capabilities don't exceed, or don't exceed by much, those of the most talented past, present, or future humans, or those of any human aided with the best 2026 computing technology, or those of humankind as a collective throughout all past and future history, or whatever. Since the bound we hold fixed is an *upper bound* on the learning resources of the class of learners we're interested in, there is no risk of context-dependence, because circumstances where members of the class of learners we're interested in (humans, in this case) breach the upper bound are never taken into account.

As I said in Chapter 3, the more concrete the upper bounds we want to impose on our model, the messier and more complicated it gets. Some constraints should be empirically informed, and this means we might pose them incorrectly due to mistakes or ignorance about the actual empirical situation. What this shows is that when you want to make correct epistemic judgments specifically about humans, or about complex empirical phenomena, the lower you want your bounds to be, the harder you need to work. It doesn't show that anything in these bounds is conventional or blurry, and there is no inherent restriction to doing it in a context-invariant way.

Perhaps more importantly, if 'ought to φ' implies 'can φ', then 'cannot φ' implies 'not-ought to φ', or 'permissible not to φ'. While it's indeed hard (given our non-ideality) to say positively how an ideal human *should* reason, it is much easier to examine more ideal models (i.e., with higher upper bounds)[23] to be able to say negatively what *cannot* be epistemically expected from any human. If the bounds we pose are upper bounds for sure, there is nothing conventional in saying that bounded as we are, we cannot be reasonably expected to breach them. In this regard, epistemology has better prospects of revealing our limits than of telling us what we can know and how.

For instance, if no possible physical learner can compute the solution to a given problem (say, find the ideal strategy for chess), then surely no human can do it, so no human can be reasonably expected to do it, or to play chess according to it. Some problems might be physically computable, but no human, aided with the best 2026 technology, can solve them (at least in the next several years). For instance, it is believed that some encryption schemes like RSA cannot be cracked by our best 2026 computers, but can be cracked by a quantum computer. The best physically possible learner might be reasonably expected to treat a 4096-bit-RSA ciphertext exactly like they would treat the plaintext, but humans in 2026 cannot be reasonably expected to do so, because cracking the ciphertext requires computational resources that safely exceed the bounds of any human in 2026.[24] Humans in 2026 can only be reasonably

[23] We use our idealised model to represent the best possible member of the class of learners in question (i.e., the upper bound of the class), so using more ideal models means increasing the upper bound.
[24] Again, as far as we know.

expected to adopt ways of learning that give good results using the computational resources available to *them*.

Now, I grant Carr's point that if we seek normative robustness, then *some* idealisation is required. If we take the actual capabilities of each specific learner as determining which circumstances are relevant when asking what this learner ought to do, then, since what an individual is said to be able to do depends on context, we can hardly make any context-invariant normative judgments. However, the fact that we need *some* idealisation doesn't mean that we must take only ideal unbounded learners as our model.

If we focus on ideal physical learners, or ideal human learners, or other ideal models of imperfect learners, what these ideal imperfect learners *can* do does not depend on context, because these ideal models function as an upper bound for their class in every context. Whether some things they can do are better than others need not be conventional, because all reasonable orderings can agree on that. Whether a given limitation applies to the model is not a matter of convention, but rather a matter of whether this limitation applies to all members of the class of learners we're interested in or not. I conclude that to do epistemology in a normatively robust way, one need not restrict one's focus only to ideal unbounded learners.

7. Which 'Can' Is Implied by Which 'Ought'?

Carr's second argument for the non-robustness of the modal base concerns which sense of 'can' is implied by the epistemic 'ought'. According to Carr, there are reasons to think that the 'can' implied by the 'ought' of epistemology of ideal unbounded learners does not correspond to something significantly more substantive than metaphysical possibility. Epistemology of less ideal learners should provide an 'ought' that corresponds to a more substantive sense of 'can'. The 'can' of less ideal epistemology, according to Carr, must be sensitive to individual variations in ability. However, there is no normatively robust way to decide which actual variations in ability should be idealised away and which should not. In Carr's words:

> "A sociological observation: the 'can' implied by the traditional non-ideal epistemologists' 'ought' is not sensitive to individual variations in ability. It makes no special allowances for particular dispositions toward biases: no dispensations for somatoparaphrenics[25] or Alex Jones-style conspiracy theorists. And yet the following seems equally obvious: if there were a non-arbitrary, normatively privileged 'can' for epistemic OIC [Ought-Implies-Can], it would be sensitive to individual variations in ability. This is a bare intuition. But we do accept this bare intuition for the 'can' of OIC in ethics and practical rationality" (Carr, 1154).

If the 'can' that is implied by the non-ideal 'ought' is sensitive to individual variations in ability, she continues, then "[t]he 'can' implied by the non-ideal epistemic 'ought' draws an arbitrary, normatively non-robust distinction: some people's limitations constitute irrationality; others' are treated as faultless" (ibid.).

I would like to suggest an alternative picture. Carr's intuition is not at all obvious. The 'can' in epistemology of imperfect learners *is* substantive, and it is indeed, contrary to Carr's intuition, not sensitive to *individual* variations in ability. Rather, it is based on the abilities of *classes* of learners. Each class is associated with a different sub-section of the *epistemic landscape*, that is, the set of all worlds ranked in terms of epistemic ideality. Focusing on a class lets us restrict the epistemic 'ought' to this sub-section of the epistemic landscape, and this restricted 'ought' implies a corresponding restricted 'can'. It is not about "ideal" vs. "non-ideal" epistemology. The classes of learners that give rise to normatively robust 'ought's are all idealised. However, some classes of learners pick sub-sections of the epistemic landscape that are more relevant to the judgments we usually make.

[25] Somatoparaphrenics are people who suffer from the delusion that their limbs belong to someone else.

The following example of an ability of a class rather than individuals might help. It is within human ability to run 100 meters in 9.58 seconds. We know humans *can* do it because Usain Bolt did it. I can’t run 100 meters in 9.58 seconds. My abilities are pathetic relative to Usain Bolt’s. However, my actual individual abilities change nothing in how fast *humans* can run.

Maybe humans can run faster than Usain Bolt, even significantly. We don’t know yet. It’s hard to say what the *least* upper bound on how fast humans can run is. However, we do know that humans can’t run as fast as cheetahs. Cheetahs' running speed can be safely used as an upper bound for human running abilities. No human, even Usain Bolt, can be reasonably expected to run as fast as cheetahs. When we say that Usain Bolt is fast, or that I’m not, we measure it according to a scale that is relevant to human running abilities. The scale is not sensitive to whether Usain Bolt was born with much greater running talent than I was or how hard we both trained, but to human running abilities in general. Compared to the speed of light or even a Ferrari, Usain Bolt’s running speed and mine are almost equally pathetic. Nothing in principle stops us from making this comparison to the speed of light, but there is not a lot to be learnt from it.

Something similar happens in epistemology, which is apparent in how epistemic praise and blame work. Epistemic praise and blame are not about one’s moral character but about one’s learning abilities. One doesn’t necessarily do something morally right or wrong when one complies with or violates epistemological norms. Rather, we praise or condemn people for being (ir)rational, (un)reasonable, or, when focusing on their general epistemic character, for being smart, wise, stupid, careless, and so on, regardless of whether they have a choice in it. Even if one’s cognitive abilities are so low that one cannot avoid being stupid, one is still considered stupid. Similarly for praise. One is epistemically praised for being smart, careful, and so on, even if one’s smart way of learning is involuntary.

Now, all bounded learners are stupid compared to ideal unbounded ones. Let's acknowledge this triviality. Moreover, we humans are all stupid compared to the best physically possible learners. Stupidity is relative, so naturally, when we talk about someone being smart or stupid, we usually measure it relative to a scale that is only a relevant sub-interval of the interval that spreads between totally stupid and unboundedly smart.

The need for different scales for different judgments is what gives rise to different variants of the epistemic 'ought'. Each variant takes a different class of learners and says what is reasonably expected from the best members of the class. The 'can' implied by each 'ought' refers to the epistemic ability of the class of learners in question. The individual abilities of actual members of the class are indeed irrelevant. However, the closer the class of learners' abilities correlate with the actual abilities of specific humans (i.e., the closer the class is to being a *least* upper bound for human epistemic ability), the closer the 'can' implied by this variant of 'ought' is to some actual people's individual abilities.

Moreover, if we are interested not in what the smartest humans can learn, but in what "smart enough" humans, say, average humans, can learn, we can define a class of learners whose bounds are determined by (say) human average epistemic abilities. The bounds of this class will not be upper bounds on human abilities. To be "smart enough" according to this model, one only needs to do at least as good as what the average human learner ought to do. The 'ought' of the average human learner implies a 'can' that corresponds to the actual abilities of many people, and doing better than the average human can be considered epistemically supererogatory.

Of course, which *scale* we are using to measure what one ought to do, or can be reasonably expected to do, does vary with context, but it doesn't mean that what the scales measure also varies with context. When we say that a person is tall compared to the average human but short compared to mountains, we don't think that the person's height is context-dependent. The epistemic landscape itself, what the scales measure, need not depend on context, even if it's beyond our abilities to grasp the totality of it. However, we do need this ability to change scales by context. When we say that a dog is smart, it's usually clear from the context that we don't judge it according to the same scale as when we say that a human is smart. A theory that implies otherwise is defective.

8. What Went Wrong with Bayesian Solutions to the Paradox of the Ravens

The discussion in this chapter helps to shed light on why Bayesian solutions to the Paradox of the Ravens fail, as I argued in Chapter 5. My argument there was a reductio. Unless we impose unreasonable restrictions on the subject's priors, we get the absurd result that the cumulative support provided by all non-black non-ravens we see is non-negligible compared to the cumulative support provided by all black ravens we see. Like any reductio argument, it shows, if successful, that at least one of the premises is false. However, I didn't say which premises Bayesian solutions rely on are false, so it's not yet clear what makes these solutions go wrong. The premises of the standard Bayesian solution seem true at a first glance, so which one is to blame?

My answer, somewhat roughly, is that all of them are. The basic reason for that is that Bayesians use the wrong idealisation. All idealisations are false, in a sense, but some are false in a benign way, which doesn't propagate to the conclusion, while others, like the one used in Bayesian solutions, are malignant. A Bayesian learner is unbounded, and as we saw, there is a difficulty with taking a model of an ideal unbounded learner as a guide for how bounded learners learn or should learn. More concretely, Bayesian solutions presume, either descriptively, that we have prior probabilities regarding ravens and non-ravens, or normatively, that to understand what we epistemically ought to believe about ravens, we should be modelled as having such prior probabilities. But bounded as we are, we don't (descriptively) have such priors, and we shouldn't be modelled as having them to infer what we ought to believe.

For the descriptive part, since the concepts *raven* and *non-raven* are not realistically learnable without seeing examples of ravens (or examples that depict ravens), a realistic learner, either idealised or actual, cannot have prior probabilities regarding ravens and non-ravens as such. A realistic learner can only form probabilities about ravens and non-ravens after seeing some examples of ravens, that is, only after already having some information about ravens. This means that descriptively, in an important sense, a realistic learner doesn't have priors about ravens and non-ravens, only posteriors (i.e., probabilities that are learnt from evidence but not learnt via conditionalisation on prior probabilities). It is incorrect to say that realistic learners have priors regarding ravens and non-ravens before seeing any ravens. Thus, it's incorrect to describe realistic learners as learning by conditionalising on prior probabilities regarding ravens and non-ravens.

Even after the learner has seen some ravens, it's unclear that the learner has the kind of credences required by the Bayesian solution. The learner needn't have any idea regarding how common ravens are, how common black things are, how common black things that are non-ravens are, or how many objects there are in the universe. I can't say I have more than a very vague idea about these matters, except regarding the number of ravens, which I checked once while writing this thesis. There must be other ways to learn about ravens that are at least as good for the purposes of realistic learners. As I said above, a realistic learner constantly gets unanticipated evidence, so updating based on prior probabilities is rarely the only means by which realistic learners learn.

Now for the normative part. Of course, *mathematically*, any posteriors a realistic learner has after seeing ravens can be modelled as resulting from some specific sets of priors by conditionalisation over the evidence the learner has. However, mathematically, this is equivalent to restricting the priors we allow the learner in our model to have post hoc to only those that result in what a realistic learner can learn *after* seeing ravens. There is no a priori reason to think of ourselves as restricted to a specific set of permissible priors this way. The only reason to restrict the permissible priors in such a way, provided that we so eagerly want to model ourselves as learning by conditionalisation, is due to the actual characteristics of ravens, which can only be learnt after seeing ravens. What does the work in restricting the permissible priors in the model is what's learnt from examples of ravens, not Bayesian considerations.

For realistic learners, conditionalisation is only one learning tool among many, and I have yet to see a good argument for why it is best for realistic learners to learn only by using this tool, no matter the circumstances. I suspect no such argument can be given. Without further argument, taking conditionalisation as the sole learning technique that a realistic learner should master and as the sole means for forming justified beliefs based on evidence is akin to conducting a quality assurance (QA) on an air conditioner by modelling it as working only in heat mode. I suspect that an air conditioner that went only through this kind of QA would not work as an air conditioner ought to. To be more exact, its heat mode would probably work as it should, so by turning the heat mode the air conditioner would do what it ought to, but it would only make things worse at this time of the year. Thank God QA engineers make sure that air conditioners work well in all modes. A learner is epistemically better off if all the learner's modes of reason work well, not just one of them, and if the learner can turn each mode of reason on and off at the right time.

Conclusion

The unifying theme that underlies this thesis is that learning should be a basic topic for epistemology, and that limitations on learners are not merely practical obstacles but essential for the analysis of learning situations. The different chapters in this thesis can be seen as exemplifying various aspects of our limitations as learners, their importance to epistemology, and the potential harm of ignoring them.

Chapter 1 showed that the apparent symmetry between *green* and *grue* gains much of its force from ignoring limitations on how we can encode data about the world. Introducing these limitations can break the apparent symmetry and dissolve the puzzle, insofar as the puzzle should pose a distinct challenge beyond Hume's riddle. Chapters 2-3 explained how bounds on computational resources can limit what learners can learn. Chapter 4 explained how bounds on learners restrict what they can learn from various kinds of data, and thus why learnable structures within the data are crucial for bounded learners. Chapters 5-6 showed again that ignoring limitations, this time on how we learn from examples, leads to an apparent symmetry between examples of ravens and examples of non-ravens. Bayesian solutions that ignore these limitations lead to implausible consequences. However, once we consider essential limitations in our learning and acknowledge the importance of learnable structures, the apparent symmetry dissolves.

Chapter 7 generalised the lesson of the preceding chapters. The thesis discussed two classical epistemological puzzles whose force, I argue, arises from a failure to acknowledge our limitations as learners. These limitations concern encodability and complexity, and the puzzles can be dissolved by acknowledging them. A natural conclusion is that our models in formal epistemology and the way we idealise in epistemology in general should reflect the importance of these limitations. This invites us to rethink the way we usually do formal epistemology and enrich formal epistemology with other kinds of models. In the debate regarding the priority of non-ideal epistemology versus fully ideal epistemology, my approach suggests a middle ground. Some idealisation is needed, but there are various levels and kinds of idealisation, and we should idealise in ways that are suitable for our specific epistemological purposes.

Taken together, these arguments suggest that we should be careful not to do epistemology in too coarse-grained a manner. When analysing the epistemic relations between our evidence and hypotheses, we should not analyse everything only at the level of propositions and predicates. Not all propositions and predicates are epistemically on a par. The way the learner can encode information about the objects to which the predicates apply, and the computations that are feasible for the learner are of great epistemological importance. We have seen this lesson recur in the cases of grue, encryption, and the ravens. It is this coarse-grained analysis that creates these apparent paradoxical symmetries, and a more fine-grained analysis that incorporates the limitations of the learner into the model that breaks them.

One can alternatively describe the thesis as an attempt to examine some implications of two plausible assumptions: that learning is a computational process and that data externalism is true. At a lower level, this also roughly captures what the thesis does. The thesis does not aim to provide a complete independent defence of the premises. Instead, it spends more effort on motivating them by showing that they are independently plausible and philosophically fruitful. One major reason for this is that people have already offered arguments for claims that are similar to these premises in other contexts. However, the most important reason is that I think that the underlying insight about the importance of our limitations as learners to epistemology is more basic and stands independently of the specific premises we adopt to capture it. One need not accept every detail of the framework presented in this thesis to be convinced that limitations of learners, and specifically limitations regarding encodability and complexity, should not be ignored.

This thesis opens various directions for future research. I believe that many other philosophical problems can be analysed in the spirit of the approach developed here. Below is a short and non-exhaustive list of some topics that arise naturally from the arguments in this thesis.

Hume's Problem

I have offered possible solutions to Goodman's New Riddle and Hempel's Paradox of the Ravens, and in both cases, I said that my goal is to show that the paradoxes don't pose a further challenge *beyond* Hume's Problem of Induction. I didn't, however, offer any idea of a solution to Hume's problem. Among the three, Hume's problem strikes me as the deepest and hardest to solve. In a sense, unlike the other two, Hume's problem is more about the world than about us and our learning processes. It asks why it is the case that we can learn about the world, including its unobserved parts, from observations, and how we can come to know that this is so. However, I believe that the framework suggested in this thesis might illuminate some novel paths to engage with Hume's problem and analyse existing solutions.

Hume's problem seems to assume that we have past observations. Without this assumption, Hume's problem arguably collapses to scepticism about the external world. With this assumption, we can conclude that our world allows encoding information. This might be an important observation about the problem. Furthermore, solutions like Van Cleve's (1984) and Reichenbach's (1938; 1949) can perhaps be presented more systematically in a computational learning framework, and in a way that evades some of their problems.

Causal Chains "of the Right Kind"

When presenting data externalism, I said that for data to encode information about physical phenomena there must be a causal chain between the phenomenon being encoded and the state of the data. I added that I take it to be a necessary condition rather than a sufficient one. Not every causal chain will do, but only causal chains "of the right kind". This thesis didn't expand on this, but a fuller account will need to include more details about the metasemantics in play here.

Encoding Logic and Mathematics

Relatedly, a fuller account will need to explain logical and mathematical learning. While a computational framework of the kind presented in this thesis will likely have little trouble with the problem of logical omniscience, there is an obvious difficulty in explaining what logical and mathematical learning are *about*, and how realistic learners can refer to logical and mathematical objects or truths. This problem is not unique to the framework presented here. However, my approach suggests a potentially fruitful and underexplored path. What we encode or refer to when we do logic and mathematics might be some structural and computational features of our data and learning processes. Learners who are reflective enough can learn about their own learning processes, and some features of their learning processes can be the kind of things that let learners learn about and refer to logic and mathematics. This is, of course, not an account of reference in logic and mathematics, but a possible starting point for developing one.

These are just three topics among many, where further philosophical progress can be made when thinking seriously about our limitations as learners. Just as Kant's dove cannot fly without air resistance, we cannot learn the way we do without our limitations. Limited learners as we are, we need some friction with the external world to be able to learn about it. Like doves, which fly not by abstracting away air resistance but by mastering their relation to it, when we understand how to use our limits, we can fly within them.

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